



Review

A survey on automated and AI-based tools for patent retrieval with a special focus on the life sciences domain

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ABSTRACT

Patent retrieval is fundamental for research and development in both the industrial and academic fields. This activity is performed for several reasons; for example, for understanding advances in technology within a specific field, or prior to filing a patent application, or even to detect if there are any intellectual property violations. However, the huge quantity of patents, consistently increasing year by year, together with complexities characterizing patent documents, pose serious challenges to the patent system and its stakeholders. Classical research is time-consuming and it is becoming quickly inefficient; automated techniques are strongly needed. This work aims to be a comprehensive review of existing techniques, from query expansion to deep learning approaches, highlighting advantages and limitations. Furthermore, a specific focus is then reserved for the Life Sciences domain, where custom needs and issues, consisting mainly of nomenclature ambiguities, polysemy of terms, and high dependency on visual representations (e.g., chemical structures), need to be addressed. The final outcome shows that current automated methods, especially in the life sciences domain, suffer from suboptimal recall and fragmentation of the information across databases, highlighting the necessity for multimodal methods integrating multiple kinds of data from different sources.

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1. Introduction

In recent years, the volume of both scientific literature and patent documents in the life sciences domain has significantly increased, reflecting the rapid development in medical devices, pharmaceutical, and biotechnology fields [1,2]. This burst of innovation associated with important growth in patent applications is posing noticeable pressure on the entire patent system and all patent stakeholders, especially for those involved in patent retrieval. This task is a particularly cumbersome, time-consuming, and sometimes, not completely efficient procedure. Indeed, it is often described as a “looking for a needle in a haystack” practice. Besides the growing number of patents, there are additional challenges: patent documents are often long, technically dense, written using a highly specific, legal, complex, and ambiguous language [3], making them hard to interpret and search, independently from domain-specific knowledge and legal and technical language understanding of experts involved. Another issue affecting patent retrieval derives from the translation problems. In detail, the growing number of patents filed in Asian countries. Indeed, according to WIPO, Asian patent applications, primarily from China, account for close to 70% of all global patents in 2023, showing an increase of around 10% from 2013 [4]. Asian patent documents translation in English or other languages is fundamental for overcoming linguistic barriers and allowing a worldwide effective patent retrieval. Multi-lingual access depends on machine translation services embedded in portals such as WIPO Translate [5], available in WIPO’s global patent database PATENTSCOPE and as a stand-alone tool to translate patent text, or Patent Translate [6] provided by EPO. However, the quality of the translation has an impact on the recall of the retrieval. Evidence from a patent-focused machine translation evaluation indicates that the percentage of disclosure (the amount of information effectively understandable for search purposes) for Chinese-English translation decreases to about 68%, compared with an average of 77%–80% for European languages [7]. Moreover, patents represent a fundamental source of information for researchers in both industry and academia, in fact, the biggest part of cutting-edge technologies is solely disclosed through patents, and not published in journals: indeed, recent studies revealed that only around the 3% of chemical structures found in patents also appear in scientific literature resources [8], thus is fundamental retrieving patents to boost research.

Traditionally, users rely mainly on keyword searches combined with boolean, proximity and truncation operators, classification codes, scanning through all the documents’ figures within a class, citation networks, iteratively refining search queries, and combining all these strategies [9]. These searches are principally conducted either on free-access or commercial databases (see Section 2), some of which provide more advanced tools, for example multilingual, sequence or chemical

structure search, particularly useful for retrieving patents belonging to the life sciences domain (search strategies for life sciences entities are further presented in Section 1.1.3 together with their forms of representations). However, these approaches are really cumbersome, time-consuming, and sometimes lead to incomplete results, overlooking relevant patents, because of the huge quantity of documents. This, especially in patentability search, conducted by patent examiners to assess whether a patent can be granted, is particularly critical. In fact, patent retrieval is considered a recall-oriented task, where losing even a single relevant prior art document could cause serious legal and economic repercussions [10]. Given the need for enhancing, simplifying, and accelerating this process, there has been a growing interest in the development of automated tools oriented to support patent attorneys, professionals, and researchers in their search activity. These methods include approaches to expand the initial query [11], oriented to overcome the mismatch between the user request and relevant documents. Other approaches leverage bibliographic, citation, and classification data [12] or rely on machine learning algorithms [13]. Latest researches have been focused on deep learning [14] and natural language processing techniques [15], which capture the semantic context of words and phrases, allowing for more accurate retrieval even in the presence of vocabulary variation or abstract terminology. Additionally, these recent methods support image-based search for retrieving patents with similar visual cues.

Over the years, diverse surveys have been presented in order to collect works regarding techniques specifically for patent retrieval [12, 12,13] or patent analysis tasks in general, covering also the retrieval task. Some of them are focused on deep learning techniques [14], others on natural language processing approaches or machine learning technologies [15,16]. However, to the best of our knowledge, to date, there is a lack in a complete organization of the patent retrieval methods (including both traditional and innovative approaches, starting from basic query expansions to advanced deep learning methods) in a single comprehensive work, as well as very little focus is given to specific methods for the life sciences domain. Consequently, this work aims to fill this gap, presenting both a synthesis of approaches developed for retrieving patents in general and an overview of techniques specifically developed for biomedical, chemical, and biotechnological patents. Moreover, a new taxonomy for categorizing automated patent retrieval approaches has been presented.

In the following, brief hints about patent general related information and specific retrieval issues currently in the state of the art are presented, with a further focus on the life sciences domain.

For ease of reference, a list of acronyms used throughout the manuscript is provided in Appendix A. Moreover, for completeness, background material on patent essentials and non-retrieval-related patent analysis tasks is provided in Appendices B and C.

1.1. Retrieval-related patent analysis task

1.1.1. Types of search

The activity of patent retrieval aims to identify relevant patent documents in order to get specific information. Depending on the purpose of the patents search and the phase of the patent life-cycle, this activity is known by various names: state-of-the-art search, prior art-patentability-novelty search, freedom-to-operate search, opposition search, invalidity or infringement search [13].

The state-of-the-art search acts as an essential process before research and development, helping inventors and scientists understand advances in technology within a specific field [13]. It is basically a survey of all relevant patents and documents, such as scientific articles, theses and dissertations, the press, and other material as instruction manuals and promotional literature, published in a specific field [17].

Generally, before submitting a patent application, a preliminary search is carried out by industries, patent attorneys, and experts in the field. This search, once a patent application has been filed, is also conducted by patent officers during the examination phase to verify if the invention is novel, has an inventive aspect, has industrial and practical applicability, and does not fall in the non-patentable area, which includes, for instance, discoveries and mathematical formulas [17].

Since patents confer territorial and exclusive rights over an invention, so that third parties are prevented from making, using, or selling it for a limited period of time, prior to launching a product, a freedom to operate search is conducted. The aim of this task is to ensure that it does not violate existing and not-expired patents. It is also used to identify countries in which no patent protection exists or to determine when an existing protection is about to cease [17].

Depending on the jurisdiction, usually during the search-examination-grant process (in some cases until the grace period from the date of grant), third parties (“opponents”) can search and verify if there exist documents that may not have been considered or may have been overlooked by the examiner, that block invention from receiving a patent [17].

Later, during the patent lifetime, rivals may conduct infringement (or invalidity) searches to detect any potential intellectual property violations [13].

1.1.2. Traditional (non-automated) retrieval methods

Current approaches for patent retrieval are typically time-consuming and require a considerable amount of effort. Manually sifting through a huge amount of patents has additional drawbacks: the research could lead to poor and incomplete results. Patent examiners and patent attorneys usually depend on searches conducted on platforms/databases such as Espacenet (by European Patent Office), TotalPatent (by LexisNexis), or PatSnap, that can be paid services [18–21] or free-to-use [22–24] tools (a list of databases can be found in Section 2). In this process, users apply boolean, proximity, truncation, and positional operators, usually combined with classification codes, utilized for refining the query and optimizing it to specific needs. Field-specific filtering (e.g., within titles, abstracts, claims, assignees, and jurisdictions), citation networks, advanced cross-lingual functionalities and translation tools (e.g., the ones provided by WIPO or other public or commercial databases) are also leveraged. Among public patent platforms, Espacenet stands out for enabling direct querying of automatically translated English versions of patent documents originally published in non-English languages, thus facilitating multilingual patent retrieval. Once the most relevant IPC or CPC classes are identified, users may examine the patents within those classes to acquire deeper domain knowledge about the state of the art in the field of the invention and, then, iteratively refine their queries. Patent figures can also provide visual insights into the invention and are often used for sifting through documents within a class [9,17]. Patent experts have to manually and iteratively formulate the search query through the combination of the right keywords, selected knowing the

field [25], in order to get pertinent patent documents they need [26]. This approach is based on matching the exact words in the query formulated by the user and the textual parts, such as titles, abstracts, claims, and full descriptions, of candidate patent documents [27]. As a result, if a document uses another terminology or a more technical expression like “myocardial infarction” instead of the more general term “heart attack”, the system may fail to find that document. This issue is particularly relevant in the patent domain, where are frequently employed abstract, generalized, and highly technical and specialized terms to maximize the legal coverage of claims [26,28]. Consequently, even if done with most professional care, this research may produce suboptimal results, falling into two types of errors that have different consequences: missing of relevant documents (false negatives) and the presence of unrelated documents (false positives). Whereas false positives increase the workload for patent examiners, who must manually filter out non related patents, false negatives are far more critical. In fact, they may result in the wrongful grant of a patent, with serious issues for the patent applicant [26].

Despite all these strategies deployed by the users, together with the “federated search”, allowing for a simultaneous search on many diverse resources [29], patent search remains a really hard, cumbersome, time-consuming, and sometimes not efficient process, no matter the expert’s skills, experience, and deep knowledge of the field.

Considering these challenges, several automated methods have been developed with the final scope of accelerating, simplifying, and improving this task, in order to help patent professionals, researchers, and individuals interested in patent search. Automated methods have evolved from basic keyword and boolean to more advanced approaches based on deep learning and Natural Language Processing (NLP) [13]. They will be discussed in Section 4.

1.1.3. Life-sciences information in patents and how to retrieve it

Following the overview of common strategies applied with the aim of retrieving patents in general, it is worth noting that some technological domains, such as the life sciences, are characterized by additional challenges due to the heterogeneity of data types beyond plain text.

First, nucleotide and amino-acid sequences are disclosed in specific *sequence listings*, which, for applications filed on/after July 1 2022, are reported in a structured XML file, compliant with WIPO Standard ST.26 [30] (thanks to the desktop tool called “WIPO Sequence” [31]) with explicit and standardized elements and attributes for sequence IDs, residue types, organism names, sequence lengths, and the sequences. For patents filed before that date, sequence listings follow WIPO Standard ST.25 [32], being represented via plain-text files with tags (e.g., < 210> sequence ID, < 211> length, < 212> type, < 213> organism, < 400> sequence) and amino acids are written as three-letters abbreviations. These listings can be found in the “Sequence Search” section of WIPO PATENTSCOPE [33], where these files are uploaded weekly, but the website does not provide a dedicated sequence-search tool. For sequence-specific searching, specialized tools as the one on Lens PatSeq [34], can be used to find relevant sequences given an input sequence of interest. It leverages BLAST+ [35] (Basic Local Alignment Search Tool) for finding regions of local similarity between biological sequences, such as protein or nucleotide sequences. Another example is the commercial tool Derwent SequenceBase [18], which enables the search on the manually annotated GENESEQ database [36], a global patent sequence database produced by Clarivate (containing DNA, RNA, and protein sequences indexed from 50+ worldwide patent-issuing authorities, including WIPO, USPTO and EPO), by performing exact matches and sequence similarity using, respectively, proprietary engines as GETSEQ and GETSIM, as well as other algorithms like BLAST and Smith-Waterman.

Second, novel chemical compounds and small-molecules can be represented in patents as (or as a combination of) images, text, structures, often accompanied by digital files. Indeed, the extraction and research of these information require advanced tools and solutions for

handling the variety and complexity of the representations. They can be represented in several different ways (numerous chemical names, trade names, generic names, lab codes, CAS registry numbers, image representations, tables etc.), that can be mainly grouped in textual, numerical or graphical forms [37,38]. Moreover, a patent may describe individual specific molecules or even entire classes of molecules, described by the generic Markush structures [37], for claiming families of compounds. Among “machine-readable” formats, including systematic nomenclature (after parsing), linear notations, and connection tables, can be found two common systematic nomenclatures that are those developed by the International Union of Pure and Applied Chemistry (IUPAC) [39] and by Chemical Abstracts Service (CAS) [40]. CAS provides also a universally recognized unique identifier for a specific chemical substance, called CAS Registry Number (CAS RN), commonly used to link together all available data and research about that substance [41]. Linear notations consist of strings of alphanumeric characters that provide a full textual description of the molecule topology. Examples include the SMILES (for Simplified Molecular Input Line Entry Specification) notation developed by Daylight Chemical Information Systems Inc. [42] and the International Chemical Identifier (or InChI), and its condensed version (InChIKey: a hashed version of the full InChI) by IUPAC [43]. In other cases, since linear notations also lack in providing clues regarding a molecule's topology, connection tables, as MDL MOL developed by MDL Information Systems Inc. [44], are utilized for providing a complete and explicit description of a molecule's topology. Moreover, sometimes chemical structures are even represented without being named at all, meaning that keyword searching alone may not be sufficient for retrieving all relevant documents, and structure-searching is conducted on open-source or commercial databases for this reason [38]. Chemical structure searching can be performed through multiple strategies, depending on the aim of the search. These approaches include: exact structure search (searching for identical compounds), family structure search (including all of the exact matches, comprising also salts and mixtures), substructure search (looking for molecules containing the queried structure), Markush search (to determine if a queried compound is included within a broader generic claim defining chemical families) and similarity search (searching for compounds that have a structural likeness to the query) [38]. Among open-source databases, can be found PubChem [45], ChemSpider [46], SureChEMBL [47], WIPO Patentscope [48], which provide free access to chemical data extracted from patent literature (and, sometimes, not only for PubChem and ChemSpider). Additionally, there also exist structure-searching tools developed by chemical companies (e.g., Sigma-Aldrich [49]) to search products sold by themselves. Since searching on open source databases may lead to missing key patents, commercial databases such as Chemical Abstracts (CAS) [50], Reaxys/Beilstein [51], Derwent World Patents Index [52] are commonly used to conduct a more detailed research.

The following sections of this work are organized as follows. Section 2 illustrates the main databases containing patent and scientific literature, while in Section 3, the principal patent retrieval evaluation metrics are reported. A comprehensive review of traditional and automated approaches utilized for retrieving patents in general is presented in Section 4 while Section 5 provides techniques specifically developed for patents belonging to the life sciences domain. Finally, conclusive considerations are presented in Section 6.

2. Main patent databases

Databases containing patent and scientific literature are the first sources consulted by patent examiners, researchers, and legal professionals to retrieve industrial property information. These databases provide access to millions of patent documents and life sciences-related entities around the world, containing metadata, as well as textual and visual information related to inventions and biochemical entities. According to their scope, structure and function, they can be grouped

into three main categories (that are respectively presented in Sections 2.1, 2.2, 2.3): (i) general-purpose patent databases; (ii) thematic or small patent databases for testing scopes; (iii) scientific institutional and commercial databases covering chemical/biological data.

2.1. General-purpose patent databases

In this Section are listed general-purpose (both free-to-use and commercial) patent databases that are continuously updated and allow for retrieval by keywords, operators, classifications, and, in some cases, biochemical queries.

- PATENTSCOPE (WIPO - World Intellectual Property Organization) [48], a free access database developed and maintained by WIPO containing more than 123 million patent documents, including over 5 million published international patent applications (PCT). It supports advanced search tools, such as cross-lingual search and modules for retrieving chemical and biological sequence information [53];
- Esp@cenet (EPO - European Patent Office) [22], a free online service for searching patents and patent applications developed by the European Patent Office (EPO) and the member states of the European Patent Organization. This database presents over 150 million patent documents from over 100 countries;
- USPTO (United States Patent and Trademark Office) [54], providing Unites States full text patents from 1976;
- DEPATISnet (Deutsches Patentinformationssystem, German Patent and Trade Mark Office) [55], giving access to German, European, and extra European patents;
- Google Patents [23], a search engine that aggregates patent publications from 17 patent offices around the world, including the USPTO, EPO, and WIPO. It allows non-patent literature indexed in Google Scholar to support prior-art search;
- Derwent World Patents Index [52], by Clarivate, a commercial database, which comprises patent applications and grants from 44 of the world's patent issuing authorities.

2.2. Thematic or small patent databases for testing scopes

Smaller or thematic patent databases have been created for research and evaluation for the tasks of patent retrieval, classification, translation, sometimes specifically for bio-chemical patents:

- CLEF-IP (Conference Labs of the Evaluation Forum — Intellectual Property) [56] patent collections, provided for CLEF-IP initiatives [57], organized from 2009 to 2013 for encouraging research in patent retrieval, in which patents from USPTO, EPO, and WIPO in three main European languages (English, French, and German) are gathered;
- NTCIR (Japanese National Institute of Informatics Testbeds and Community for Information access Research project) [58] collections, between 2002–2013, dedicated to patents tasks, such as patent retrieval, classification, and translation;
- TREC-Chem (Text REtrieval Conference - Chemical IR Track) collections [59], developed for chemical tracks of TREC for motivating research on chemical patent retrieval;
- CHEMDNER corpus [60], developed within the BioCreative V Workshop (2015–2016), on the automatic recognition of chemical and biological entities within medical patents and scientific texts [61].

2.3. Scientific institutional and commercial databases covering chemical/biological data

In this Section, data sources that can be consulted while searching for life sciences entities, such as biological sequences or chemical compounds extracted from scientific and patent literature, are presented. They are listed below:

- BindingDB [62], a public web-accessible database, containing about 3,156,460 binding data for 11,367 proteins and over 1,380,881 drug-like molecules, extracted from the literature and patents;
- Patent Lens (or The Lens) [63] is a free online database of patents and literature, provided by Cambia (a non-profit organization in Australia). This database provides a dedicated module, Lens PatSeq [34], that can be used to find relevant sequences for retrieving nucleotide and protein sequences disclosed in patent documents, given an input sequence of interest.
- PATENTSCOPE [48] biological and chemical retrieval modules (respectively, the Chemical Compound search module [64] and the Sequence Search module [33]) allow search via chemical compounds and DNA and amino-acid sequence listings filed with PCT Applications [53]. Moreover, for larger-scale sequence data by year can be downloaded from the same page with the “ftp anonymous download service” [53].
- Chemical Abstracts Service (CAS) SciFinder-n [65], a commercial platform, which provides access to the world’s largest collection of chemistry and related scientific information, allowing research of substances, reactions, sequences, structures (including Markush structures).
- The National Center for Biotechnology Information (NCBI), a public research center of the U.S. National Institutes of Health (NIH), provides a comprehensive ecosystem of databases and computational pipelines for the management, integration, and retrieval of biological data. The infrastructure includes numerous databases for molecular and genomic information, including GenBank [66] for DNA sequences, Gene [67] for gene-specific information, Protein [68] for protein sequences, and PubChem [45] for chemical compounds and bioassays. Notably, NCBI has made substantial efforts to extensively cross-index patent documents with genes, proteins, and chemical entities, enabling integrated exploration of scientific and patent literature;
- The European Bioinformatics Institute (EMBL-EBI), a part of the European Molecular Biology Laboratory family, which is focused on bioinformatics research and services, maintains the world’s most comprehensive range of freely available and up-to-date molecular data resources [69]. Among the databases can be found: ChEMBL [70], containing bioactive molecules with drug-like properties, SureChEMBL [47], containing compounds extracted from the full text, images, and attachments of patent documents, UniProt [71], for protein sequence and functional annotations.;
- ChemSpider [46], a free database of more than 120 million chemical structures, properties, and associated information. ChemSpider integrates data from hundreds of sources and includes additional properties, related information, and links back to original data sources.
- GENESEQ [36], a global patent sequence database produced by Clarivate, containing DNA, RNA, and protein sequences indexed from 56 worldwide patent-issuing authorities, including WIPO/PCT, USPTO, Europe (EPO), Japan (JPO), South Korea (KIPO), India (IP India), and China (CNIPA).
- Other examples of life sciences related databases are the biological issues regularly published by the journal Nucleic Acid Research [72] since 1980s.

3. Metrics

The effectiveness of patent retrieval methods can be measured by diverse metric, including:

- Recall: the fraction of retrieved relevant documents over the total relevant documents. It is defined as:

$$Recall = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of relevant documents in the collection}} \quad (1)$$

- Precision: the fraction of retrieved relevant documents among the total of the retrieved documents.

$$Precision = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of retrieved documents}} \quad (2)$$

- F-score (F-measure): is the generalization of the F_1 -score, that is the harmonic mean of precision and recall, where the two metrics evenly weighted. It is computed as:

$$F_\beta = (1 + \beta^2) \cdot \frac{Precision \cdot Recall}{\beta^2 \cdot Precision + Recall} \quad (3)$$

Where β is a weighting factor, used to emphasize either precision or recall.

When $\beta = 1$, precision and recall are equally weighted (F_1 -score), when $\beta > 1$, recall is given more importance, and if $\beta < 1$ precision is emphasized. In this work, the performance results are always reported as F1-score.

- Mean Average Precision (MAP): is the mean of the Average Precision (AP) computed over a set of queries (Q). It reflects how well the system ranks relevant documents higher in the retrieval list. These two metrics are computed as follows:

$$AP(q) = \frac{1}{R} \sum_{k=1}^n P(k) \cdot rel(k) \quad (4)$$

Where R is the total number of relevant documents for the query, $P(k)$ is the precision at rank k , n is the number of retrieved documents, and $rel(k)$ is a binary value that is equal to 1 if the document at position k is relevant or to 0 otherwise.

$$MAP = \frac{1}{Q} \sum_{q \in Q} AP(q) \quad (5)$$

Where Q is the number of queries.

- Normalized Discounted Cumulative Gain (nDCG): is a normalization of the Discounted Cumulative Gain (DCG) measure. DCG, by applying a logarithmic discount to lower-ranked documents, favors retrieval systems that prioritize highly relevant documents in top ranks, while it penalizes the score for relevant documents that are ranked down in the list. Given a ranked list of documents, the DCG is computed as:

$$DCG = \sum_{i=1}^p \frac{rel_i}{\log_2(i+1)} \quad (6)$$

Where rel_i is the graded relevance score of the document at rank i .

Since DCG is not directly comparable across different queries, it is normalized by the Ideal DCG (IDCG), that represents the maximum possible DCG obtainable by perfectly ranking all relevant documents.

$$nDCG = \frac{DCG}{IDCG} \quad (7)$$

- Patent Retrieval Evaluation (PRES) score. This metric has been specifically designed to evaluate the performance of patent retrieval systems. It was proposed for the first time by Magdy and Jones in [73] and it combines the recall (R) of the system with the quality of rankings (r_i : rank of the i -th retrieved relevant document) of the retrieved documents within the documents to be checked by the user (N_{max}). PRES ranges from 0 to 1, where higher values indicate both high recall and early concentration of relevant results. The PRES score is computed as follows:

$$PRES = 1 - \frac{\sum r_i - \frac{n+1}{2}}{N_{max}} \quad (8)$$

$$\sum r_i = \sum_{i=1}^{nR} r_i + nR(N_{max} + n) - \frac{nR(nR-1)}{2} \quad (9)$$

Where R is the Recall (relevant retrieved documents in the first N_{max} documents) and n is the number of relevant documents,

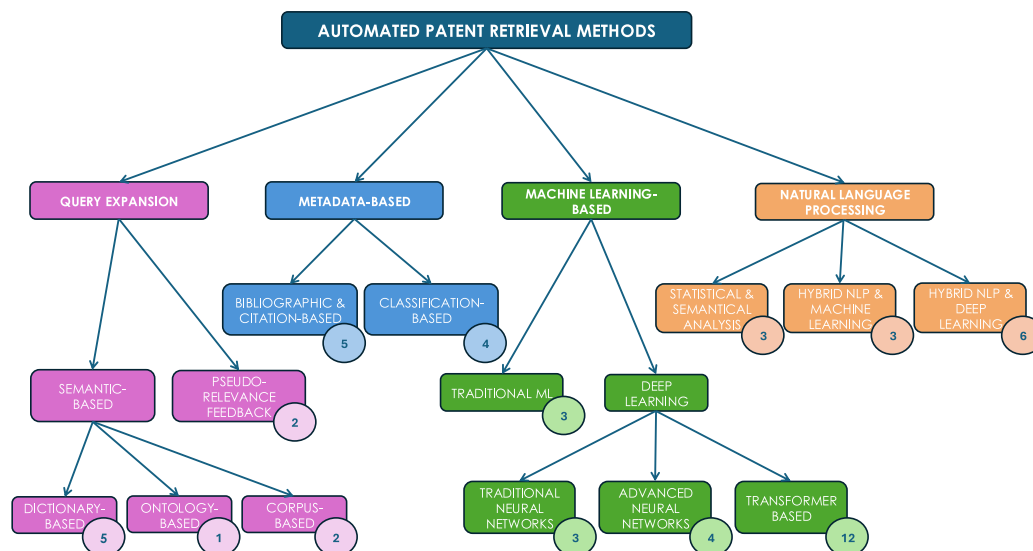


Fig. 1. Overview of the automated methods for patent retrieval. The scheme represents the main categories, with their further subcategories, of automated approaches for retrieving patents in general: query expansion methods, metadata-based methods, machine learning-based methods, and natural language processing-based methods. Additionally, numerical values indicating the number of studies analyzed within the present work per category are reported.

thus, nR is the total number of relevant documents retrieved. If some of the documents considered relevant are not retrieved within N_{max} , it is assumed that they are ranked with a value greater than N_{max} .

4. Automated approaches for patent retrieval

In this chapter, automated and AI-based approaches developed for patent retrieval are first presented in general, regardless of the domain. Although many of these approaches are technically applicable to the life sciences domain (such as pharmaceuticals, biotechnology, and medical technologies), they are not always optimized for its specific characteristics. For this reason, a separate section (Section 5) is dedicated to methods that have been specifically developed for patent retrieval in the life sciences domain. In this framework (depicted in Fig. 1), four main categories are presented, ranging from earlier approaches to more advanced and effective methods: query expansion, metadata, machine learning, and natural language processing-based methods.

4.1. Query expansion techniques

Most research regarding the improvement of patent search, for overcoming mismatch between search queries and relevant patents, has focused on query expansion techniques for automatically extending and enriching initial search queries, in order to include related terms that have similar meanings to the original query terms. They can mainly be divided into semantic-based techniques and methods based on pseudo-relevance feedback.

The results, in terms of metrics, of the proposed query expansion-based methods (when provided) are summarized in Table 1.

4.1.1. Semantic-based techniques

Semantic-based techniques improve patent retrieval via query expansion, by leveraging external sources to find terms related to those contained in the initial query. Depending on the source used, they can be grouped into dictionary-based, ontology-based, and corpus-based approaches.

- **Dictionary-based.** The first category consists of dictionary-based techniques. These methods retrieve synonyms, hypernyms, or hyponyms from either generic, technical, or patent-specific dictionaries for query expansion. For instance, Magdy and Jones [11]

test WordNet [80] (a large lexical database of English, built at Princeton University) on patent documents, for expanding query terms with their synonyms and hyponyms. However, their approach leads to poor results in the retrieval process due to the large number of expanded terms. They also propose an approach involving the automatic generation of a SynSet, a probabilistic synonym dictionary, from English and French translations of patents' titles and claims sections, through GIZA++ (a statistical word-based machine translation toolkit [81]). In order to assess the effectiveness of their SynSet dictionary on patent retrieval, they conduct two experiments: one using the probability associated with the SynSet entries as a weight for each expanded term in the query (Wsynset), the other one using uniform weighting for all synonyms (Usynset). Despite their developed approach showing to be superior over WordNet, the results by using the SynSet technique do not overcome baseline, with a decrease of 0.7% and 1.7% in terms of PRES for Wsynset and Usynset, respectively.

A proximity-based framework for query expansion that leverages a domain-specific lexicon as a source for query terms is built by Mahdabi et al. in [74]. The lexicon is derived from IPC classification, and consists of a set of different terms attributed to every IPC class, and serves for addressing term disambiguation. In order to choose the right expansion terms from the list of candidates, terms are given weights based on their textual proximity to the original query terms, using kernel functions to assign a weight to the distance between the terms. After that expansion terms from the IPC-lexicon have been identified, the authors propose four strategies to reformulate the query: (1) including terms from the lexicon that appear in the query document, (2) including all IPC-lexicon terms weighted considering proximity, (3) combining the original query terms with the ones from the IPC-lexicon, (4) using terms retrieved from top documents after a first search (as in Pseudo Relevance Feedback, discussed in 4.1.2) whose proximity to query terms is high (Proximity-based Pseudo Relevance Feedback). The strategies are tested by the authors on CLEF-IP 2010 and CLEF-IP 2011 datasets. Results show that the method achieving the best performance is (2), leveraging all terms in the developed lexicon, with a recall of 65.95%, even higher than a classical Pseudo Relevance Feedback model. However, the MAP score is quite low, probably due to noise created by the introduction of non-relevant IPC terms. Furthermore, relying solely on an IPC-lexicon

Table 1

Performance Comparison of Query Expansion Methods for Patent Retrieval. Only studies reporting the shown metrics are included. The table also specifies the patent fields used to build the retrieval corpus, allowing a comparison across studies based on the searchable content.

	Dataset	Patent fields of the retrieval corpus	Precision	MAP	Recall	PRES	F1-score
[11]	CLEF-IP 2010	bibliographic data, abstract, description, claims	–	0.1440	–	0.485	–
[74]	CLEF-IP 2010/2011	bibliographic data, abstract, description, claims	–	0.1050	0.6595	0.5540	–
[75]	Indian Patent Database	abstract	0.0513	0.1632	0.944	–	–
[76]	Espacenet (EPO)	not specified	0.13	–	1	–	0.23
[77]	NTCIR-6 USPTO	not specified	–	0.5434	0.8497	–	–
[3]	CLEF-IP 2010	title, abstract, description, claims	–	0.126	0.564	0.461	–
[78]	CLEF-IP 2010	title, abstract, description, claims	–	0.158	–	0.558	–
[79]	Indian Patent Database	abstract	0.0611	0.214	0.93	–	–

introduces the risk of retrieving irrelevant patents, due to its limited granularity compared to the more recent and fine-grained systems such as CPC (see Section 4.2.2 for a detailed discussion of patent classification systems). In fact, the Cooperative Patent Classification System was introduced to provide better specificity and better allocation of emerging technologies. This limitation highlights the need for an auxiliary disambiguation mechanism. A domain-specific lexical database to expand the search query, called PatNet, is developed by Tannebaum et al. in their work [82]. This database is created by downloading real query log files of USPTO patent examiners from Google as described in their previous work [83], in order to extract synonyms used by patent examiners using the Boolean operator “OR” and the proximity operator “ADJ”. PatNet for every single term of the query suggests: single synonyms, synonym phrases, and single terms, which in combination with the query term create a key-word phrase and suggests a synonym phrase. For the single terms PatNet finds, on average, 11 expansion terms. The authors apply three different query expansion strategies: synonyms are incrementally suggested (1) in five ranked batches (top-5 to top-25), (2) considering the USPTO patent classes, (3) performing word sense disambiguation (WSD), with a query context size of 20 words. However, in this study, the authors solely verify if the PatNet expansion terms correspond to the ones proposed by patent examiners, and they do not evaluate the synonyms’ impact on downstream patent retrieval tasks.

In order to find conceptually similar patents, Sharma et al. [75] use WordNet as an external source to expand the search query. The authors restrict this work to solely patents filed in Computer and related technologies fields. Their work includes these steps: (1) filtering patents’ abstracts from the computer technology field on IPC basis, (2) processing the abstracts to obtain keywords, (3) expanding the query, (4) building parent and concept vectors, and, lastly, (5) calculating similarity between terms. The abstract is used as a query, and IPC as metadata. Patents’ abstracts are indexed by removing stop words, undergoing stemming, and term weighting (term frequency–inverse document frequency, tf-idf). The query expansion step is made by searching synonyms, hypernyms, and hyponyms for every term in the search query and then assigning weights to terms. In particular, the original term is given a score of 1, while extended tokens receive a score between 0 and 1 that depends on the hierarchical distance between them and the original term. Thus, the expanded query becomes a vector with weighted terms, which is used to compute the similarity between query and candidate abstracts, by means of cosine and Wu-Palmer [84] similarities. The authors evaluate their method using mean recall, mean precision, and MAP, comparing it to a search carried out without query expansion. Results show that the method significantly increases the recall (from 81.1% to 94.40%), meaning it retrieves more relevant patents that would otherwise be missed. However, this comes at a cost: MAP drops from 25.11% to 16.32%, reflecting reduced precision because of some

expanded terms that may not be contextually relevant. The mean precision remains low (about 5%). Term weighting refinement or domain-specific ontologies integration could improve precision in [75].

As observed in previous studies, despite its widespread use, WordNet does not deliver high-quality results. For this reason, Sharma et al. in [79], leverage both WordNet and Wiktionary [85] (a lexical and multilingual dictionary which is constantly updated) to find keywords for enriching the query. In this study, the query is generated from a patent abstract, after that the text undergoes tokenization, stop words removal, stemming, term weighing, and tf-idf application, that gives terms a weight based on their presence in the abstract. Then, the query expansion step is performed by extracting terms such as hypernyms, hyponyms and synonyms from WordNet and Wiktionary, and keeping only words under a certain path length distance. Once the query is expanded, two different models are applied to find similar patents: cosine similarity (or vector space model), which computes the angle between vectors, and Wu-Palmer measure, which calculates similarity between the nearest shared parent concept and the distance from the root of the hierarchy. As a result of the experiments conducted on the Indian Patent Database, the combined use of WordNet and Wiktionary leads to a greater number of retrieved patents with respect to the individual use of them, and to an improvement of the performance both in terms of precision and recall, achieving a 93% of recall and 21% of MAP. However, the usage of two sources for query expansion does not address the problem of multiple meanings of words, which could worsen the precision of the method if irrelevant terms are chosen.

• **Ontology-based.** The second category of semantic-based methods consists of ontology-based approaches. Among the first of these methods, a custom-made ontology, called Graph-embedded Tree, was created by Phan et al. in their study [76], aimed to integrate patent classification hierarchies and the textual contents of the patent itself. This is a graph-based ontology characterized by nodes representing classes (of USA Patent Classification - USPC, a now discontinued classification system, which has been replaced by the CPC), documents, phrases, terms and relationships such as similarity, specificity, and hierarchical links. Their work includes (1) the development of a sentence similarity algorithm, (2) the construction of a Graph-embedded Tree (GeT-In), and (3) the implementation of a search method using the developed ontology. In (1), the sentence similarity algorithm first defines word meaning in the phrases by extracting the possible meanings (synsets from WordNet) of each word in the phrase and selecting the most relevant meaning by considering the remaining words in the phrase. The algorithm then measures the similarity between words, considering the length of the shortest path between two words on WordNet and the depth of the words in the hierarchy, and computes the word order similarity. Finally, the algorithm measures the semantic similarity between

two phrases. In the second phase (2), they construct a Graph-embedded Tree (GeT-In) search ontology, in which each node can be of four categories: Class Type (generated from a classification scheme), Document Type (from information of set of documents), Phrase Type (noun phrase from titles/abstracts), Term Type (single keywords). The ontology presents several relationships, among which: HAS_CHILD, IS_SPECIFIC_OF, IS_SIMILAR_TO, HAS_TERM. In (3), they then implement a search method using the developed ontology. The user query is firstly enriched by extracting nouns and noun phrases by identifying the correct meaning of the terms and utilizing HAS_TERM, IS_SIMILAR_TO, and IS_SPECIFIC_OF relationships. Then, the phrases are ranked by the most common occurrence, and relevant documents are searched. For the evaluation, the authors use a collection of 600 patents, in the G08Q10/08 class, related to Logistics and Transportation, from Espacenet, corresponding to a 20% of the total amount of patents in that class, due to limited resources and systems. They utilize Python to implement the algorithms and use WordNet to find synonyms and MongoDB [86]. Despite the work in [76] bringing better 15% in terms of F-Measure when compared with traditional approaches, as they claim, their work still needs improvement, because the noun and noun phrase-based extraction could not be sufficient (e.g., excluding verbs and adjectives). Moreover, hierarchical relationships (HAS_CHILD), based only on USPC classification, should be enriched by semantic meaning, and, furthermore, for only 600 patents the system is characterized by 62K nodes; thus, with a higher number of patents, the system would be very inefficient, reducing the scalability of their developed method. It is also worth noting that, since this approach relies on a now outdated classification scheme, its applicability to more recent datasets is limited.

- *Corpus-based.* The third category of query expansion approaches consists of corpus-based techniques, which, by analyzing large text corpora, automatically extract semantically related concepts to query terms which can be used for query expansion. Among these approaches, some methods exploit structured textual corpora like Wikipedia. For instance, Al-Shboul et al. [77] propose an approach which utilizes WordNet for single-word expansion and Wikipedia semantic annotations (i.e., categories) for query phrase expansion. The method first extracts query terms from patents by removing stop words (using KEA [87]/InQuery/Lemur [88] lists), then applies Stanford POS tagging [89] to identify nouns, verbs, and adjectives. Keywords and keyphrases are captured via RegEx. When the query is generated by extracting key terms from a query patent document, it is divided into a Bag Of Words (BOW) and a Bag Of Phrases (BOP), expanded via WordNet and Wikipedia, respectively. In particular, for phrase expansion, the method creates Wikipedia “surrogate pages”: every relevant Wikipedia page is converted into a semantic structure which includes primary categories and secondary categories of the page, titles of the pages under the primary and secondary categories belonging to the page. Then, the query phrases are compared to surrogate pages and receive a similarity score, extracting phrases found in 60% or more pages among the top 5 surrogate pages. Finally, n-grams covered by other longer n-grams are removed to avoid redundancy and ambiguity. The expanded query is utilized for retrieving patents (using Indri, an indexing and retrieval tool of the Lemur Project, and Okapi BM25 as a default retrieval model) that are re-ranked considering the IPC information associated with them. A patent is considered relevant (ground truth) for a query at one or more of the three levels (i.e., Sub-category (SC), Main Group (MG), or Sub-group (SG)), if its IPC matches with that of the query at least partially. They compare their method with other approaches, considering as a main baseline the Relevance Model (RM) [90] and results show that the proposed approach improves patent retrieval precision (MAP: + 2.55% (SC), + 6.38%

(MG), + 12.95% (SG)) over RM on NTCIR dataset by leveraging Wikipedia phrases to disambiguate queries, decreasing topic drift caused by RM and maintaining a similar recall.

Wang et al. present in their work [3] a domain lexicon-based query expansion approach that uses custom domain lexicons built from patents grouped by IPC classification. Their approach consists mainly of two stages. In the first stage (the domain-specific lexicon construction), the key concepts are identified by firstly filtering out generic and patent-specific stopwords that appear frequently, and then ranking the remaining terms by their importance in the domain (how frequently a concept appears in a domain, how representative those documents are for their IPC class). In the second stage (the domain lexicon-based query expansion), the proximity degree between candidate expansion concepts and query concepts is computed, considering distance and the co-occurrence frequency between them. Their approach, evaluated on the CLEF-IP dataset, indexed through Indri, outperformed the pseudo-relevance feedback and the Wikipedia-based expansion methods by improving the retrieval metrics, for instance, PRES by 13.5% and 7.2%, respectively.

4.1.2. Pseudo-relevance feedback method

Other approaches to enrich the query rely on the pseudo-relevance feedback method (PRF). With this method, the first k search results from an initial query are used to find keywords to expand the search query [78] or to reduce it by removing the least similar keywords [91]. The fundamental assumption of PRF is that the top-ranked documents contain useful words that could be utilized to identify relevant documents. However, not all the top-ranked documents are always relevant and, for this reason, noise words could be used to expand the query, leading to poor results, causing the query-topic drift [77,78,92]. To address this issue, in the study [92] by Bashir et al. a new strategy based on a clustering (cluster-LFT, clustering with Local Frequent Terms) approach to capture dominant relevant documents is presented. This approach begins with the classical pseudo-relevance feedback by retrieving the first top 35 documents and then clustering them, considering the most frequent terms that appear in each cluster. Only validated clusters (that contain at least 5 distinctive terms that are shared by 6/10 documents belonging to that cluster) are used to expand the original query. Although the results show (through the Gini coefficient) that this approach reduces the retrievability bias, making documents more equally retrievable, the lack of standard metrics limits direct comparison with other approaches. Also Mahdabi et al. in their work [78], aim to address the issue related to non-relevant primary retrieved documents, using for query expansion solely pertinent feedback documents selected by a predictive model. In particular, for every top ranked patent, features such as Relevance Score (the similarity score between the selected document and the whole feedback set), Clarity (how much a document is different from the collection), Topical Clarity (how closely the document is focused on the same topics as the query), and IPC-based Clarity (similarity between the patent and the other from the same IPC-class) are computed. These features are used as predictor variables (input features) for the Stochastic Gradient Boosting Trees (SGBT) model that predicts the true Average Precision (AP) of each feedback document, thus determining which patents can be used for query expansion. Both documents and queries are stemmed using the Porter Stemmer algorithm, while stop-word removal is performed by the standard Terrier stop-word list, and all the formulas and numeric references are removed. The authors use a five-fold cross-validation, and their method achieves a 53% MAP improvement over standard pseudo relevance feedback method and 15% over query reduction baselines by considering only pertinent documents among the top-ranked ones. However, despite these techniques improving fairness and retrieval coverage, the effectiveness of pseudo-relevance feedback methods still depends on the quality of the feedback signal: some or

Table 2

Performance Comparison of Metadata-Based Methods for Patent Retrieval. Only studies which provide results in terms of the presented metrics are reported. The table also specifies the patent fields used to build the retrieval corpus, allowing a comparison across studies based on the searchable content.

	Dataset	Patent fields of the retrieval corpus	MAP	Recall	PRES	nDCG	F1-score
[95]	NTCIR-6 USPTO	claims	0.0811	0.3991	–	–	–
[96]	CLEF-IP 2010/2011	bibliographic data, title, abstract, description, claims, citations	0.1250	0.6470	0.5363	–	–
[28]	CLEF-IP 2011	bibliographic data, abstract, description, claims	0.105	0.579	0.481	0.251	–
[97]	USPTO	title, abstract, inventors, citations	–	0.812	–	–	0.726
[99]	NTCIR-4	claims	0.2257	–	–	–	–
[98]	TREC-Chem'09	title, abstract, claims, description	0.0626	0.2585	–	0.2844	–
[101]	MAREC	title, abstract, description, claims, citations, ipc	0.097	0.423	–	–	–

even all of the retrieved documents may be irrelevant [93], introducing noise expansion terms [3].

In summary, the methods analyzed in the present Section highlight how query expansion techniques improve recall by reducing the vocabulary mismatch between the search query and patent documents. However, the introduction of noisy terms from general resources or irrelevant top-k documents leads to a reduction in terms of precision. Semantic-based techniques leverage external sources, yet their effectiveness heavily depends on domain specificity, while pseudo-relevance feedback methods are really sensitive to the quality of the first retrieved documents.

4.2. Metadata-based methods

Other automatic approaches developed to enhance patent search involve the usage of metadata, including bibliographic data and citations, and classification codes. Bibliographic data include details such as patent title, inventors, assignees, filing, publication, and priority dates, and patent number. While some studies rely solely on bibliographic data [94], the majority exploit citation-based methods [95,96], often combined with bibliographic metadata [28,97], and classification codes (e.g., IPC, CPC) [98–100].

Bibliometric methods can be divided into approaches leveraging bibliographic and citation data, and classification-based methods.

The results, in terms of metrics, of the proposed query metadata-based methods (when provided) are summarized in Table 2.

4.2.1. Bibliographic & citation-based

These methods leverage structured information, such as bibliographic data (e.g., inventors, assignees, classification codes) and citation networks to discover hidden relationships between patents that textual content alone might miss. In the patent domain, each document cites prior patents (backwards citations) and may be cited by subsequent patents (forward citations), creating in this way a citation network. The citation-based approaches leverage the principle that patents that cite or are cited together are often related to similar technological fields, even if they differ lexically, overcoming also vocabulary mismatch. Moreover, usually the citations are created by patent examiners and inventors, providing semantically meaningful connections.

Wang et al. in [94] propose an algorithm that utilizes the assignee's data, called the Keyword Snowball Algorithm (KSA), oriented to find the “design-around” synonyms (unusual technical terms used by patent writers for avoiding the infringement of other patents or for reducing the discoverability of the patent from other competitors). This method is tested on DVD3C (organized by Sony and Philips) and DVD6C (formed in 1999 between Toshiba and other five companies) patent pools [102]. In particular, the KSA algorithm computes the Inverse Assignee Frequency (IAF) for each term from an initial set of significant-frequent terms (T_{SF}), so that if a term is used by few assignees the IAF is high (it is rare, and it is potentially a design around synonym). The significant-rare coefficient (WSR) is then calculated by

considering the co-occurrence of the considered term with the others from the initial set of commonly used technical keywords and the IAF. This process is then iteratively repeated (in a “snowball” manner) for all the terms in T_{SF} , selecting only the ones with WSR over a certain threshold manually defined by experts, depending on the dataset. A limitation of this study could be that this method requires human intervention to validate the results. Moreover, in this study, the method is tested only on two pools of patents with their specific characteristics. A limit of this approach is that not knowing the semantic meaning, the algorithm identifies uncommon terms, which could also be typos.

In [95], the improvement of the retrieval, using a combination of claim text-based (Okapi BM25 [103]) and citation-based methods, is demonstrated using USPTO (1993–2000) patents. In particular, Fuji et al. propose two citation-based retrieval methods, both of which are combined with the previously applied text-based retrieval, computing the ranking score by a formula that multiplies text-based and citation-based (whose relative weight is controlled by a constant α) scores for each document. One of the citation-based retrieval is “static” and adapts PageRank [104] (originally designed for ranking web pages) to patent citations, treating the citations as “votes” and applying a fixed score to a patent depending on how many times it has been cited. The other is a topic-sensitive method, using only citation links between the N-top patents retrieved by the text-based method and assigning a weighted score (if a patent cites n other patents, every cited patent receives a $1/n$ score). Their experiments show that combining text with citation data improves recall and precision, particularly when citations are restricted to relevant patents (topic-sensitive approach), with a Recall@100 of 0.2171 and a MAP of 0.0811. However, the proposed approach includes a parameter that needs to be optimized by preliminary experiments.

While the previous study relies on static PageRank [95], a later work by Mahdabi et al. [96] addresses its bias against recent patents (a very old patent would have a high score even if it is irrelevant for a recent technology) by introducing a time-aware random walk (temporal decay), giving more importance to newer nodes. Moreover, their method weights the edges not only by citations and temporal proximity, but also via textual similarity, contextual similarity (common classification codes, common inventor, and common applicant). They first build a text-only baseline performing a query-based search, using a bigram model, and taking the top-k documents (“root set”) and, then, including documents citing or cited by any element of the root set. The baseline approach was compared to the proposed method based on a time-aware random walk on a weighted network of patent citations. The developed approach is tested on CLEF-IP 2010 and CLEF-IP 2011 patent datasets, from which they extract title, abstract, description, claims, patent publication date, classification codes, and bibliographic data, such as inventor names and applicant names for all documents. Only citations made by the patent examiner on the search report are considered (initial citations by the patent applicant excluded). Their method achieves a 26.9%–10% improvement in MAP, recall, and PRES over the baseline approach. Their strategy, which also considers IPC information, could be improved by considering the

hierarchical similarity, instead of relying solely on counting the amount of shared IPC codes.

In their work [28], Mahdabi and Crestani propose a citation-based approach for query expansion. Firstly, they build a topic-dependent citation graph, starting from the initially retrieved set of feedback documents and utilizing citation links of feedback documents to expand the set. They then identify relevant patents by means of citation analysis. The score of the edges between patents is determined by the sum of the votes from other documents in the citation graph, through a modified version of PageRank. The main assumption of their work is that documents with more importance in the citation graph are more likely to be relevant and thus term selection from them is more effective. Finally, they select the relevant terms from the cited documents by means of the computed relevance of the documents. They test their method on CLEF-IP 2011 patents, and use the Terrier Information Retrieval System to index the CLEF-IP 2011 collection (using the default stemming and stop-word removal), also removing numbers. Despite results showing that using the link-based structure of the citations leads to improvements over a strictly textual-based method with a MAP and recall of 0.105 and 0.579, temporal bias has not been considered, giving more importance to older more cited patents.

While early studies focus on individual data types, recent advances demonstrate that integrating multiple data sources, such as citations and inventors networks lead to better retrieval performances. In fact, Siddharth et al. in their work [97] utilize inventor information, together with textual data (titles and abstracts) and citation data, to retrieve patents relevant to an initial set of patents, enhancing the retrieval task. This method leverages Sentence-BERT for semantic text embeddings and TransE-based knowledge graph embeddings for inventor (<inventor, write, patent>) and citation (<patent, cite, patent>) networks. The target patents are retrieved using mean cosine similarity. Through the open-source data platform called PatentsView are gathered title, abstract, citations and inventors information of almost 8000000 utility patents granted by USPTO from 1976 to 2021. They analyze the results for diverse combinations of text (A), citation (B) and inventor(C) embeddings with concatenation and addition operations, finding that combining these approaches is better than considering them alone. In particular, the best combination, that overcomes the text only embeddings approach (A) consists in the concatenation of A, B and C, that leads to an increase of 12%, 2.5%, 11% in terms of precision, top-10 accuracy, and F1-score, respectively. Moreover, the authors apply their method to a set of patents. In particular, in the case study, they use a seed set of 10 patents previously identified by Song and Luo [105], obtained by keyword search (“rolling toy” and “rolling robot”) to retrieve 36 relevant patents in the domain of spherical self-propelled robots, finding that some of the retrieved patents do not contain the initial keywords, neither share inventors or citations, but they are still relevant, going beyond the traditional keywords method. This could be due to the use of Sentence-BERT that captures semantic similarity (the concept of a spherical robot) and TransE that finds indirect citations.

4.2.2. Classification-based

Patent documents are classified into mechanical, electrical, and chemical categories according to their technological field. In fact, patent examiners assign classification codes to all patent documents: these codes can be considered as topic labels about the content of the patent [99] and can be utilized in patent retrieval. In fact, diverse classification-based methods have been implemented in order to address issues related to the use of keyword searching, including the use of vague or insistent terminology, a variety of synonyms, typos, acronyms and abbreviations and the different meaning of terms depending on the considered field [98]. Among the classification systems, the International Patent Classification (IPC) is the most widely adopted, and it represents the internationally standardized taxonomy maintained by the World Intellectual Property Organization (WIPO),

providing a shared and harmonized representation of the technological domains across jurisdictions at the international level. In addition to IPC, some patent offices utilize more detailed classification systems, such as the Cooperative Patent Classification (CPC), jointly developed by the European Patent Office (EPO) and the United States Patent and Trademark Office (USPTO). This system provides more detailed and granular technological subdivisions, and more frequent updates with respect to IPC, by preserving the overall structure of IPC (with Sections A-H), while extending it through additional subgroups and a new Section (Section Y) for emerging technologies [106]. Prior to the establishment of the CPC, EPO and USPTO maintained independent classification systems: the ECLA and USPC, respectively, that are no longer applied. The CPC was introduced with the aim of providing a harmonized, up-to-date, and detailed classification scheme [107].

Among the earliest classification-based approaches, Kang et al. in [99], investigate the use of IPC classification codes within cluster-based language models for patent retrieval. In particular, the authors develop two distinct IPC-driven statistical approaches: a smoothing oriented language model and a topic-oriented language model. The first approach adjusts term weights by combining the term frequency in the specific patent, in patents which have the same IPC code, and in the entire collection. The second model considers IPC clusters as distinct technological topics and blends document-specific and cluster-based term distributions. The authors evaluate the retrieval effectiveness of their methods on NTCIR-4 (1.7 M of Japanese patents) patent test sets on the invalidity search task (excluding any documents filed after the filing date of the target patent). From the claim sections, character bi-grams have been extracted as index terms and query terms and they use TREC_EVAL program to evaluate the performance. They find out that an excessive smoothing of document models with the collection models could be harmful, as patent claims are long but highly specific and verbose. They also get an improvement (albeit marginal, about 3% of MAP) over the baseline language model.

Chen and Chiu in [100] propose an IPC-based VSM (vector space model) to generate indexing vocabulary to address the weaknesses in traditional VSM in patent search (typically used in Information Retrieval tasks): the indexing vocabulary varies when changes occur in: (i) the document set, (ii) the vocabulary selection algorithm (tf-idf, entropy, chi-square), (iii) parameters of the algorithms, or (iv) language evolution. With the implemented method, indexing vocabulary remains unchanged, reaching a higher accuracy compared to other two traditional methods, entropy and chi-square. The approach consists in 5 phases: (1) collecting of patent documents (title and abstract) from the (USPTO) database, (2) text pre-processing (syntax tagging, word stemming, and stop-word elimination, keeping nouns and verbs), (3) generating category-term vectors (each category code is represented by a vector of terms, which are keywords or key-phrases), in which the weight of the terms are computed with a modified tf-idf formula, (4) transpose them into term-category vectors, and (5) generate document-category vectors. The method is tested on titles and abstracts of patents from USPTO database, considering in the same category only patents whose section, class, subclass, and group codes are entirely the same. The comparison between the IPC-based method and the entropy and chi-square methods is realized by a manual (10 experts evaluate lists of similar documents retrieved by the three methods, giving a score from 1, low relevance, to 5, excellent) and automatic (computing the cosine similarity) evaluations. The proposed method receives a score of 3.5 over 5 with respect to 2.17 (entropy method), and 3.03 (chi-square method), with the manual evaluation. The automatic evaluation is performed considering that documents in the same IPC branch would have a higher similarity score than those in different branches. In this case, the IPC-based method reached a higher accuracy than the other two methods (e.g., on average, acc@10 for the proposed method (IPC) equal to 0.963, 0.803 (entropy method), 0.906 (chi-square method)). The authors also studied the stability of the vocabulary due to set changing, by expanding the document set, adding to the baseline set

(D_I) a half ($D_I + 1/2D_{II}$) and an entire other (and not overlapped) set of documents ($D_I + D_{II}$). The results show that for both the entropy and chi-square methods, no more than 80% and 70% of attributes were common between D_I and $D_I + 1/2D_{II}$, and between D_I and $D_I + D_{II}$, respectively, while for the IPC the score is 100% (no changes), proving the stability of their method. However, this method does not use the entire IPC hierarchy, stopping at the group level and ignoring the deeper level (sub-groups), and this could affect the precision.

In [98], Harris et al. explore how the hierarchical structure of IPC classification can be utilized in patent retrieval, using patents from the chemistry domain. The authors test four hypotheses: (1) hierarchical IPC relationships improve patent retrieval; (2) weighting IPC's 'Group' component more heavily than 'Class' boosts performance; (3) using all IPC codes outperforms primary-code-only searches; and (4) the primary IPC code alone still significantly improves over their baseline method. They first develop a machine-readable IPC classification representing each code by its full hierarchical path (unavailable in raw patent documents), adding it to the XML-formatted TREC-Chem'09 dataset that they used in this study. The IPC is split into two weighted components: Section/Class/Subclass and Group/Subgroup, to assess their impact on ranking. The distribution of IPC codes in their dataset is not uniform: the majority of IPC codes is linked to only a few patents, whereas some codes are tied to over 100 patents. To verify their hypotheses, they build a baseline approach retrieving a ranked set of up to 1000 prior art patent documents for each 100 patent applications from the PA-Small task of TREC-Chem '09 [108] using the methods discussed for TREC-Chem Run 1 in [109], that does not consider classification codes, but is based on text searches utilizing patent's title, abstract, claims and description parts. They use Indri (a search engine and an open source component of the Lemur Language Modeling Toolkit for information retrieval that uses language modeling with an inference network [110]) to index (organize in a structured database) patents and compare documents for similarity based on their IPC code fields, retrieving relevant candidates and ranking them by their similarity. The results show that the classification hierarchy increases the ability to retrieve relevant patents (MAP, Recall@100, and nDCG scores increase by 29%, 37%, and 27%, respectively). Additionally, their findings indicate that there is no substantial difference between the class part and the group part of the IPC code (both are important), but the best score is generated from a ratio between Class:Group 1.6:1. Moreover, while relying solely on the primary IPC code already improves performance over the baseline, incorporating all available IPC codes leads to even better results.

In [101], a prior art retrieval method that combines keyword phrase extraction with IPC classification vectors propagated through citation network is explored. In particular, the authors use TextRank*idf, a modified version of TextRank, that not only considers contextual relations, based on co-occurrences, but also promotes rare (but relevant) terms, in order to extract query terms from patents. Then, query and corpus patents are converted into binary vectors (with all zeros and a 1 if a patent presents a certain code) by means of IPC classification and citations information, and cosine similarity is utilized to retrieve similar vectors. The retrieved patents are subsequently re-ranked using the top 20 words in the query, scored by TextRank*idf. The experiments are performed on patents from the MAREC [111] dataset (contains over 19 million patent applications and granted patents from EPO, in English, German, and French, and expanded by documents from the WIPO), and data are indexed by Lemur [88] toolkit. The authors try three different methods: (1) COS (cosine similarity applied only to IPC classification) that leads to a high recall (0.600) and to a very low precision (0.049), thus finding a big amount of patent of the same category, but including also irrelevant ones; (2) TextRank*idf to extract words from the query patent, obtaining low recall and precision (0.399, 0.055, respectively); and (3) a combination of (1) and (2) resulting in a comparable recall 0.595 and a slightly higher precision (0.057). Then, the addition of citations to these three approaches worsen the MAP of COS (citation push upward less relevant documents) and elevates the performance

of the TextRank*idf approach, demonstrating that the introduction of citation data can be profitable, if well combined.

Among the reviewed metadata-based approaches, citation networks allow for a retrieval of conceptually related patents, while classification systems provide a representation of technological domains. These methods present some limitations as the temporal bias favoring older patents and a limited semantic signal deriving from a not exact correspondence between metadata and the actual technical content of the invention. Indeed, even if patents are connected by the citation networks, they may not necessarily address the same technical solution. Therefore, metadata information should be leveraged in combination with text rather than in isolation, because relying solely on it could lead to reduced precision.

4.3. Machine learning based techniques

In this Section, machine learning-based techniques, mainly categorized into traditional machine learning and deep learning methods, are presented. The results, in terms of metrics, of the proposed query machine learning-based approaches (when provided) are summarized in Table 3.

4.3.1. Traditional Machine Learning techniques

Traditional Machine Learning (ML) techniques have also been applied in patent retrieval tasks, particularly in classification, topic modeling, and result ranking. In [112], Kreuchauf and Korzinov develop a Support Vector Machine (SVM) learning method, used as a classifier to decide whether a patent belongs to a service robotic category or an industrial robotics one. They retrieve 228 patents from the 2013 April version of 'EPO Worldwide Patent Statistical Database' (PATSTAT) and split them into 205 for training and 23 for testing. They are selected from the IPC class B25J or contain keywords like 'robot*' in their titles or abstracts. Two groups of experts defined whether a patent belongs to one of the aforementioned categories. To transform patent texts into structured data, titles and abstracts are combined in one body, and the resulting strings are split into single terms, stop words are removed, and stemming is applied. Then, n-grams of term combinations (up to 3 words in one) are built, and normalized word and n-gram frequencies for each document are derived, constructing the matrix given to the machine learning algorithm. The algorithm is implemented in Python, using the scikit-learn tool for machine learning. While their model shows good performance in terms of precision (85%) and recall (83%), as the authors transparently acknowledge, there are some limitations: for instance, textual features are limited (claims are excluded) and there are computational scalability constraints (due to the manual validation of the 228 patents and the processing of 30,987 features, reduced to 1206 post-selection).

In [113], an AI-assisted patent retrieval system, which combines traditional machine learning and patent examiners' expertise, is developed. This system firstly classifies patents into three main domains (civil engineering, computing, and transportation) through SVM, obtaining an accuracy of 100%, then extracts the most significant keywords from the application and applies LDA for topic modeling and Word2Vec/WordNet for query expansion. Then, the system retrieves documents via Elasticsearch [127] and ranks them. However, the system achieves a low precision (between 30% and 50%), concluding that whereas ML techniques help in patent documents classification and topic modeling (that could help both applicants and patent examiners to explore technological domains or by data analysts to identify global innovation trends), fully automating prior art search requires more semantic understanding beyond traditional machine learning techniques capabilities.

In [29], two machine learning-based methods are proposed for merging results in patents federated search and tested on patent data (CLEF-IP). The authors create a Centralized Sample Index (CSI) containing document samples from distributed patent databases (e.g., Google

Table 3

Performance Comparison of Machine Learning-Based Methods for Patent Retrieval. Only studies which provide results in terms of the presented metrics are reported. The table also specifies the patent fields used to build the retrieval corpus, allowing a comparison across studies based on the searchable content.

	Dataset	Patent fields of the retrieval corpus	Recall	Precision	MAP	PRES	F1-score	Acc
[112]	PATSTAT	title, abstract	0.83	0.85	–	–	–	–
[113]	PATSTAT, UK IPO, EPO, USPTO	title, description (2000 words)	–	0.50	–	–	–	–
[29]	CLEF-IP	title, inventor/applicant information, abstract, description (500 words), claims	0.3468	–	0.0777	0.3358	–	–
[114]	Freepatent	patent figures (technical drawings, chemical structures, program codes, gene sequences, flowcharts, diagrams, mathematical formulas, tables)	–	–	–	–	–	0.30
[115]	DeepPatent	patent figures (design patent drawings)	–	–	0.376	–	–	0.691
[116]	DeepPatent	patent figures (design patent drawings)	–	–	0.622	–	–	–
[117]	DeepPatent	patent figures (design patent drawings)	–	–	0.712	–	–	–
[118]	DeepPatent2	patent figures (design patent drawings)	0.586	–	0.691	–	–	–
[119]	DeepPatent2	patent figures (design patent drawings)	–	–	0.458	–	–	–
[120]	WIPS	title, abstract, first claim	0.9429	0.7174	–	–	0.8148	–
[121]	EPO	patent figures (technical drawings, diagrams, graphs)	–	–	0.715	–	–	–
[97]	USPTO	title, abstract, inventors, citations	0.812	0.659	–	–	0.726	0.979
[122]	DeepPatent	patent figures (design patent drawings)	–	–	0.856	–	–	–
[123]	ClaimCiteRetrieval	first claims of patent applications, descriptions of not/relevant prior art (from search reports)	–	–	–	–	–	0.4966
[124]	USPD, KPRIS	full text (first 100 tokens)	–	–	–	–	–	–
[125]	–	description, claims	0.4046	–	–	–	–	–
[126]	DeepPatent	patent figures (drawings)	–	–	0.57	–	–	–

Patents, Espacenet). Once submitted, a query is sent to the selected resources, which will return a ranked list of documents with their score, and to the CSI. The overlapping documents shared between the resources and the centralized index are then utilized as training data for machine learning models (such as SVM, decision tree, random forest, linear regression), which learn to predict CSI global scores, which enable comparisons between documents. The authors test their methods in two different environments: a cooperative environment, where resources provide document scores, and an uncooperative environment, where only rankings are available, and they assign artificial scores. They use two methodologies: the first method consists of training one different model per resource (MMs, Multiple Models), and the other method uses the same ML model for all the resources for a single query (GMs, Global Models). In order to simulate a federated environment, the authors partitioned the CLEF-IP patent collection into 632 IPC-based virtual resources (allowing document overlap across resources). In the study, their method is compared with the state-of-the-art methods CORI, SSL, and SAFE (commonly used for results merging), achieving noticeable improvements. In particular, the best performance is given by MMs random forest, outperforming the state-of-the-art methods by 60% in terms of recall in the cooperative environment. Although, while this study demonstrates that ML models can be used to substitute traditional methods for federated search, the simulation of the environment through IPC-based partitions may not fully capture the heterogeneity of real-world patent databases, that should be investigated in the future.

4.3.2. Deep learning

- *Traditional Neural Networks.* Most patents contain images, which are fundamental for disclosing an invention. Ranging from technical drawings to chemical structures, they provide complementary information to textual data and are used in patent retrieval for identifying relevant documents. Given the need to accelerate the activity of patent search, machine learning methods, and in particular neural networks, have emerged as interesting tools for automating the retrieval of these images. Among the earliest

attempts to automate patent image processing, there is Kravets et al. work [114], in which a custom CNN (3 convolutional layers and 8-neuron output) is trained on patent images (obtained subsequently to a removal of images consisting entirely of text, containing parts of text or considered “broken”) downloaded from Freepatent [128] and saved in folders organized by IPC codes, in order to classify the images in the 8 main classes of the IPC hierarchy. The results of this attempt could be improved (accuracy during training is 51% and during the testing is only 30%), leaving a lot of space for improvement. Considering the outcome, the authors suggest the need for more data to train the model (only 2400 patent images were used to train the neural network because of the limited hardware), the change of layer parameters for identifying the optimal structure for this task, and even the use of another CNN architecture.

Despite their advancements, other authors, Kucer et al. in their study [115], highlight persistent challenges in patent image retrieval. They introduce DeepPatent, a large dataset consisting of 350,000 drawings specifically curated for recognition and retrieval of design patent drawings. They also develop a deep learning model, called PatentNet, for patent drawing retrieval. Their network adapts ResNet50 [129] with two modifications: replacement of standard average pooling with Generalized Mean (GeM) pooling [130], and L2-normalization of feature vectors. A three-stage training protocol is applied for PatentNet: (1) initialization with ImageNet-pre-trained weights (they demonstrate that features learned from natural images lead to better results than self-supervised feature learning on patent data), (2) fine-tuning using patent IDs as labels, and (3) retrieval optimization via triplet loss. They compare their method with other deep learning-based approaches: Sketch-a-Net [131] (specialized on free-hand sketches), and Sketchy-Resnet [132] (trained for the purpose of sketch-to-image retrieval), obtaining the best performance in terms of MAP (0.376) and accuracy (top-1 acc: 0.841). However, the authors note limitations, including scalability issues: going

from a 38k-image database to a 350k-image database, the performance in terms of MAP and top-1 accuracy decreases by 30% and 20%, respectively.

Another study [133] uses a novel convolutional neural network (CNN) architecture, called Dual-VGG, with the aim of deriving vector representations of patent images, enabling image cosine similarity calculations for retrieval. This method represents patent images embedding not only considering superficial visual information but also technology-related knowledge, combining two tasks: image-type categorization (e.g., drawing, flowchart, graph, chemical structure, table, DNA, math formula, computer program, and symbol) and IPC (International Patent Classification) section-label prediction, so that visual characteristics and technology-related knowledge are integrated. Two approaches for retrieving images are proposed: an image-to-image retrieval (given a query image, the most similar images are retrieved), and a text-to-image retrieval (a text query can be used to retrieve relevant patents through the USPTO website, Google Patents, PatSnap, Incopat, or other patent search tools and their images are used as queries for the dual-VGG model). For their experiments, the authors use three datasets: CLEF-IP 2011 (where all images have already been labeled into nine type-categories), USPTO as IPC section-label data set, and a custom curated evaluation data set, by downloading images from PatSnap, that contains patents in four different technological domains related to mechanical design: flywheel, milling machine, aircraft transport, and robotics. The authors compare their method with ResNet50, DenseNet121 (ImageNet pre-trained CNN), as well as XRCE's method (based on Fisher Vectors), reaching a 95.37% accuracy in image-type prediction and 54.32% IPC section prediction, overcoming the other models. However, performance varies significantly across technology categories, highlighting the lack of high-resolution images, the need for more balanced datasets, and the integration of textual information. Moreover, the image representation is heavily related to the IPC system, and may not be fully accurate if the code lacks the appropriate granularity to reflect the specific technological domain. Additionally, since the same IPC-based label is given to all images in the patent documents, there is an intrinsic issue in acquiring a sparse image representation [116].

- **Advanced Deep Learning Methods.** In recent years, more advanced neural architectures have been developed in order to address the challenges of the patent retrieval task, with a particular focus on image-based patent retrieval. While early studies rely on basic convolutional networks, recent works have introduced advanced and more sophisticated neural networks leveraging deep metric learning, multimodal learning, and large language models. For instance, Higuchi et al. in [116] introduce a cross-entropy-based metric learning framework for patent image retrieval, substituting the conventional Triplet loss with InfoNCE [134] and ArcFace [135], addressing the Triplet's limit of handling one positive pair and one negative pair at the same time. In particular, InfoNCE uses many Anchor-Negative pairs for each Anchor-Positive pair, and ArcFace maximizes variance among different classes and compacts representations of the same class. The authors use the DeepPatent dataset previously published by Kucer et al. [115] and the MAP score to evaluate their system. They test their methods (EffNet+InfoNCE and the EffNet+ArcFace) against the baseline ResNet+Triplet and EffNet+Triplet. Additionally, Hard Negative Mining [136] is utilized for Triplet training. The proposed method achieves a score of 0.622 in terms of MAP, realizing a superior performance on the DeepPatent dataset with respect to the baseline. Additionally, they develop a web application for patent drawings search using the indexing method, Faiss [137], allowing users to search only by image query. However, despite the positive comments from the practitioners, as the authors also

suggest, the MAP and the accuracy of the system need to be enhanced for professional use.

Wang and Zhang [117] advance in this field by proposing a lightweight yet efficient model for patent image retrieval, consisting of a backbone network, that leverages EfficientNet-B0 [138], neck structure (that comprises a fully connected layer and a consecutive batch normalization layer), and a training head (characterized by a classifier and ArcFace loss). They implement data augmentation without scaling, because patent drawings present a relatively fixed line width and thus objects do not vary a lot in size. This method outperforms the approach proposed in [115], among previous methods, by a significant margin (33.5% of improvement in MAP). The authors also confirm their hypothesis that the scaling transformation for data augmentation is detrimental to performance in the patent drawing field. Additionally, they identify that random rotation also dramatically damages the outcome.

While Wang et al. [117] rely solely on Patent IDs as labels (one ID label assigned to all images from the same patent), limiting the retrieval to intra-patent matching, Lo et al. [118] enrich the context by incorporating textual descriptions, enabling more semantically-aware retrieval. Furthermore, they aim to address the long-tailed class distributions imbalance (few dominant classes and many rare classes) problem. In particular, they propose a multimodal framework that combines: Large Language Models (LLM), to generate textual patent image descriptions, Vision Language Models (VLM), to learn visual features aligned with their textual descriptions, and “distributions-aware” loss functions to handle the long-tailed classes problem. They pre-process the patent images in input using data augmentation techniques tailored for patent images (flipping, random cropping [139], random erasing [140], and gridmask [141]), and apply CNN backbones to extract features from the augmented visual data (using a Multi-Layer Perceptron, MLP, to handle mismatches in feature dimensionality). For the textual component, they leverage patent images, their original descriptions, object name, Locarno classification, and perspective to generate, through an LLM, a fine-grained description, whose features are extracted by a frozen text encoder. Then, their models leverages a contrastive learning approach on three levels, using the conventional VLM contrastive (fine-grained) loss L_{clip} for exact image-description pairs, a class-wise loss (L_{cls}) to match pairs from the same class, and a category-wise loss (L_{cat}) for broader semantic categories (head or tail classes). This approach allows learning not only what is identical, but also what is semantically near, enhancing the retrieval performance and the generalization on imbalanced data. During the query phase, patent images are encoded into visual embeddings and stored in a vector database, and the image retrieval is performed by ranking embeddings with cosine similarity. For their experiments, they use DeepPatent2 [142] data from 2016 to 2019, obtaining a MAP score of 0.780, 0.691, and 0.616 in the head classes, all classes, and tail classes, respectively, obtaining a significant improvement for the tail classes (approximately +20%). Furthermore, a user study conducted by 15 patent agents reported a greater satisfaction of the users along with a reduced search time, underscoring the potential of advanced neural networks to revolutionize patent search workflows.

More recently, another interesting multi-modal approach has been proposed by Kavimandan et al. in [119], consisting of a hierarchical multi-positive contrastive framework for patent image retrieval. The system exploits the hierarchical structure of the Locarno Classification system for assigning increasing similarity weights to positive pairs of figures belonging to the same class, subclass, and patent. Thus, in this approach, each image is trained with multiple pairs weighted according to their hierarchical proximity. The authors conduct the experiment on

the DeepPatent2 dataset with ResNet, ViT, and CLIP backbones, showing that hierarchical multi-positive contrastive learning improves the outcome of the retrieval, evidencing an increase in terms of MAP, nDCG, and MRR@K. This promising hierarchical framework can also be extended to other taxonomies, such as the CPC system, in order to be adapted to diverse patent domains.

- *Transformer-based.* In recent years, transformer-based approaches have emerged as a cornerstone for patent retrieval, with a particular focus on Bidirectional Encoder Representations from Transformers (BERT) [143] model and its variants. These models are capable of comprehend and represent in an efficient way the semantical complexity and the length of patent documents, overcoming traditional methods, going beyond also query expansion and metadata based approaches.

In [144], Kang et al. suggest a user-friendly patent search system based on one of the first multimodal frameworks, combining BERT's linguistic capabilities with data from patent drawings. The authors propose an explorable visualization model for prior art search that shows similarity between search results. The system consists of a textually oriented architecture, using BERT to create precise embeddings for elements such as title, abstract, claims, and a vision-oriented framework to represent drawings or flowcharts.

In a later work [120], the authors present an application of BERT for prior art search, in particular dealing with a binary classification task of distinguishing relevant patents from noise patents. Their approach fine-tunes BERT on three patent parts including title, abstract and claims, and their combinations. They conduct the experiment in Google's 'Colaboratory' environment, fetching the dataset from the WIPS database by March 2016. In particular, each patent in the dataset is characterized by nation code (US/CN/EP/PCT), title (64 tokens), abstract, first claim (only the first 128 tokens), application number, main IPC code and a binary number (0 or 1), indicating if a patent is relevant (1) or not (0). The authors evaluate their method on two different datasets: a non-uniform dataset (with 9093 patents, containing only 4.4% of relevant documents), and a uniform dataset (artificially created balancing 402 relevant patents and 402 noise patents), achieving better results on the balanced dataset. In particular, the best performance is obtained by using the title and the first claim with a recall value of 0.9429. However, these results are reached on a curated balanced dataset, which diverges from real-world conditions: performance drops to 0.4524%, in terms of recall, on the natural imbalanced dataset. Moreover, claims are truncated at 128 tokens, potentially discarding useful information, and the dataset is domain-specific, extrapolated from results to a search query.

In another study [121], Pustu-Iren et al. propose an innovative system for semantic patent image search, combining patent images and their textual descriptions. For image feature extraction, they use the Contrastive Language-Image Pre-training (CLIP) [145] model (pre-trained on 400k image-text pairs) while, for textual feature extraction, the authors realize OCR (Optical Character Recognition) by leveraging the CRAFT (Character Region Awareness For Text detection) [146] model for text block detection and, subsequently, a four-stage Scene Text Recognition (STR) [147] framework to extract text, in particular numbers. In fact, by these numbers, relevant sentences in the XML file of the patent document are identified and then embedded through a RoBERTa [148] model, in order to create an average vector over all related sentences to represent a patent image. This approach is evaluated on 30,379 patents (from 2007 to 2020) of the autonomous driving and wind power field from the EPO dataset, obtaining a MAP@50 score of 0.683, 0.696, and 0.715 by leveraging, respectively, only textual features, solely visual features, and combining the features. Hence, despite the results suggest

that combining textual and visual data is beneficial for patent retrieval, future research should consider also the descriptive text (e.g., Compound X, Carboxyl group), especially in chemistry and pharmaceutical patents, in patent images, that here has been ignored.

In [149], Freunek et al. leverage BERT in novelty searches by fine-tuning the model on concatenated descriptions-claims pairs in order to identify novelty-relevant descriptions for patents. To this aim, due to BERT's maximum input length limitation of 512 tokens (about 200-300 words), patent descriptions are split into chunks of 100-200 words. Claims (only the first claims, also referred to as independent claims) are then concatenated with either their corresponding description segments (Label = 1) or random descriptions from other patents (Label = 0). Specifically, the authors use the Bert For Sequence Classification model, training it on input sequences of the form: [CLS] (Classify Token) + Claim + [SEP] (Separate Token) + Description Chunk + [SEP] + Label (0 not relevant, 1 relevant). The model achieves an F1-score of 0.936, performing very well in matching claims with their respective descriptions during pre-testing, but during the final test, it exhibits difficulties in ranking prior art. In fact, some highly relevant patents are ranked correctly top-scoring, but others are ranked very poorly (e.g., 345th over 1011 patents), perhaps indicating, as the authors suggest, the need for the inclusion of dependent claims and patents from all the classes. Furthermore, the authors also recognize that the model has been trained particularly on granted patents, while novelty search is usually conducted on patent applications, and, additionally, dependent claims tend to have more features than independent claims, which are the only ones considered in this study.

In their subsequent work on Freedom To Operate (FTO) analysis [150], the same authors invert the approach by training BERT to identify independent claims for a description of an invention, in order to discover if an invention or a product is already protected by a patent. Thus, in this work, the input structure is reversed: [CLS] + Description Chunk + [SEP] + Claim + [SEP] + Label, where Label=1 indicates that the claim belongs to the same patent as the description, otherwise Label=0. This method is tested using 5 reference patents (inventions), whose abstracts are concatenated with all the claims in the test set (2577 patents from USPTO, EPO, GB, and AU). As a result, for all five patents, their associated claims have been ranked first. However, these two works by Freunek et al. present some limitations, such as the exclusion of dependent claims, which limits the completeness of the information, the training phase performed only on granted patents, and the lack of objective metrics, thus leaving space for improvement.

Siddhart et al. [97] combine transformer-based text embeddings with knowledge graph embeddings for retrieving patents related to an initial set of patents. This method concatenates text embeddings from Sentence-BERT, applied to titles and abstracts, to citation and inventor embeddings obtained from TransE-trained knowledge graphs. The hybrid approach, combining concatenated text, citation, and inventor embeddings, outperforms the individual embeddings.

Higuchi and Yanai, who previously proposed a patent drawing retrieval method based on cross-entropy using CNNs [116], introduce in [122] a novel deep metric learning architecture, achieving a higher accuracy than prior studies. They utilize cross-entropy-based methods such as InfoNCE [134], a widely used contrastive training loss, and ArcFace [135], which can train many negative pairs at the same time, for metric learning, instead of relying on the common function Triplet loss. The authors apply their method on the DeepPatent dataset, obtaining a MAP score equal to 0.856, higher than state-of-the-art models' values.

Vowinkel et al. introduce in their work [151] SEARCHFORMER, a BERT model specific for patent language, optimized for creating embeddings for prior art search. They use GP-BERT [152] as a base model, pre-trained on full patents in the English language. As training data, they use the first claim of the application as an anchor sentence, the X-cited text as a positive sentence, and a non- or A-cited text as a negative sentence. In particular, regarding the negative sentences, they use: random paragraphs from any other document, random paragraphs from random documents with the same CPC subclass as that of the anchor's, non-cited paragraphs from cited documents, and A-cited paragraphs. Moreover, in their experiments, a patent application is compared to a specific subset of potential prior art (PPA) documents instead of the complete corpus, in order to reduce the computational burden. The authors measure the search performance in terms of Rank of the First Relevant (RFR), which measures how early the system retrieves the first truly relevant prior art document (X/Y-citations) in the ranked list. The model excels at distinguishing random irrelevant texts (98.04% accuracy) and same-field patents (90.36%), but struggles with non-cited passages from cited documents (66.31%) and A-citations (53.83%). Moreover, SEARCHFORMER's achieves state-of-the-art performance with a RFR = 52.83, significantly outperforming the state of the art model BM25 (72.95).

Also, the effectiveness of ChatGPT and Google Gemini in the field of prior art search has been evaluated [123], in particular for the task of finding paragraphs in previously granted patents that could affect the novelty of claims in patent applications. To this aim, Chikkamath et al. firstly create a custom dataset based on EPO search reports, conducting 49 manual examinations to compile the dataset and focusing on the patent G06F IPC class (digital data processing). Then, the authors give in input to ChatGPT and Google Gemini nine candidate paragraphs (including both cited and non-cited) from prior art and a claim of a patent application (query), aiming for predicting the top 3 most similar paragraphs that could affect novelty of the claim of the patent application. Then, the predicted Top 3 paragraphs are compared with the real examiner-cited paragraphs, computing the accuracy of the models. The results on the dataset developed by the authors (ClaimCiteRtrieval) based on European examiners' search reports, show that both ChatGPT (particularly) and Google Gemini outperform PatentSBERTa in novelty passage retrieval, achieving 49.66, 46.94, and 44.21 scores in terms of accuracy. However, despite their success, the two models still face challenges in retrieving relevant novelty passages.

More recently, Lee and Bai present in their work [124] a novel framework, named PAI-NET, for similarity patent retrieval, based on a Retrieval-Augmented Generation (RAG) system and contrastive learning. The framework leverages expert domain knowledge by exploiting prior art relationships to minimize the distance between cited prior art documents and maximize the distance from non-cited patents, thereby learning to position technically similar patents closer in the embedding space. This framework, trained on USPD and KPRIS datasets, achieves a 15% of improvement in Mean Reciprocal Rank (MRR), which measures how well relevant documents are positioned at higher ranks by the model, over SOTA models such as PatentBERT and SentenceBERT. Moreover, PAI-NET, by integrating expert knowledge embedded in citation networks, reduces the human effort required for constructing and maintaining domain-specific knowledge bases. However, relying solely on citation networks for contrastive fine-tuning may not always be optimal, especially in cases where patents have few or noisy citations, which could be underrepresented or misplaced in the latent space. Future improvements could involve integrating other similarity cues, such as those derived from the CPC classification system, or integrating

into the system multi-modal retrieval, incorporating image-based features.

A Graph Transformer-based dense retrieval method for patent searching is developed by Daniell et al. in their work [125]. Each patent is represented as a graph, where nodes correspond to components and edges encode relationships among them, and for each patent document, three different graphs (one containing only the first independent claim, another containing all the claims, and one containing both the claims and the full description) are created. The Graph Transformer model, which receives as input patent graphs, is trained, with a triplet loss, on prior art citations in order to capture domain-specific similarities, going beyond diverse descriptive terminology, achieving a recall of 0.4046 and an nDCG of 0.5564.

Another notable contribution has been made by Awale et al. [126], who developed iPATENT, a web-based multimodal patent image retrieval system, allowing for image-to-image, text-to-image, and text-conditioned image-to-image search, going into the most promising direction, which leverages both textual and figurative information within patent documents. This framework uses the CLIP model for unimodal, cross-modal, and multimodal retrieval. In particular, the images are encoded using CLIP and are then indexed offline in a vector database together with their associated metadata. With this system, the user can query image, text, or a combination of them, and then visualize the results obtained via a cosine similarity, thanks to the interactive interface. Moreover, the system offers another way of visualizing results, going beyond the classical grid view, which consists of a cluster-based representation: images are grouped in clusters, and, additionally, each cluster provides a synthetic title and a description, generated by a Large Vision-Language Model (LVLM). Additionally, the developed platform also provides fine-grained retrieval capabilities, such as sub-figures or components search, enabled by an interactive cropping tool. The CLIP model, evaluated on the image-to-image retrieval task, on the DeepPatent database, achieves a MAP of 0.57 at rank $k = 1$, without any fine-tuning or domain adaptation, demonstrating the practical utility of this model in multimodal patent retrieval.

In summary, among the machine learning-based retrieval approaches analyzed within this Section, traditional machine learning-based methods demonstrate good performance when used for retrieval-related tasks such as classification, topic modeling, and result ranking in federated search. However, due to their limited semantic understanding, these methods are better suited as supporting tools and present limitations for fully automated patent search.

On the other hand, deep learning-based methods substantially advance patent retrieval by learning complex semantic representations and introducing image-based search. The earliest neural approaches, particularly CNN for patent image processing, demonstrated the feasibility of patent image-based retrieval; however, they reported limited performance mainly due to small datasets, simple architectures, and constrained computational resources. Consequently, subsequent works introduced larger datasets and improved architectures, allowing better retrieval performances. More recent and advanced models leverage large-scale datasets, contrastive learning, and multimodal representations, leading to further improvements. Despite these advancements, challenges, such as scalability issues and dependence on image quality, still remain.

Transformer-based models also significantly improve patent retrieval by better capturing the semantic meaning of patent texts. However, these methods also report limitations, such as constraints related to patent text length and input token limits.

Considering the analyzed studies, future research directions emerge. First, approaches combining textual and visual data, together with metadata, particularly classification codes, should be further explored

Table 4

Performance Comparison of NLP-Based Methods for Patent Retrieval. Only studies that provide results in terms of the presented metrics are reported. The table also specifies the patent fields used to build the retrieval corpus, allowing a comparison across studies based on the searchable content.

	Dataset	Patent fields of the retrieval corpus	Recall	Precision	MAP	PRES	F1-score	Acc
[153]	USPTO	bibliographic data, title, abstract	–	–	0.74	–	–	–
[154]	USPTO	title, abstract	–	0.18	–	–	–	–
[155]	–	patent figures, figures' captions	0.7180	0.9069	–	–	0.7895	0.9565
[156]	CLEF-IP	full text	0.631	–	0.285	0.544	–	–
[157]	PatentMatch	claims of patent applications, descriptions (paragraphs) of not/relevant prior art (from search report)	–	–	–	–	–	0.83
[158]	–	descriptions, claims	0.827	–	–	–	–	–
[159]	CLEF-IP, MS MARCO	full text	0.2880	–	0.0812	0.2110	–	–
[160]	Google Patents	abstract	0.9494	0.9331	–	–	0.94	0.94

to capture all information from patent documents and, therefore, allowing for an enhanced patent search. Furthermore, future works should address scalability issues and propose strategies for handling the long texts characteristic of patent documents in order to include all information without cutting the input text.

4.4. Natural language processing based techniques

NLP techniques have also been explored in modern patent retrieval as a way to overcome the limitations of traditional methods. In this Section, diverse NLP-based methods, from statistical and semantic methods to more advanced hybrid systems that integrate machine learning and deep learning, are explored. The results, in terms of metrics, of the proposed NLP-based approaches, when provided, are summarized in Table 4.

4.4.1. Statistical & semantical NLP

Statistical and semantical approaches in NLP rely on representations such as tf-idf and Word2Vec, Latent Dirichlet Allocation, and topic modeling, to analyze text, extract features, and rank documents. Supraja et al. [153] develop an engine for patent search which displays the relevant patents given the user input query. The developed framework uses LDA to cluster patents into topics and BM25F, which combines inverse document frequency (idf) and term frequency (tf), to rank documents. In particular, BM25F weights term relevance across titles, abstracts, and recency. For their experiments, the authors use US granted patents from USPTO, and evaluations show a MAP equal to 0.74. A limitation of this work is its reliance on solely uni-gram queries, which could severely limit the practical use of this framework, since real-world patent searches often require multi-term phrases and boolean operators. In fact, the authors themselves acknowledge this limitation and plan to add, at least, a bi-gram search model in future work.

In [161], Buckley et al. combine tf-idf similarity search with domain-specific topic modeling to develop PatentExplorer, a system for patent search. The system allows users to retrieve similar patents and then refine the results by filtering topics. In particular, the system creates the embeddings of both the query text and the documents through tf-idf weighted sparse representation, and then, before computing the cosine similarity between them, the framework leverages approximate nearest neighbor search in order not to compute the similarity for every document. Then, the system's output consists of a ranked list of the N top-scoring patents. Additionally, for topic modeling, LDA [162] and non-negative matrix factorization (NMF) [163] are used. However, three main limitations could be observed in this work: tf-idf embeddings do not capture semantic relationships (e.g., synonyms) and the context, results can be filtered, but the system is not so intuitive (min probability and max rank need to be manually set), and evaluation metrics are not provided.

Sarica et al. [154] create a semantic knowledge graph, trained on USPTO patent documents, by applying NLP and semantic analysis techniques. In particular, thanks to the developed system, the user, by means of keywords, retrieves a first set of patents that contain any of the keywords in their titles or abstract. Then, the method extracts words and phrases from the retrieved texts and selects those contained in the knowledge graph. These words are subsequently ranked according to semantic similarity to prior query terms. To build the knowledge graph, the authors use titles and abstracts of more than 5.7 million utility USPTO patents (from 1976 to 2017). After a pre-preprocessing (lowercasing and punctuation removal) phase of text data, the authors extract key phrases using statistical co-occurrence patterns and train word embeddings (Word2Vec) on this corpus to capture semantic relationships between technical terms. In a case study, the authors demonstrate that their method overcomes the one proposed by Song and Luo [105] by achieving a higher precision (18% vs 9%), in less time (a few minutes vs a few days), suggesting that the method developed in this study may support researchers in finding non-obvious keywords to enlarge their search queries. However, this approach is not ready for a real-world application, because, even if the precision is higher with respect to the other method, it is still not sufficient. Furthermore, this approach does not manage polysemy of words with multiple meanings, and the research of new candidate keywords is done by considering only titles and abstracts, ignoring descriptions and claims which may contain critical terms.

4.4.2. Hybrid NLP & machine learning

Several hybrid approaches that combine traditional NLP techniques with machine learning models have been proposed to improve patent retrieval. For instance, in [155], a patent image retrieval system is proposed. It combines visual and textual analysis for concept detection. This framework, which utilizes Adaptive Hierarchical Density Histograms (AHDH) [164] for visual features and bag of words with tf-idf for text, trains SVM classifiers to detect global concepts (for example "ski boots"). Going into the details of visual features extraction, they use the AHDH algorithm specifically developed by Sidiropoulos et al. [164] for binary patent images (black and white) that recursively divides images into regions based on geometric centroids, creating hierarchical density histograms in order to capture structural patterns. Whereas, for textual features extraction from image-associated descriptions, the authors use a bag of words approach to model each figure with a vector and compute the weight of each term with tf-idf, creating a 100-dimensional vector per image. SVM is also utilized, given the visual and textual features, to understand if an image represents the written concept. The authors test their method on patent images of A43B and A63C IPC subclasses, extracted from around 300 patents, which contain parts of footwear. Furthermore, patent image segmentation and the matching with their respective descriptions are made manually. Moreover, this study adopts one classifier per concept (in total, 8 concepts: cleat, ski boot, high heel, lacing closure, heel with

spring, tongue, toe caps, roller skates), where individual SVMs are trained for detecting specific patent image categories (e.g., “ski boots”) and the approach is tested for visual, textual and a combination of both features on PatMedia [165] search engine. The best performance is reached by the hybrid approach, with 95.65%, 90.69%, 71.80%, and 78.95% of accuracy, precision, recall, and F1-score, respectively. While underlying the importance of the combination of visual and textual features for capturing the global essence of objects, this work presents scalability limitations due to its reliance on manual annotations and figure segmentation. Additionally, it covers a limited taxonomy, testing the method on only eight concepts, implying re-training necessity, and it is capable of recognizing objects in a global manner, not local.

In [156] Andersson et al. investigate query generation methods for patent passage retrieval, using NLP techniques, in order to catch domain specific concepts depicted as multi-word terms, MWTs, (e.g., composite cell, cross-linked polymer). The authors also propose a novel approach to recognize domain specific terms from other ones more general. The study leverages the CLEF-IP 2013 collection and evaluates the retrieval at both document level and passage level on 50 English patent topics (consisting of one or more claims). The authors create two different indices, one for documents and the other for paragraphs, with white space tokenization, stop word filtering, and a particular stemmer, namely Emin [156]. They evaluate three retrieval models: Vector Space Model, BM25, and Language Model with Jelinek-Mercer smoothing method (LMJM). To automatically identify MWTs, the authors develop a machine learning approach (Random Forest) to classify phrases into technical and generic, creating a manually-labeled dataset of 4400 phrases with features such as term frequency, IPC code distribution, phrase length, and Mutual Information. The query generation process is structured into four stages: (1) choosing between a statistical or NLP-based approaches, (2) extracting either from claims only or expanded documents using cosine similarity, (3) applying methods for identifying MWTs, and (4) employing two re-ranking methods: SUM (summing passage scores per document) and MERGED (combining document and passage indices). The best performance in terms of PRES and recall (0.544 and 0.631, respectively) is achieved by the NLP approach combined with LMJM, using the merged re-ranking method, overcoming state-of-the-art methods. Although this method achieves statistically significant improvements, it is evaluated on solely 50 patent topics in the English language, thus, further studies should include more data.

Another work [166] proposes a patent retrieval method based on a double-classification to support knowledge transfer for innovative electromechanical product design. The system combines a first-stage classification that uses primary functional verbs (e.g., “stabilize”) to categorize patents in functional macro-categories, and a second stage that leverages topic modeling. For the first stage classifier, which is trained utilizing titles and abstracts of 2000 patents from different IPC classes, whose features are mapped into a Vector Space Model (VSM), the feature selection algorithm CHI2, the term weighting algorithm TF-MI, and the classification algorithm SVM are selected because of giving the best results. The second stage utilizes the LDA topic modeling algorithm to extract technical features from patent subsets. Topic functional verb sets, using WordNet’s synsets/hyponyms, are then constructed to label LDA extracted technical topics; however, to do this, human intervention is required to filter electromechanical specific terms. Subsequently, the authors build a Word2Vec similarity model (500-dimension) trained on 300 US patents’ titles and abstracts to compute semantic similarity between patents and topic functions. Patents are then ranked by cosine similarity. Despite the topic functions of the second stage are defined “dynamic” because they come from patents’ content thanks to LDA, and if the patent database is expanded, the LDA algorithm extracts new topics from updated patents, the first stage is still static. In fact, the functional verbs belong to a predefined list that does not automatically change. Thus, the system presents a structural limitation that prevents it from really adapting to new documents. This issue is recognized by the authors themselves and, for

this reason, they decide, while comparing the framework with other static functional approaches, to only demonstrate the superiority of the secondary classification stage over other methods, computing the semantic similarity between assigned functions and patent contents, obtaining an average similarity of 0.6-0.8 for the dynamic method and 0.4 for the static one. These results prove the higher adherence of the topic functions to patent content, whereas the authors do not provide the classical metrics to evaluate their method.

4.4.3. Hybrid NLP & deep learning

Patent retrieval methods combining NLP with Deep Learning have been developed for enabling the extraction of richer semantic representations from patent texts. Helmers et al. [26] adopt entire patent texts for similarity comparison with existing patents and evaluate various feature extraction methods, such as bag-of-words, Word2Vec, and Doc2Vec, and comparison methods. The similarity between two documents is computed via the cosine similarity. The authors test their methods on two datasets: (1) one containing more than 100000 patents from the CPC category A61 (medical or veterinary science and hygiene) consisting of 2500 target patents, all cited patents (labeled as “cited”) and 1000 not cited randomly selected patents (labeled as “random”), in order to assess if the approach is able to distinguish cited (potentially relevant) patents from random ones (non-relevant); (2) a subset of the first dataset, characterized by solely 10 target patents and random patents with either low, medium, or high similarity (computed with cosine similarity on tf-idf BOW features), manually targeted as “relevant” or “irrelevant”. This last dataset is built to find out which of the cited patents is truly relevant and is important prior art is not included in the search reports. Results of this study show that by computing the cosine similarity between feature vectors created from full patent texts, the method can distinguish cited from non-cited patents, and that the best performance is achieved by computing the cosine similarity using tf-idf BOW features. Furthermore, the official citations showed low precision (43%) and recall (78%), with many false positives/negatives, indicating the limitations of the classical keyword-based search. On the other hand, the proposed automated method correlated better with expert judgment than the search report citations, discovering also missed prior art.

In [157], Chikkamath et al. propose a patent novelty detection framework, in the form of a binary classification problem, evaluating machine learning, deep learning models, and diverse word embeddings (e.g., Word2Vec [167], glove [168], fasttext [169], and domain-specific embeddings [170]) on the PatentMatch dataset [171]. This dataset, which derives from EPO search reports, contains pairs of patent application claims and corresponding text passages from cited patent documents, with binary labels indicating if the text passage invalidates the novelty of the claimed invention (category X, 0) or not (category A, 1). They evaluate the performance of diverse machine learning algorithms (Support Vector Machine (SVM), Naive Bayes (NB), k-Nearest Neighbor (KNN), and Random Forest (RF) and various combinations of them), deep learning architectures (LSTM, GRU, 1D CNN, and also some hybrid nets), leveraging all the selected kind of word embeddings. Key findings reveal that the combination of Naive Bayes and Support Vector Machine (NBSVM) with Word2Vec gives the best results, outperforming deep learning models, achieving 83% of accuracy with a significantly lower computational cost (9 s for every epoch, while deep learning models take at least 108 s/epoch).

In [172], Kumaravel et al. propose a novel framework, named PQPS (Prior-art Query-based Patent Summarizer), for retrieving and summarizing prior art patents, that addresses the vocabulary mismatch problem of the classical keyword-based method through query expansion with knowledge bases (domain ontology and WordNet). Furthermore, the framework leverages topic modeling and citation information for enhancing retrieval outcome and utilizes restricted Boltzmann machine (RBM) and bi-directional long short-term memory (Bi-LSTM) models to summarize the content of the retrieved patents.

Specifically, the PQPS system extracts noun phrases from textual fields such as title, abstract, and description of a novel patent application document given in input, and builds with them a query based on term frequency-inverse field frequency (TF-IFF) scoring. This query is, then, semantically enriched with ontology and WordNet and utilizes on the Google Patent Search System to retrieve patents. In order to remove irrelevant patents, an LDA filter (on titles, abstracts, and descriptions of the retrieved patent set) is applied, keeping solely patents whose topic clusters are close from one another. After filtering, a citation graph is constructed to retrieve missing relevant documents, identifying patents with a number of common citations over a certain threshold. A cosine similarity filtering to remove possible irrelevant patents retrieved during the last phase is then applied. The retrieved relevant patents are then ranked, assigning ascending weights to title, abstract, and description similarities to the query. Regarding the patent summarizer, RBM-EPS creates individual summaries of documents by selecting sentences from them, and then, the Bi-LSTM-Based summary generator paraphrases each summary cluster. The authors test this framework on a dataset that includes 753 smartphone patents, 478 smartwatch patents, and 421 smart-home patents collected from Google Patents. The efficiency of the three retrieval subsystems, consisting of Google Patents, ontology-based query expansion system, and WordNet-based query expansion system, is evaluated in terms of MAP and recall, and it is found out that queries expanded with domain ontology improve relevant document retrieval in terms of recall by circa 28% and 56%, respectively, over WordNet-based query expansion system and Google prior-art search system. It is also discovered that LDA filtering reduced irrelevant patents (e.g., from 1642 to 1074 for a sample query), and that the bibliographic coupling retrieves 263 patents adding to the other 1074. However, the authors do not provide a metric-based evaluation comprehensive of the entire framework, preventing its comparison with other developed approaches. Furthermore, the author tests their method on a restricted field, thus further analysis should include patents from other fields.

Björkqvist and Kallio, in their work [158], present a patent search engine, based on a graph neural-network trained using citations compiled by patent examiners in the search reports in order to learn how similar inventions relate to each other. The usage of a graph structure is chosen by the authors specifically to represent patent contents in a concise way. The search set is created by transforming the patent documents into a graph through the parser and, then, embedding them into vectors with the graph neural network, and also the user search queries follow this procedure. Once the query has been created, the system, through a nearest neighbor algorithm, performs the search. The parser is characterized by three main parts: (1) a part that detects the structure of the document thanks to a pre-trained NLP pipeline provided by the SpaCy [173] library, (2) a relation extraction phase, and (3) a graph creation phase. The graph neural network, whose structure is a modified version of Tree-LSTM [174], is trained using a triplet loss function on citations extracted from EPO search reports. The evaluation of the methods is carried out on a custom dataset consisting of approximately 44,000 citations that invalidate novelty (X citations) and point to around 27,000 unique cited documents, constituting the search set. As a result, the implemented method leads to a 0.827 score in terms of recall, overcoming previous methods.

In [159], Stamatis et al. propose a novel architecture for patent documents re-ranking, combining lexical (BM25) and semantic (BERT) techniques. The aim of their work consists of overcoming limitations of lexical models, which fail to capture semantic relationships, missing relevant documents that describe concepts in a different manner, and semantic models (BERT). The last ones manifest suboptimal results in the patent domain due to input length constraints (512 tokens for the BERT model), forcing truncation and information loss. Moreover, relevance depends on a single section of the patent (e.g., description, claim...), while BERT analyze the entire document. Thus, to overcome these issues, the architecture of this work introduces a segmentation

of both the query and candidate documents, which are full patent documents, into their principal sections (Abstract, Description, and Claims), each containing a maximum of 500 words. To compute the relevance score for every pair of query and candidate section, the method utilizes the BERT-for-patents model to extract embeddings and compute cosine similarity for semantic scoring, and BM25 for lexical scoring. Finally, the system combines these two scores. In particular, it uses AI models (Multilayer Perceptron, MLP, with two hidden layers along with LambdaMART) to merge section-level scores, adding dynamic weights to give more importance to rare terms (BM25 focused) or broad concepts (BERT focused). For the experiments, the authors use two datasets: CLEF-IP, converted to standard TREC format, and MS MARCO, to assess the broad applicability of their method, to check if it works on shorter, generic web documents. The proposed approach shows a better performance to BM25 baseline, on CLEF-IP, in all the metrics, MAP, PRES, and RECALL, by 30%, 27%, and 28%, respectively, while on the MS MARCO a lack of improvement, revealing the patent domain specificity of the developed approach.

In a more recent work [160], which draws inspiration from [149, 171], a new approach utilizing concatenated abstract and top terms (extracted from title, abstract, description and claims) of patent documents to generate a dataset used to train a BERT for patents model is proposed. The dataset is composed of 2722 patents (CPC: A43C1/04) retrieved from Google Public Patent Database, pre-processed with the Natural Language Toolkit (NLTK). More precisely, the publication number, title, cpc code, abstract, and set of top terms are used to build the dataset. Data are labeled by pairing each abstract with all possible top term sets (both its own and those of other patents) and assigning a label equal to 1 when the abstract matches its own top terms or to 0 otherwise. This obtained dataset is used to further fine-tune the already pre-trained BERT for Patents model. This model achieves 94% of accuracy, an F1-score of 0.94, a precision of 0.9331, and a recall of 0.9494, overcoming both the other two methods [149,171]. The authors of this work also assess if their model is able to identify X document (relevant prior art) cited in search reports, starting from an abstract of a document under investigation. To this aim, 5 patents are chosen, and the model correctly recognizes only 2 of them, probably because the top terms alone are not sufficient to represent the exact scope or novelty boundaries that are legally defined in the claims section.

Across the NLP-based approaches analyzed within this Section, a progression emerges from statistical and semantical methods toward hybrid models that integrate NLP approaches with ML and DL techniques. Statistical and semantical methods leverage tf-idf, Word2Vec, and topic modeling for feature extraction and topic discovery. However, limitations such as weak semantic understanding and difficulties in handling polysemy arise. Approaches combining NLP with ML demonstrate to improve the retrieval performance by integrating visual and textual features over single-feature type approaches. However, limitations still characterize these works. Indeed, some approaches rely on manual annotations, which limit their scalability, on limited concept taxonomies, and small datasets. More recent approaches combine NLP with DL to allow richer semantic representations from patent texts; however, they also seem to be limited in terms of generalizability, and, in certain scenarios, may be outperformed by simpler models.

5. Automatic and AI-based approaches for life sciences patent retrieval

In this Section, automated and AI-based approaches developed for the retrieval of life sciences patents, overcoming challenges faced by traditional search methods, are presented in Section 5.1. The results, in terms of metrics, of the proposed methods are summarized (when provided) in Table 5. Additionally, this Section includes three tasks for life sciences patent retrieval: Chemical, Biological and Biomedical Named Entity Recognition (Section 5.2), Chemical Structure Extraction (Section 5.3), and Biological Sequences Retrieval (Section 5.4).

Table 5

Performance Comparison of Methods for Life Sciences Patent Retrieval. Only studies which provide results in terms of the presented metrics are reported. The table also specifies the patent fields used to build the retrieval corpus, allowing a comparison across studies based on the searchable content.

Method	Dataset	Patent fields of the retrieval corpus	Recall	Precision	MAP
[175]	TREC CHEM	title, abstract, description, claims, noun phrases, chemical and biomedical entities	–	–	0.2336
[176]	TREC	title, abstract, claims	–	–	0.182
[177]	Oracle DB by Novartis	title, abstract, claims	–	0.0676	0.0422
[178]	PatCID	patent figures (chemical structures)	0.56	–	–

5.1. Life sciences patents and information retrieval

In life science patent retrieval, the extraction of structural features (such as chemical compounds, diseases, and genetic markers) from text is important to improve the accuracy and efficiency of the system [179]. However, challenges arise from nomenclature ambiguity: drugs present several synonyms, systematic, traditional, formulaic, commercial, or generic names, and multiple nomenclatures (e.g., the medicine “acetaminophen” presents 17 additional names in the Medical Subject Headings (MeSH) terminology [176]). Private enterprises, aiming to identify chemical names in patents, have developed databases such as the Chemical Abstracts Service (CAS) [40] or the Reaxys [51] database from Elsevier. Also, different initiatives, such as TREC-Chem [180] and PatOlympics [181], have arisen during the years to promote the development of specialized retrieval systems. In particular, the 2009 and 2010 Tracks of TREC-Chem were focused on chemical information retrieval in general and specifically on chemical patent retrieval, proposing Prior Art search (searching documents that can invalidate given test patents) and Technology Survey (aiming to find sets of documents that best answer certain questions) tasks, while the 2011 edition presented a special emphasis on bio-chemistry. Among the tasks proposed in 2011 were prior art search, technology survey, as in previous years, and a new task: image-to-structure. During the same period, PatOlympics were also organized (as a part of the IRF Symposium 2011), which included a specific competition named ChemAthlon, focused on chemical structure recognition.

Among the systems proposed at the TREC-Chem competition, the framework by Gurulingappa [175] represents a notable contribution. The authors present a system for retrieving patents that invalidate a query patent. NER and normalization of chemical names, genes, proteins, diseases, and pharmacological terms are performed in this study to overcome challenges related to synonyms, acronyms, and morphological variants in query expansion. These entities are then indexed in SCAIView [182], a semantic search engine, obtaining an index containing not only tokens, but also noun phrases, chemicals, and biomedical entities. For retrieval, the study evaluates multiple querying strategies: baseline token-based searches using BM25, noun phrase enhanced queries, and hybrid noun phrase with entity queries. Results show that the best performance is reached by the approach leveraging noun phrases together with entities and co-citation networks, achieving a MAP of 0.2336.

The study [176] by Pasche et al. presents TWINC (To WIN ChemAthlon), a web-based application oriented to chemistry-based patent search to aid life and health specialists. This tool allows for two search modalities: an ad-hoc search, to retrieve documents pertinent to a short keyword-based query; and a related patent search, to identify relevant documents to patents' paragraphs. This system also includes a chemical query expansion feature and a relevance feedback feature. For both the search modalities, patents are normalized using MeSH terminology. Then, the Terrier engine with BM25 is utilized to retrieve a set of relevant documents, and results are, subsequently, re-ranked by considering co-citations (a patent highly cited by other patents will be re-ranked higher). For the retrieval, a set of patents provided by TREC campaign (including EPO, USPTO, and WIPO documents) is utilized. In order to expand the query, Oscar-3, a Named Entity

Recognition (NER) tool, is utilized to recognize chemical terms, then a MeSH categorizer normalizes these terms and maps them to unique identifiers, and, finally, the query expansion is performed by adding terms from different thesauri, such as MeSH or PubChem. The system is evaluated during the ChemAthlon by Intellectual Property experts, achieving a MAP of 8.9% with MeSH normalization and 18.2% of MAP with query expansion. However, the performance of the system developed in [176] is still not sufficient for real-world applications.

In [177], the authors develop a retrieval engine, in the Novartis framework, called PATentPilot, to search information in patent collections in the pharmaceutical domain. For their experiment, the authors select a random set of more than 1M patents from the 13M patents stored in an Oracle Database Management System by IBM Almaden for Novartis, which is supposed to cover all fields of applications relevant for the life and health sciences. The implementation of the search engine is based on four steps. The first step consists in the pre-processing of the collection by an ontology-driven normalization, using MeSH for entities related to anatomy, chemical compounds, devices, disorders, genes, geography, populations, procedures and species, Gene Ontology (GO) for entities describing biological processes, cellular components and molecular functions, and Caloha terminological resource [183] for cells and tissues. The second step includes the indexing of the collection by the Terrier Information Retrieval platform [184], using baseline settings with Porter stemming. The third phase involves patents retrieval, and it is carried out using the Terrier Information Retrieval platform, testing two different models: Okapi BM25 and PL2 [185] (a model based on Poisson estimation from randomness). Fourth, the authors assess the use of co-citation networks in enhancing results by re-ranking patents, favoring more cited patents. Despite the interesting attempt of the authors, in the study [177], in focusing the patent retrieval on the pharmaceutical field, the results of the optimized system, removing the description field, utilizing BM25, leveraging co-citation network and without IPC-codes, reaches solely a top precision of 6.76% and a MAP of 4.22%, leaving a lot of space for improvement.

A federated search pipeline specific for biomedical patents, integrated into the @Note2 [186] biomedical text-mining framework, is presented by Alves et al. in their work [1]. The developed system is capable of retrieving metadata and full-text documents from patents deriving from different sources (USPTO, EPO, WIPO, PubChem), converting PDF documents with Tesseract (an OCR algorithm from Google) along with the Apache PDFBox library to extract machine-readable texts from retrieved documents, structuring the output in a corpus for downstream biomedical text-mining tasks. The authors validate their framework through two approaches: one using the BioCreative V CHEMDNER dataset and downloading 99.5% of the files, and a real-world case study searching for “vanillin” related patents through the different sources of the federated search, retrieving 7215 patents. While the “vanillin” example in the study [1] demonstrates the capability of the system, the authors do not provide standard patent retrieval metrics, making the comparative performance with other studies difficult.

In [187], Taduri et al. introduce an ontology-based computational framework that can aid in the retrieval of relevant patents, documents, and scientific publications from the bio-medical domain. The system integrates BioPortal [188], which includes a variety of ontologies (for

example National Cancer Institute (NCI) Thesaurus [189], Medical Subject Headings (MeSH) [190], Gene Ontology (GO) [191], and National Drug File (NDF) [192]) for semantic query expansion, and Neo4j database [193] (an open-source, NoSQL, graph database) for enabling cross-references between patents, scientific literature, basic regulatory documents and judicial decisions related to “erythropoietin” (that the authors, previously, downloaded from the USPTO database). Despite the integration of multiple kinds of documents, which could be useful particularly for patentability search, the system proposed in [187] faces critical scalability challenges. In fact, each new topic of interest would require manual construction of the domain-specific ontology and a download of related documents.

In [194], an open-source system for retrieving relevant pharmaceutical patents for a gene of interest, via chemicals or biological modulators, is presented. Specifically, the system maps target genes to standard identifiers in ChEMBL [195] (a manually curated collection of bioactive compounds exhibiting drug-like properties) to extract high-activity modulators (chemicals or biological agents with pChEMBL > 6) in ChEMBL database. These compounds are then queried in SureChEMBL [196] database to retrieve associated patents. The authors provide also a case study, using Orphanet [197], to test PEMT on rare disease genes. The system identifies 135 patents connected to 10 target genes through 97 compounds. Despite the innovative use of bioactivity data of PEMT [194], this system remains limited to ChEMBL and SureChEMBL annotated elements, which may not contain innovative compounds and related patents, consequently missing relevant documents.

A recent work by Shimizu et al. [198] leverages automated prior art to generate novel pharmaceutical molecules that do not infringe existing patented compounds but that are characterized by a high “drug-likeness”. To this aim, the authors first collect, in a SQLite database, drug-related (from A61K and A61P IPC classes) patent compounds from the SureChEMBL database and Google Patents, gathering 10,720,835 compounds, saving the chemical structures information by means of InChIKeys [199] (character strings that represent the chemical structures) and SureChEMBL identifiers. This database is also used as a training source for the generative model based on Recurrent Neural Networks (RNN), which employs two reward functions, which incentivize the creation of non-patented molecules. Then, to assess if AI-generated molecules have already been patented, the authors implement, through Python using RDKit, a search system based on InChIKeys. However, the method in [198] solely relies on molecules from the SureChEMBL database, not considering patents’ text information (e.g., descriptions or claims). Moreover, approximately 2.7 million of OCR-derived structures in the SureChEMBL database have been removed in the study [198] due to inaccuracies in chemical structure recognition, potentially omitting relevant structures.

Morin et al. in their work [178], create PatCID, an open-access dataset of chemical structures extracted from patent documents. This collection allows for patent document retrieval given a query molecule, and retrieving molecules covered by a specific patent. The dataset covers documents from the United States (USPTO), Europe (EPO), Japan (JPO), Korea (KIPO), and China (CNIPA). PatCID is created by using three modules: patent document segmentation in order to find and extract chemical images (DECIMER- Segmentation [200]), image classification (MolClassifier) to recognize molecular structure images, and molecular recognition to get chemical structures (MolGrapher [201]). In order to test the retrieval performance of the developed pipeline, the authors build two datasets: D2C-RND (Document to Chemical Structures, Random, with average quality of images) and D2C-UNI (Document to Chemical Structures, Uniform, with more challenging cases coming from molecules images of older patents and using a less common display style). The results show that the implemented method is able to correctly retrieve 56% (47.6%) of the molecules from the D2C-RND dataset (D2C-UNI, respectively), demonstrating better performance than SureChEMBL (23.5%), Google Patents (41.5%), Reaxys

(53.5%), and SciFinder (49.5%). However, the study [178] does not consider textual data, but only molecular structure images to find relevant patents, missing text-based chemical descriptions. Moreover, also Markush structures (that use a generic chemical structure, summarizing several structural formulas), are not utilized in the pipeline. Thus, these limitations highlight the need for hybrid approaches, leveraging both text and image information, in future retrieval systems.

A recent contribution is provided by Kosonocky et al. [202], who develop an LLM- and embedding-based method for extracting functional information of molecules (from SureChEMBL) from patents and literature with the aim of creating a Chemical Function (CheF) dataset, containing 100k molecules and their 631K functional labels derived from patents. In particular, the functional labels are generated by ChatGPT (gpt-3.5-turbo) starting from patent titles, abstracts, and the first 3500 characters of the descriptions. In order to group similar concepts, the functional labels are then semantically embedded and clustered using OpenAI’s text-embedding-ada-002 [203], creating a molecular-functional landscape. Despite this work proposing a non-retrieval method per se, it opens the way to functional-based patent retrieval and drug discovery, where functional labels can be used to find molecules with similar biochemical effects.

On a macro scale, Geissler et al. [204] propose a patent data-driven method able to identify innovation trends and their scientific drivers. This approach, despite not being directly aimed at specific document retrieval, could be integrated in patent retrieval systems to guide the search process in the most innovative and emerging fields.

Recently, research in biomedical literature retrieval has proposed AI-based approaches for mapping and integrating biomedical concepts into document and query representations to improve the performance of the search. For instance, Frihat and Fuhr [205] have developed a method where entities such as genes, diseases, and chemicals are automatically recognized and normalized to standard identifiers in biomedical ontologies such as MeSH [206] for diseases or NCBI Gene ID [67] for genes. Transferring and adapting such approaches to the patent domain could enhance the retrieval of life sciences patents, reducing terminological ambiguity and heterogeneity in biomedical vocabulary, allowing for an improved recall and reducing false negatives during the research.

While, to the best of our knowledge, after the TREC-Chem and PatOlympics initiatives, no major studies have been exclusively conducted on life sciences patent retrieval. Indeed, research efforts have shifted to two subtasks fundamental for patent search: chemical, biological and biomedical named entity recognition (NER), and extraction of chemical structures from patent documents. The two following sections will discuss these topics.

5.2. Chemical, biological and biomedical Named Entity Recognition (NER)

Biological, chemical and biomedical Named-Entity Recognition (NER) entails the identification of specific terms, such as compounds, diseases, genes, tissues, cells, or anatomical parts names to allow for knowledge discovery and development in this domain. It is fundamental for these algorithms to have corpora, sources of knowledge, expert-annotated textual data to train on [207]. In literature, several collections have been created to support chemical and biological NER. The ChEBI gold standard [208] has been expanded by Grego et al. in their work [209] for recognizing chemical entities through a machine learning approach based on Conditional Random Field. Beyond chemical entities, other linguistic and semantic elements have been annotated in patent texts. For instance, Kiss et al. [210] created a corpus of pharmaceutical patents belonging to the class A61K. And, at a larger scale, the BioCreative V - CHEMDNER [211] Patents Track provided annotations for chemical and gene names across 21000 medicinal patent abstracts [207].

Over the years, several approaches for biomedical NER have been developed, ranging from simpler to more sophisticated ones. Early systems relied primarily on dictionary-based techniques leveraging terms

from vocabularies, for instance, Jochem [212] (a dictionary for the recognition of small molecules and drugs in text, that merges information from UMLS, MeSH, ChEBI, DrugBank, KEGG, HMDB and ChemIDplus) and DrugBank [213] (a dictionary for drugs), to identify the entity occurrences in texts. These systems determine if a word or a group of words match exactly or approximately terms in dictionaries [214]. Despite these systems providing high precision, the recall is hindered by spelling errors in text. Moreover, these methods are expensive due to the amount of work necessary to keep the dictionary updated.

Other simple approaches are rule-based systems that rely on hand-crafted rules, sometimes combined with dictionaries [215] to enhance recall and precision. An example consists of suffixes like “-ase” enabling the recognition of enzymes. As dictionary-based approaches, these techniques are costly to maintain.

The introduction of machine learning-based approaches marked an advancement; in fact, over the years, the previous methods have been surpassed by machine learning models. These systems leverage statistical models through training them on labeled data and applying patterns to new texts. Traditional ML approaches for biomedical NER rely on supervised learning, based mainly on Maximum Entropy Markov Models (MEMMs) [216], and Conditional Random Fields (CRFs) [217–219], emerging as a gold standard. In fact, this last model overcomes the label-bias problem, that leads to over-predicting common labels, which characterizes MEMMs [214]. However, these approaches require meticulous feature engineering, which includes handling orthographic patterns (e.g., capitalizations, hyphens), morphological cues (e.g., “-ase” for enzymes), and their performance remains constrained by the quality of annotated data. For this reason, unsupervised and semi-supervised learning methods have also been investigated, but they have not been widely used in NER.

For chemical entities, approaches such as ChemSpot [220], which recognize drugs, abbreviations, molecular formulas, and IUPAC entities, have also demonstrated how hybrid methods, combining the pros of the previous approaches, could achieve good performance [214]. Another hybrid approach, combining dictionary-based and statistical methods, has been developed by Akhondi et al. [221] on CHEMDNER-patents data.

Not requiring extensive manual feature engineering, deep learning methods represent the direction the current state of the art has shifted to. In fact, in recent years, deep learning methods, capable of uncovering hidden features from non-labeled data, mainly Long Short Term Memory (LSTM) and BERT-based models, have been proposed, outperforming the state-of-the-art approaches [222–225]. For instance, Saad et al. in their work [226], develop a Long Short Term Memory (Bi-LSTM) based model for biomedical NER, evaluating and comparing different models for detecting biomedical named entities. Their model outperforms CRF and Bi-LSTM using solely word-level embeddings significantly. Mahendran et al. [227] evaluate both bidirectional long short-term memory (BiLSTM) and bidirectional encoder representations from transformer (BERT)-based methods. Copara et al. [228], and Malakordi et al. [229] also proposed a method for chemical entities identification in patents.

5.3. Chemical structure extraction

Chemical and pharmaceutical patents are full of images depicting the structures of protected compounds. Patent examiners, during the retrieval process, analyze these images to assess if a patent document is relevant, for example, to determine novelty or infringement. While humans can look at the image and intuitively understand the object of the invention, automatic retrieval systems need these images to be converted into machine-readable formats, such as the commonly used SMILES [42] and Molfile [44], which contain all necessary information to build the complete molecular graph. To this aim, several tools, both rule-based (e.g., OSRA [230]) and AI-based (e.g., machine

learning-based, such as chemoCR [231], or deep learning-based, such as ChemGrapher [232]), for identifying graph structures from chemical compound images have been developed. Among these tools, ChemReader [233] and CLiDE Pro [234] can also be found. Since these tools struggle with bond multiplicity, stereochemical configurations [232] and low-quality data, other studies have been carried out to enhance chemical structures extraction systems. For instance, Staker et al. in their work [235], developed a deep learning approach for first segmenting molecular structures from documents and then predicting SMILE chemical structures from the segmented images, to avoid the necessity for hand-crafted features, and to allow representation of structures from low-resolution images. Oldenhof et al. [232] utilize a deep neural network for predicting the graph structure of chemical compounds, consisting of a segmentation model and models for classifying atom locations, bonds, and charges.

Since chemical patents contain two main kinds of information, text and image, a system able to join them and leverage them to exploit all available chemical knowledge it is needed. For this reason, aiming to bridge text mining and optical chemical structure recognition in an automated way, Beard and Cole [236] propose ChemSchematicResolver, a tool for converting chemical schematic diagrams into machine-readable formats such as SMILES. This system leverages ChemDataExtractor [237], for text data, and OSRA [230], for image data, for the automatic assignment and extraction of diagram-label pairs and to resolve generic R-group structures, obtaining image-label-SMILES associations.

More recently, extending the concept of bridging text and image data to the patent domain, Wang et al. [238] developed a multi-modal framework, named CIRS (Chemical Information Reconstruction System), for identifying chemical entities from the text and images of chemical patents and aligning them to enrich near-drug space and boost drug discovery. This information extraction system could pave the way for developing an intelligent retrieval technology of chemical knowledge.

5.4. Biological sequences retrieval

The amount of biological sequence data, including DNA and RNA sequences, amino acid sequences of proteins, and three-dimensional structures of proteins, is consistently increasing over the years [239]. In order to search and compare biological sequences, sequence alignment tools have been developed, among which the Basic Local Alignment Search Tool (BLAST) [240] represents the most used approach in bioinformatics [241]. This algorithm, through a comparison between query sequences and sequences in databases, is able to discover similar sequences to a subject sequence. The output of the algorithm can be a variety of formats, which include plain text, HTML, and XML. There are also diverse versions of the algorithm specifically developed for determined sequences searches: BLASTn, that offers nucleotide to nucleotide searches, tBLASTn, for searching protein sequences into nucleic acid databases, BLASTx, for comparing a nucleotide query sequence against a protein database, BLASTp, to compare protein sequences [242], or tBLASTx, for looking for nucleic acid sequences into nucleic acid databases (further information regarding BLAST, its procedures and functions are reported in [35]). BLAST algorithm provides a fast search among sequences, although, at the same time, it overlooks biologically relevant but weak similarities that can be found thanks to PSI-BLAST [243]. There are also other sequence alignment tools, such as FASTA [244], provided by EBI, which includes others algorithms such as FASTX, FASTY, SSEARCH, GGSEARCH, GLSEARCH [245], and BLAT [246], provided by UCSC Genome Browser, Clustal Omega, and ClustalW/ClustalX [247] for multiple sequence alignment. Although there exist several biological sequence-search tools, the retrieval of relevant nucleic acid and protein sequence information from patent publications remains a growing challenge for R&D organizations and patent stakeholders. Indeed, sequence information in patent documents

is disclosed in diverse formats and in various sections within the document: it can be found in text (e.g., description, abstract, claims), in figures, in tables, and/or sequence listings [248]. Such listings, which were formatted as an ASCII plain text according to the WIPO ST.25 standard, after July 2022, must follow the ST.26 standard, which introduces a fully XML-based representation of sequences. Although this last format, which was introduced with the aim of increasing the machine-readability and management of these kinds of data, improves the standardization and the interoperability, a huge amount of pre-2022 patent literature still contains ST.25 sequence-listings, or even non-standard representations dispersed in descriptions, figures, or PDF attachments.

This fragmentation in biological sequence representation has led to the need of extracting, normalizing, and curating biological sequences from patent publications. Moreover, information regarding patented sequences is also fragmented among patent databases. Indeed, an early work by Xu et al. [249] concludes that a database comprehensive and inclusive of all patented sequences does not exist, and, thus, a combination of searches conducted across multiple databases is needed for getting a comprehensive view of the nucleic acid and amino acid sequences patent landscape. A later work by Li et al. [250] describe the construction of a free non-redundant patent sequence databases developed jointly by EPO and EMBL-EBI, structured on two-level clusters: the first level groups sequences that are 100% identical over their full length and the second level clusters further subdivide these groups by patent family, ensuring that identical sequences coming from distinct inventions are treated separately. This database has been further extended in a subsequent work by Li et al. [251], by introducing enriched biological annotations, correcting patent data, and increasing the coverage. However, these databases have not been updated anymore since 2013. Other initiatives have been carried out to aggregate non-redundant patent sequence data, such as Patgen [252] and Patome [253], although, Patgen is no longer accessible, while Patome handles redundancy solely at the patent-number level, and, consequently, identical sequences published under different patent numbers are treated as distinct [251]. Jefferson et al. [254] further analyzed worldwide practices for biological sequences disclosure, and highlighted the lack of a real global standardization (different patent offices handle the representation in a non unified way, and, moreover, from 2001 until 2007, most international applications included sequences in tiff or pdf files, containing non ASCII binary data), incomplete and heterogeneous access to published data (some patent offices offer bulk downloads or searchable listings, others provide only limited formats, and several do not publish sequence data at all), extensive redundancy and inconsistency among databases (identical sequences often appear under different patent numbers or families, and major public repositories, such as ENA/EBI, GenBank/NCBI, DDBJ, are not fully synchronized), and the analysis of these data remains challenging, making the construction of an integrated patent-sequence landscape slow, costly, and inaccessible to many users. Commercial infrastructures have also attempted to extract and normalize biological sequences from patents: Patent Lens or GENESEQ provide big sequence collections derived from sequence listings and full-text disclosures of patent documents. However, descriptions of their algorithms are not publicly documented and accessible for academic use, and data coverage may vary across jurisdictions. Despite these efforts, the academic literature offers a limited and fragmented coverage of approaches for retrieving biological sequences from patents, and further research is needed in this area.

Overall, life sciences patent retrieval encompasses a wide range of approaches that leverage named entity recognition and ontology-driven normalization to chemical structure extraction, in order to deal with the challenges posed by this field, such as nomenclature ambiguity and the need to capture both textual and non-textual information. However, current systems, when evaluated using retrieval metrics, often achieve low performance, making them not ready for real-world

applications. Moreover, many existing approaches present limits in terms of scalability, because they rely on manual ontology construction or on specific databases. Another common issue consists in the lack of consistent evaluation practices, since some studies do not report quantitative results or utilize standard metrics, making the comparison across studies difficult. Furthermore, despite improvements in chemical structure recognition, some systems still rely on partial information due to imperfections in the structure extraction, and, sometimes, textual information is not leveraged with non-textual data. Additionally, the fragmentation and heterogeneity of biological information across different formats, patent sections, and databases pose an additional challenge for the retrieval task. These limitations suggest the need for retrieval frameworks capable of jointly modeling textual, molecular, and biological information. In this direction, recent efforts to construct landscapes linking molecules to functional labels appear promising, as they may enable function-aware patent retrieval.

6. Discussion and conclusions

Patents are fundamental resources for research and development activities in both academic and industrial communities. In fact, before investing large amounts of money, organizations conduct prior art searches in order to determine whether an invention has already been protected. Among the documents examined during this process, patents are crucial as representing the earliest exposure of new technologies; indeed, solely a tiny fraction of new compounds, in the pharmaceutical field, is disclosed in the scientific literature and, for those that do, they are commonly exposed in patents 1-3 years earlier [255].

However, independently of the expertise of patent examiners, who typically rely on traditional keywords or boolean search, finding relevant patents is a cumbersome and time-consuming task [171]. Even though these methods are combined with several additional strategies, that include proximity, truncation and positional operators, field-specific filtering, classification-based exploration, citation networks, and advanced features (e.g., multilingual tools), sometimes specifically provided for searching bio-chemical entities, these methods still require extensive manual effort, time and knowledge. The difficulties of this task arise from several factors, including the consistent and exponential growth of patent documents worldwide, the complex technical and legal language adopted, particularly in the claim sections, and the multimodal nature of these data, consisting of text and images. In fact, when seeking pertinent documents, patent experts also look at visual cues, such as technical drawings or diagrams. Thus, to create systems able to partially replace or, at least, assist experts in their work, any supporting tool must account for these aspects. Furthermore, for patents belonging to the life sciences domain, there are additional points to consider, such as ambiguities in biomedical nomenclature, polysemy of terms, and the high dependency on visual representations, such as chemical structures not described in text, but that are fundamental in representing the scope of the invention.

Over the years, several automated and AI-based techniques both general-purpose and life sciences domain-specific have been proposed to overcome the given challenges. These approaches range from more traditional methods, such query expansion, leveraging external knowledge sources and top-k retrieved documents to obtain expansion terms, to metadata-based methods and more recent techniques, based on machine learning and natural language processing. Query expansion methods aim to enhance recall, but, by introducing noise terms from too general external knowledge or from irrelevant retrieved documents, they often suffer from reduced precision. Similarly, metadata-based methods present some limitations: several methods solely rely on certain metadata types, such as classification codes or citations, possibly disregarding relevant patents due to possibly inaccurate citations by examiners that, due to the huge amount of literature, overlook important prior art, or due to classification codes that may not precisely capture new or interdisciplinary technologies [25].

In life sciences domain specifically, these limitations are even more pronounced: several methods have been proposed by leveraging ontologies (e.g. MeSH, Gene Ontology), using semantic normalization and, despite they have demonstrated promising results, these methods, reaching low performance (e.g., rarely exceeding 20% and recall is around 50%), are still far from real-world applications, where finding all relevant documents is crucial.

Machine learning-based approaches have also been explored for patent retrieval, however they are often limited in catching the complex nature of patent documents, consisting in texts and images [16]. At the same time, recent developments in deep learning architectures, specifically transformer-based models, able to process contextual information in these text, have improved significantly the retrieval task performance. Nonetheless, these methods also come with limitations. For instance, models like BERT have an input length limit of 512 tokens, which means that long texts, very common in patent documents, should be segmented and combined in an optimal manner, so that the risk of losing information could be reduced.

Another recurring issue that characterizes all methods is the sub-optimal recall. This parameter is critical in the patent domain, where missing even a single relevant document could lead to serious legal and economic consequences, as well as wasted time and effort spent in research and development.

Looking ahead, multimodal approaches are emerging as a promising direction. In fact, since many patents include essential non-textual content, consisting of technical drawings, sketches, diagrams, and chemical structures, that is not always disclosed in patent texts, future systems should aim to integrate these diverse kinds of data. Some recent methods have already begun to explore this path, either by integrating visual cues to text-based retrieval or by developing specific systems intended to individually retrieve images. In the life sciences domain, these information are still leveraged separately; in fact, even if systems like PatCID achieve higher performance in extracting chemical structures from patent documents, they do not integrate these information with textual data and Markush structures, which are heavily used in patent claims to define the boundaries of the protected invention.

Recent advances in foundation models for chemistry and biology have shown that it is feasible to learn representations of molecules and biological sequences. These representations can be reutilized thanks to transfer learning for accomplishing other tasks. Among these models, on the chemical side can be found molecular language models such as ChemBERTa [256] or the more recent ChemML [257], while on the biological side, protein/genic language models such as DNABERT [258] and ProtBERT [259] that provide contextual embeddings of amino acid and genomic sequences. Building on these models, multimodal architectures such as Text2Mol [260] or MolLM [261] have been developed, which show how chemical structures and natural language descriptions can be mapped in the same latent space, enabling cross-modal retrieval (text-to-molecule and molecule-to-text), or encoding SMILES strings and textual information to support downstream tasks. Furthermore, at a broader level, collections of open models for drug discovery and development, such as TxGemma [262,263], have demonstrated how domain-specific representations can be leveraged in hard life sciences tasks, such as generating predictions, classifications, or text from therapeutic-related data. These architectures could be inspiring for developing multimodal patent retrieval systems in the life sciences domain, that go even beyond the text-image retrieval, but includes protein, chemical structure search from natural language and vice versa. Moreover, classification codes, such as CPC, could be complementarily leveraged to improve LLM-, VLM-, and multimodal patent retrieval systems, as well as for benchmarking and performance analysis. Classification information may indeed provide a useful reference for assessing recall and precision, especially in the life sciences domain, where terminological ambiguity, heterogeneous disclosure practices across applicants, and the frequent use of visual cues pose important challenges. Further research is needed for improving biological

sequence retrieval. Sequence information, such as DNA, RNA, and proteins, is disclosed in a variety of formats and locations within patent documents (descriptions, claims, tables, figures, and sequence listings). Even though the transition from the old WIPO ST.25 to ST.26 has increased the machine-readability and standardization of new patent documents, a large volume of patents still relies on old representations. To address this issue, initiatives aiming to construct non-redundant patent sequence databases, to aggregate and normalize sequences from multiple offices have been carried out. However, information remains fragmented across databases, and many extraction algorithms and pipelines are not openly available for academic use.

Another area that should be explored is federated search to integrate information from multiple data sources. This is because a patent document alone may not be sufficient to determine novelty of a filed invention application, as public disclosures can also be conveyed by means of scientific publications, technical sheets, or even web sources. Consequently, novel systems should be able to access and synthesize data across diverse sources, in order to reduce the risk of overlooking relevant prior art.

Another aspect that deserves consideration concerns the level at which patent retrieval systems operate, namely individual patent documents or patent families. While patent family members usually share the technical content of the invention and inventors, they often differ in terms of language, claims, classification codes, and prosecution stages. Consequently, document-level processing enables precise modeling of legal protection and linguistic variations across jurisdictions, but, at the same time, it inflates dataset size, resulting in an increased computational complexity, redundancy, and bias, possibly distorting retrieval and ranking outcomes. Conversely, family-level aggregation reduces redundancy and noise, but it may overlook technical or legal differences across national filings. Therefore, these aspects should be considered throughout all stages of data extraction, metric definition, and methodological design.

In summary, there are still several factors that prevent the development of general all-purpose systems. These include the high semantic ambiguity and variability of biomedical terminology, the limited performance of existing systems, the lack of integration of multimodal data, the incomplete database coverage, and the high fragmentation of biological sequence data across both multiple databases and formats. Additional challenges arise from the length and complexity of patent documents, and the need to search on multiple databases to obtain a more complete overview of the prior art.

Moving forward, to advance in the field of patent retrieval, in particular in the life sciences domain, diverse future actions could be done. Among them, the development of truly multimodal architectures able to jointly handle text, images, technical drawings, chemical structures, and biological sequences could enhance the recall of retrieval systems. In this direction, future systems should be able to align different kinds of representation, such as text-image, text-structure, and text-sequence. Particularly referring to textual information, improved strategies for handling long and complex documents are needed. Furthermore, the development of large-scale multimodal public patent datasets would allow reproducible benchmarking across the different developed patent retrieval methods, facilitating the comparison. The use of classification codes would also be useful, in particular for improving retrieval systems and benchmarking, as well as for developing category-specific approaches for the heterogeneous variety of inventions belonging to the life sciences field. A unified system containing all patent literature would be beneficial for users seeking complete information and needing to search across different databases.

Going beyond future long-term research directions, practitioners working with life sciences patents have multiple choices at their disposal. For research predominantly conducted via textual queries, transformer-based methods, and the utilization of vocabularies such as MeSH, Gene Ontology, or ChEBI can enhance the recall compared to traditional keyword search combined with operators. In the

chemical domain, practitioners can combine keyword-based searches and structure-based strategies (e.g., exact match search, similarity search, Markush search, and other approaches further explained in Section 1.1.3), conducted on both open-access resources and commercial platforms, which provide a wider coverage. For biological sequence information involving DNA, RNA and proteins, research conducted leveraging sequence similarity algorithms such as BLAST embedded on multiple databases remains fundamental due to the fragmented and heterogeneous nature of sequence disclosures. Relying on a single repository, nowadays, is insufficient, and practitioners should search across multiple sources (e.g., GenBank, GENESEQ) and consider both exact and approximate matches.

In conclusion, despite over the years progress has been made in automated methods for retrieving relevant patents both in general and within the life sciences domain, several challenges still characterize the state of the art systems. These challenges highlight the need for systems able to integrate multiple kinds of data from multiple sources, exploiting the information in patent documents, including text, chemical structures, biological sequences, and technical figures. Recent developments in multimodal architectures, chemical and biological language models, and federated search systems indicate that more integrated and efficient solutions are becoming available. By combining these emerging methods with the well-established practices for text, structure, and sequence search, it will be possible to design retrieval systems aimed at improving efficiency that can support users from both industry and academia.

CRedit authorship contribution statement

Sara Poce: Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Gianni Cerro:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. List of acronyms and abbreviations

This section provides a list of the main acronyms and abbreviations used throughout the manuscript in order to improve readability and facilitate comprehension for the reader (see Table A.1).

Appendix B. Patent essentials

Patents are legal instruments that confer to applicant or the inventor exclusive and territorial rights to exclude others from making, using, or selling a particular invention for a limited period of time, which is usually 20 years from the date of filing [264]. In order to be patented, an invention must meet technical and legal requirements, which include novelty, inventive step (non-obviousness), industrial applicability, subject matter eligibility, and lawfulness, and sufficient disclosure [265]. Patent documents typically present: bibliometric data (such as the title of the invention, patent number, filing date, publication date, classification codes, citations and applicant, inventor(s) and assignee(s) information), abstract, background and state of the art of the invention, a brief and a detailed descriptions of the invention, figures and their descriptions, and claims. Among them, hierarchical classification codes are assigned by patent issuing bodies (e.g., EPO, WIPO, USPTO) in order to streamline search and indexing. The main systems are the International Patent Classification (IPC) [266] and the Cooperative Patent Classification (CPC) [106]. The IPC, established

Table A.1

List of acronyms and abbreviations used throughout the manuscript.

Acronym	Full name
WIPO	World Intellectual Property Organization
EPO	European Patent Office
USPTO	United States Patent and Trademark Office
KIPO	Korean Intellectual Property Office
CNIPA	China National Intellectual Property Administration
IPC	International Patent Classification
CPC	Cooperative Patent Classification
FTO	Freedom-to-Operate
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
NER	Named Entity Recognition
SVM	Support Vector Machine
KNN	k-Nearest Neighbors
RF	Random Forest
NB	Naive Bayes
NBSVM	Combination of Naive Bayes and Support Vector Machine
RBM	Restricted Boltzmann Machine
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory
GRU	Gated Recurrent Unit
XML	eXtensible Markup Language
WSD	Word Sense Disambiguation
LDA	Latent Dirichlet Allocation
tf	term frequency
tf-idf	term frequency–inverse document frequency
Word2Vec	Word to Vector
Doc2Vec	Document to Vector
GeT-In	Graph-embedded Tree
BOW	Bag Of Words
BOP	Bag Of Phrases
LFT	Local Frequent Terms
SGBT	Stochastic Gradient Boosting Trees
IAF	Inverse Assignee Frequency
CSI	Centralized Sample Index
MM	Multiple Models
GM	Global Models
IR	Information Retrieval
BM25	Best Matching 25
BM25F	Best Matching 25 with Fields
BERT	Bidirectional Encoder Representations from Transformers
LLM	Large Language Model
VLM	Vision-Language Model
LVLm	Large Vision-Language Model
CLIP	Contrastive Language–Image Pre-training
ViT	Vision Transformer
RAG	Retrieval-Augmented Generation
TransE	Translation-Based Knowledge Graph Embedding
VSM	Vector Space Model
CLS	Classify Token
SEP	Separation Token
SOTA	State of the Art
AP	Average Precision
MAP	Mean Average Precision
DCG	Discounted Cumulative Gain
nDCG	Normalized Discounted Cumulative Gain
IDCG	Ideal Discounted Cumulative Gain
P@k	Precision at k
R	Recall
R@k	Recall at k
PRES	Patent Retrieval Evaluation Score
MRR@k	Mean Reciprocal Rank at k
RFR	Rank of the First Relevant
OCR	Optical Character Recognition
CRAFT	Character Region Awareness For Text detection
STR	Scene Text Recognition
NCBI	National Center for Biotechnology Information
EBI	European Bioinformatics Institute
BLAST	Basic Local Alignment Search Tool
BLASTn	Basic Local Alignment Search Tool for nucleotides

(continued on next page)

Table A.1 (continued).

tBLASTn	translated Basic Local Alignment Search Tool for nucleotides
BLASTx	Basic Local Alignment Search Tool for translated nucleotides
tBLASTx	translated Basic Local Alignment Search Tool for translated nucleotides
PSI-BLAST	Position-Specific Iterated Basic Local Alignment Search Tool
FASTA	Fast Alignment Search Tool Algorithm
FASTX	Fast Alignment Search Tool for translated nucleotide to protein comparison
FASTY	Fast Alignment Search Tool with frameshift alignment
SSEARCH	Smith–Waterman Search
GGSEARCH	Global–Global Search
GLSEARCH	Global–Local Search
BLAT	BLAST-Like Alignment Tool
SMILES	Simplified Molecular Input Line Entry Specification
ASCII	American Standard Code for Information Interchange
OSRA	Optical Structure Recognition Application
InChI	International Chemical Identifier
CAS	Chemical Abstracts Service
MeSH	Medical Subject Heading
NCI	National Cancer Institute
EMBL-EBI	European Molecular Biology Laboratory – European Bioinformatics Institute
GO	Gene Ontology
NDF	National Drug File
UMLS	Unified Medical Language System
KEGG	Kyoto Encyclopedia of Genes and Genomes
HMDB	Human Metabolome Database
DDBJ	DNA Data Bank of Japan
ENA	European Nucleotide Archive

with the Strasbourg Agreement in 1971 [98], is the most popular system used worldwide in more than 100 countries [267], and structures technologies into Sections (from A to H), Classes, SubClasses, Groups, and SubGroups (e.g., A61B17/11). The CPC system, developed by EPO and USPTO as an extension of IPC, is primarily used in Europe and the U.S. It includes Section A-H (with a finer subdivision of IPC) and Section Y for emerging technologies.

In the life sciences domain, the most commonly used IPC/CPC codes fall within IPC Sections A, B, C, and G. Examples include the following Classes (e.g., A61) and Subclasses (e.g., A61B) and are not limited to [266,268]:

- A01 – Agriculture: A01H new plants; A01N biocides and plant protection products;
- A61 – Medical Science and Health: A61B diagnosis and surgery; A61C dentistry; A61D veterinary; A61F implants and prostheses; A61G patient transport/accommodation, operating tables/chairs; A61H physical therapy devices; A61J medical containers; A61K medical preparations; A61L sterilization and biomaterials for bandages, dressings, absorbents; A61M devices for introducing or removing media to/from the body; A61N electro/magneto/radiation/ultrasound therapy; A61P therapeutic indications;
- B01 – Physical or Chemical Processes or Apparatus: B01L laboratory apparatus;
- C07 – Organic Chemistry;
- C12 – Biochemistry, Microbiology, Enzymology, Genetic Engineering: C12N microorganisms, enzymes, genetic engineering, culture media; C12P fermentation or enzyme-using processes; C12Q testing/assays;
- G16 – Informatics: G16B bioinformatics; G16H healthcare informatics;

In addition, the CPC scheme provides finer refinements by introducing new subgroups relative to IPC. Typically, users rely on IPC classes delimiting the technological domain, whereas use CPC subgroups for operative refinement.

At the European Patent Office (EPO), patent examiners conduct a prior art search for every patent application. The result of this process is a document, called *search report*, which lists relevant documents

retrieved during the research accompanied by comments/opinions regarding how each document affects the claims of the filed patent. Cited documents can mainly fall into three different categories: (i) “A”, the closest prior art that does not prejudice novelty or inventive step, (ii) “Y”, documents that combined prejudice novelty or inventive step, and (iii) “X”, documents that prejudice novelty or inventive step [151].

Appendix C. Non-retrieval-related patent analysis tasks

Due to the intricate legal and technical aspects that characterize patent documents, various tasks have been developed to support their analysis and exploitation. These include patent classification, patent surveillance and monitoring, patent landscaping and technology forecasting, patent quality assessment and market value prediction, and patent generation [14,16].

C.1. Patent classification

Patent classification refers to the process of assigning one or more standardized codes (e.g., IPC, CPC) to a patent document, in order to allocate the invention to a specific group of technologies. This process of organizing patents into structured categories is fundamental in facilitating patent search and analysis, allowing for efficient management. This procedure is undertaken almost exclusively by patent experts and patent examiners. Due to the huge increase in patent applications, classification techniques based on automatic methods, such as machine learning and natural language processing, have been developed to support and accelerate this process [267,269,270].

C.2. Patent current and future trends

Patent surveillance and monitoring consist of the continuous tracking of new patented inventions within specific technological domains, competitors, or geographic regions. Patent landscaping and technology forecasting involve the extrapolation of information from large volumes of patent data oriented to the identification of technological trends, emerging fields, and innovation hotspots. This is particularly important for companies for decision-making, investments, and research and development planning.

C.3. Novel patent generation

In general, patent drafting is a cumbersome procedure carried out by patent attorneys, who have a deep knowledge of both the law and the patent system. It is also a really expensive task: the cost to draft a patent of average complexity in US is about \$10000 [271]. Hence, in recent years, a new automated patent task is emerging: patent generation. It involves the use of computational methods, often LLM-based, to assist in the drafting of patent applications, helping in writing parts of patents' document, such as claims [271,272]. Despite in several studies, such as [273] where the application of LLM to biomedical patent claim generation was assessed, it was demonstrated that these models can generate patent claims with appropriate writing styles and technical contents if adequately fine-tuned, patent experts remain irreplaceable since they are not able to operate fully autonomously.

Data availability

No data was used for the research described in the article.

References

- [1] T. Alves, R. Rodrigues, H. Costa, M. Rocha, Development of an information retrieval tool for biomedical patents, *Comput. Methods Programs Biomed.* 159 (2018) 125–134, <http://dx.doi.org/10.1016/j.cmpb.2018.03.012>.
- [2] European patent office - healthcare and life sciences, 2020, <https://report-archive.epo.org/about-us/annual-reports-statistics/statistics/2020/healthcare-and-life-sciences.html>. (Accessed 21 June 2025).
- [3] F. Wang, L. Lin, Domain lexicon-based query expansion for patent retrieval, in: 2016 12th International Conference on Natural Computation, Fuzzy Systems and Knowledge Discovery, ICNC-FSKD, 2016, pp. 1543–1547, <http://dx.doi.org/10.1109/FSKD.2016.7603405>.
- [4] World Intellectual Property Organization, World intellectual property indicators 2024: Highlights - patents highlights, 2024, <https://www.wipo.int/web/publications/world-intellectual-property-indicators-2024-highlights/en/patents-highlights.html>. (Accessed 03 November 2025).
- [5] World Intellectual Property Organization (WIPO), WIPO translate – breaking language barriers with AI, 2025, <https://www.wipo.int/en/web/ai-tools-services/wipo-translate>. (Accessed 03 November 2025).
- [6] European Patent Office, Patent translate, 2025, <https://www.epo.org/en/searching-for-patents/helpful-resources/patent-translate>. (Accessed 03 November 2025).
- [7] A.A. Larroyed, Machine translation and disclosure of patent information, *IIC-International Rev. Intellect. Prop. Compét. Law* 49 (7) (2018) 763–786, <http://dx.doi.org/10.1007/s40319-018-0736-9>.
- [8] T.-F. Zhu, R. Qian, X. Wei, A.-P. Lu, D.-S. Cao, PatentNetML: A novel framework for predicting key compounds in patents using network science and machine learning, *J. Med. Chem.* 67 (2) (2024) 1347–1359, <http://dx.doi.org/10.1021/acs.jmedchem.3c01893>.
- [9] Y.T. Demey, D. Golzio, Search strategies at the European patent office, *World Pat. Inf.* 63 (2020) 101989, <http://dx.doi.org/10.1016/j.wpi.2020.101989>.
- [10] A. Khode, S. Jambhorkar, A literature review on patent information retrieval techniques, *Indian J. Sci. Technol.* 10 (36) (2017) 1–13, <http://dx.doi.org/10.17485/ijst/2017/v10i37/116435>.
- [11] W. Magdy, G.J. Jones, A study on query expansion methods for patent retrieval, in: Proceedings of the 4th Workshop on Patent Information Retrieval, 2011, pp. 19–24, <http://dx.doi.org/10.1145/2064975.2064982>.
- [12] W. Shalaby, W. Zadrozny, Patent retrieval: a literature review, *Knowl. Inf. Syst.* 61 (2019) 631–660, <http://dx.doi.org/10.1007/s10115-018-1322-7>.
- [13] A. Ali, A. Tufail, L.C. De Silva, P.E. Abas, Innovating patent retrieval: a comprehensive review of techniques, trends, and challenges in prior art searches, *Appl. Syst. Innov.* 7 (5) (2024) 91, <http://dx.doi.org/10.3390/asi7050091>.
- [14] R. Krestel, R. Chikkamath, C. Hewel, J. Risch, A survey on deep learning for patent analysis, *World Pat. Inf.* 65 (2021) 102035, <http://dx.doi.org/10.1016/j.wpi.2021.102035>.
- [15] L. Jiang, S.M. Goetz, Natural language processing in the patent domain: a survey, *Artif. Intell. Rev.* 58 (7) (2025) 214, <http://dx.doi.org/10.1007/s10462-025-11168-z>.
- [16] H.H. Shomee, Z. Wang, S.N. Ravi, S. Medya, A survey on patent analysis: From nlp to multimodal ai, in: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2025, pp. 8545–8561, <http://dx.doi.org/10.18653/v1/2025.acl-long.419>.
- [17] N.S. Clarke, The basics of patent searching, *World Pat. Inf.* 54 (2018) S4–S10, <http://dx.doi.org/10.1016/j.wpi.2017.02.006>.
- [18] Clarivate, Derwent innovation, 2025, <https://clarivate.com/intellectual-property/patent-intelligence/derwent-innovation/>. (Accessed 23 July 2025).
- [19] PatSnap, PatSnap: AI-powered IP and R&D intelligence, 2025, <https://www.patsnap.com/>. (Accessed 23 July 2025).
- [20] Minesoft and RWS, PatBase, 2025, <https://www.patbase.com/login.asp>. (Accessed 23 July 2025).
- [21] Questel, OrbitIntelligence — IP intelligence software, 2025, <https://www.questel.com/patent/ip-intelligence-software/orbit-intelligence/>. (Accessed 23 July 2025).
- [22] European Patent Office, Espacenet-worldwide patent search, 2025, <https://worldwide.espacenet.com/>. (Accessed 23 July 2025).
- [23] Google, Google patents: Search and read patents worldwide, 2025, <https://patents.google.com/>. (Accessed 23 July 2025).
- [24] United States Patent and Trademark Office, Patent public search (PPUBS)–published patent application access and status, 2025, <https://ppubs.uspto.gov/pubwebapp/static/pages/landing.html>. (Accessed 23 July 2025).
- [25] J. Lee, S. Park, J. Lee, A fast and scalable algorithm for prior art search, *IEEE Access* 10 (2022) 7396–7407, <http://dx.doi.org/10.1109/ACCESS.2022.3141494>.
- [26] L. Helmers, F. Horn, F. Biegler, T. Oppermann, K.-R. Müller, Automating the search for a patent's prior art with a full text similarity search, *PLoS One* 14 (3) (2019) e0212103, <http://dx.doi.org/10.1371/journal.pone.0212103>.
- [27] Z. Xie, K. Miyazaki, Evaluating the effectiveness of keyword search strategy for patent identification, *World Pat. Inf.* 35 (1) (2013) 20–30, <http://dx.doi.org/10.1016/j.wpi.2012.10.005>.
- [28] P. Mahdabi, F. Crestani, The effect of citation analysis on query expansion for patent retrieval, *Inf. Retr.* 17 (2014) 412–429, <http://dx.doi.org/10.1007/s10791-013-9232-5>.
- [29] V. Stamatis, M. Salampasis, K. Diamantaras, Machine learning methods for results merging in patent retrieval, *Data Technol. Appl.* 58 (3) (2023) 363–379, <http://dx.doi.org/10.1108/DTA-06-2021-0156>.
- [30] World Intellectual Property Organization, WIPO standard st.26, 2024, <https://www.wipo.int/documents/d/standards/docs-en-03-26-01.pdf>. (Accessed 28 October 2025).
- [31] World Intellectual Property Organization (WIPO), WIPO sequence, 2025, <https://www.wipo.int/en/web/standards/sequence/index>. (Accessed 28 October 2025).
- [32] World Intellectual Property Organization, WIPO standard st.25, 2009, <https://www.wipo.int/documents/d/standards/docs-en-03-25-01.pdf>. (Accessed 28 October 2025).
- [33] World Intellectual Property Organization (WIPO), PATENTSCOPE — Sequence search, 2025, <https://patentscope.wipo.int/search/en/sequences.jsf>. (Accessed 29 October 2025).
- [34] The Lens, PatSeq data — Globe, 2025, <https://www.lens.org/lens/bio/patseqdata/#/globe/>. (Accessed 29 October 2025).
- [35] D. Wheeler, M. Bhagwat, BLAST QuickStart: Example-driven web-based BLAST tutorial, in: N.H. Bergman (Ed.), *Comparative Genomics: Volumes 1 and 2*, Humana Press, Totowa, NJ, 2007, http://dx.doi.org/10.1007/978-1-59745-514-5_9.
- [36] Clarivate, GENESeq, 2025, <https://clarivate.com/intellectual-property/patent-intelligence/geneseq/>. (Accessed 06 November 2025).
- [37] G.M. Downs, J.D. Holliday, P. Willett, Representation and searching of chemical structure information in patents, in: *Current Challenges in Patent Information Retrieval*, Springer, 2017, pp. 391–407, http://dx.doi.org/10.1007/978-3-642-19231-9_17.
- [38] TPR - Technology & Patent Research, Chemical structure searches explained: Types, tools and tips for getting the most accurate patent search results, 2024, <https://www.tprinternational.com/chemical-structure-searches-explained-types-tools-and-tips-for-getting-the-most-accurate-patent-search-results/>. (Accessed 31 October 2025).
- [39] International Union of Pure and Applied Chemistry (IUPAC), IUPAC–International union of pure and applied chemistry, 2025, <https://iupac.org/>. (Accessed 30 October 2025).
- [40] Chemical Abstracts Service (CAS), Division of the American Chemical Society providing authoritative chemical substance information, <https://www.cas.org/>. (Accessed 31 July 2025).
- [41] Chemical Abstracts Service (CAS), CAS registry, 2025, <https://www.cas.org/cas-data/cas-registry>. (Accessed 31 October 2025).
- [42] D. Weininger, SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules, *J. Chem. Inf. Comput. Sci.* 28 (1) (1988) 31–36.
- [43] S. Heller, A. McNaught, S. Stein, D. Tchekhovskoi, I. Pletnev, InChI-the worldwide chemical structure identifier standard, *J. Cheminformatics* 5 (1) (2013) 7.
- [44] A. Dalby, J.G. Nourse, W.D. Hounshell, A.K. Gushurst, D.L. Grier, B.A. Leland, J. Laufer, Description of several chemical structure file formats used by computer programs developed at molecular design limited, *J. Chem. Inf. Comput. Sci.* 32 (3) (1992) 244–255.
- [45] National Center for Biotechnology Information (NCBI), PubChem, 2025, <https://pubchem.ncbi.nlm.nih.gov/>. (Accessed 31 October 2025).
- [46] Royal Society of Chemistry (RSC), ChemSpider: Search and share chemistry, 2025, <https://www.chemspider.com/>. (Accessed 31 October 2025).
- [47] SureChEMBL, SureChEMBL, 2025, <https://www.surechembl.org/>. (Accessed 31 October 2025).
- [48] World Intellectual Property Organization (WIPO), PATENTSCOPE - WIPO patent search, 2025, <https://patentscope.wipo.int/search/en/search.jsf>. (Accessed 31 July 2025).
- [49] Sigma-Aldrich / MilliporeSigma, Sigma-aldrich – product database, 2025, <https://www.sigmaaldrich.com/IT/it/search/database?focus=products>. (Accessed 31 October 2025).
- [50] Chemical Abstracts Service (CAS), a division of the American Chemical Society, CAS common chemistry: An open community resource for accessing chemical information, 2025, <https://commonchemistry.cas.org/>. (Accessed 31 October 2025).
- [51] Reaxys, Expert-curated chemistry and patent database integrating beilstein, gmelin and patent chemistry data, 2025, <https://www.reaxys.com/>. (Accessed 31 July 2025).
- [52] Clarivate Analytics, Derwent world patents index (DWPI), 2025, <https://clarivate.com/intellectual-property/patent-intelligence/derwent-world-patents-index/>. (Accessed 31 October 2025).
- [53] Dr. Paul Oldham (Manchester Institute of Innovation Research at Manchester University) and One World Analytics under the coordination of Ms. Irene Kitsara (WIPO), The WIPO manual on open source patent analytics (2nd edition), 2025, <https://wipo-analytics.github.io/manual/patentscope-1.html>. (Accessed 11 November 2025).

- [54] United States Patent and Trademark Office (USPTO), USPTO patent public search, 2025, <https://www.uspto.gov/patents/search/patent-public-search>. (Accessed 31 July 2025).
- [55] German Patent and Trade Mark Office (DPMA), DEPATISnet - patent search by DPMA, 2025, <https://www.dpma.de/english/search/depatismet/>. (Accessed 31 July 2025).
- [56] CLEF Initiative, TU Wien, CLEF-IP - evaluation lab on intellectual property, 2025, <http://ifs.tuwien.ac.at/~clef-ip/>. (Accessed 31 July 2025).
- [57] F. Piroi, A. Hanbury, Evaluating information retrieval systems on European patent data: The clef-ip campaign, in: *Current Challenges in Patent Information Retrieval*, Springer, 2017, pp. 113–142, http://dx.doi.org/10.1007/978-3-662-53817-3_4.
- [58] National Institute of Informatics (NII), NTCIR, 2025, <https://research.nii.ac.jp/ntcir/permission/data-en.htm>. (Accessed 31 July 2025).
- [59] M. Lupu, J. Huang, J. Zhu, J. Tait, TREC-CHEM: large scale chemical information retrieval evaluation at TREC, in: *ACM SIGIR Forum*, vol. 43, ACM New York, NY, USA, 2009, pp. 63–70, <http://dx.doi.org/10.1145/1670564.167057>.
- [60] M. Krallinger, O. Rabal, F. Leitner, M. Vazquez, D. Salgado, Z. Lu, R. Leaman, Y. Lu, D. Ji, D.M. Lowe, et al., The ChEMDNER corpus of chemicals and drugs and its annotation principles, *J. Cheminformatics* 7 (Suppl. 1) (2015) S2, <http://dx.doi.org/10.1186/1758-2946-7-S1-S2>.
- [61] Y. Zhang, J. Xu, H. Chen, J. Wang, Y. Wu, M. Prakasam, H. Xu, Chemical named entity recognition in patents by domain knowledge and unsupervised feature learning, *Database* 2016 (2016) baw049, <http://dx.doi.org/10.1093/database/baw049>.
- [62] UC San Diego, BindingDB, 2025, <https://www.bindingdb.org/rwd/bind/index.jsp>. (Accessed 11 November 2025).
- [63] Cambia, Patent lens, 2025, <https://www.lens.org/>. (Accessed 11 November 2025).
- [64] World Intellectual Property Organization (WIPO), PATENTSCOPE chemical compounds search, 2025, <https://patentscope.wipo.int/search/en/chemc/chemc.jsf>. (Accessed 11 November 2025).
- [65] CAS (Chemical Abstracts Service), SciFinder-n, 2025, <https://www.cas.org/solutions/cas-scifinder-discovery-platform/cas-scifinder>. (Accessed 05 November 2025).
- [66] National Library of Medicine - National Center for Biotechnology Information, GenBank, 2025, <https://www.ncbi.nlm.nih.gov/genbank/>. (Accessed 05 November 2025).
- [67] National Library of Medicine - National Center for Biotechnology Information, Gene, 2025, <https://www.ncbi.nlm.nih.gov/gene/>. (Accessed 05 November 2025).
- [68] National Library of Medicine - National Center for Biotechnology Information, Protein, 2025, <https://www.ncbi.nlm.nih.gov/protein/>. (Accessed 05 November 2025).
- [69] EMBL-EBI, EMBL-EBI data resources and tools, 2025, <https://www.ebi.ac.uk/services/data-resources-and-tools/>. (Accessed 05 November 2025).
- [70] Chemical Abstracts Service (CAS), a division of the American Chemical Society, ChEMBL, 2025, <https://www.ebi.ac.uk/chembl/>. (Accessed 11 November 2025).
- [71] EMBL-EBI, UniProt, 2025, <https://www.uniprot.org/>. (Accessed 05 November 2025).
- [72] Nucleic Acids Research, Nucleic acids research - database issues, 2025, https://academic.oup.com/nar/search-results?f_TocHeadingTitleList=Database+Issue. (Accessed 06 November 2025).
- [73] W. Magdy, G.J. Jones, PRES: a score metric for evaluating recall-oriented information retrieval applications, in: *Proceedings of the 33rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2010, pp. 611–618.
- [74] P. Mahdabi, S. Gerani, J.X. Huang, F. Crestani, Leveraging conceptual lexicon: query disambiguation using proximity information for patent retrieval, in: *Proceedings of the 36th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2013, pp. 113–122, <http://dx.doi.org/10.1145/2484028.248405>.
- [75] P. Sharma, R. Tripathi, V.K. Singh, R. Tripathi, Automated patents search through semantic similarity, in: *2015 International Conference on Computer, Communication and Control, IC4, IEEE*, 2015, pp. 1–5, <http://dx.doi.org/10.1109/IC4.2015.7375692>.
- [76] C.-P. Phan, H.-Q. Nguyen, T.-T. Nguyen, Ontology-based heuristic patent search, *Int. J. Web Inf. Syst.* 15 (3) (2019) 258–284, <http://dx.doi.org/10.1108/IJWIS-06-2018-0053>.
- [77] B. Al-Shboul, S.-H. Myaeng, Wikipedia-based query phrase expansion in patent class search, *Inf. Retr.* 17 (2014) 430–451, <http://dx.doi.org/10.1007/s10791-013-9233-4>.
- [78] P. Mahdabi, F. Crestani, Learning-based pseudo-relevance feedback for patent retrieval, in: *Multidisciplinary Information Retrieval: 5th Information Retrieval Facility Conference, IRFC 2012, Vienna, Austria, July 2–3, 2012 Proceedings 5*, Springer, 2012, pp. 1–11, http://dx.doi.org/10.1007/978-3-642-31274-8_1.
- [79] P. Sharma, R. Tripathi, R. Tripathi, Finding similar patents through semantic query expansion, *Procedia Comput. Sci.* 54 (2015) 390–395, <http://dx.doi.org/10.1016/j.procs.2015.06.045>.
- [80] Princeton University, WordNet: A lexical database for english, 2025, <https://wordnet.princeton.edu/>. (Accessed 25 June 2025).
- [81] Giza++: Training of statistical translation models, 2025, <https://www2.statmt.org/moses/giza/GIZA++.html>. (Accessed 04 February 2025).
- [82] W. Tannebaum, A. Rauber, PatNet: a lexical database for the patent domain, in: *Advances in Information Retrieval: 37th European Conference on IR Research, ECIR 2015, Vienna, Austria, March 29–April 2, 2015. Proceedings 37*, Springer, 2015, pp. 550–555, http://dx.doi.org/10.1007/978-3-319-16354-3_60.
- [83] W. Tannebaum, A. Rauber, Using query logs of USPTO patent examiners for automatic query expansion in patent searching, *Inf. Retr.* 17 (2014) 452–470, <http://dx.doi.org/10.1007/s10791-014-9238-7>.
- [84] Z. Wu, M. Palmer, Verb semantics and lexical selection, 1994, <http://dx.doi.org/10.48550/arXiv.cmp-lg/9406033>, *Proceedings of ACL94*, arXiv preprint Cmp-Lg/9406033.
- [85] M. Strube, S.P. Ponzetto, WikiRelate! computing semantic relatedness using wikipedia, in: *AAAI*, vol. 6, 2006, pp. 1419–1424.
- [86] MongoDB, 2025, <https://www.mongodb.com/>. (Accessed 07 July 2025).
- [87] KEA: Keyphrase extraction algorithm, 2025, <https://community.nzdil.org/kea/>. (Accessed 07 July 2025).
- [88] Lemur, 2025, <https://www.lemurproject.org/>. (Accessed 07 July 2025).
- [89] The Stanford Natural Language Processing Group, Stanford log-linear part-of-speech tagger, 2025, <https://nlp.stanford.edu/software/tagger.shtml>. (Accessed 07 July 2025).
- [90] V. Lavrenko, W.B. Croft, Relevance-based language models, in: *Proceedings of the 24th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR '01, ACM*, 2001, pp. 120–127, <http://dx.doi.org/10.1145/383952.383972>.
- [91] D. Ganguly, J. Leveling, W. Magdy, G.J. Jones, Patent query reduction using pseudo relevance feedback, in: *Proceedings of the 20th ACM International Conference on Information and Knowledge Management*, 2011, pp. 1953–1956, <http://dx.doi.org/10.1145/2063576.2063863>.
- [92] S. Bashir, A. Rauber, Improving retrievability of patents with cluster-based pseudo-relevance feedback documents selection, in: *Proceedings of the 18th ACM Conference on Information and Knowledge Management*, 2009, pp. 1863–1866, <http://dx.doi.org/10.1145/1645953.1646250>.
- [93] H. Li, A. Mourad, B. Koopman, G. Zuccon, How does feedback signal quality impact effectiveness of pseudo relevance feedback for passage retrieval, in: *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2022, pp. 2154–2158, <http://dx.doi.org/10.1145/3477495.3531822>.
- [94] L. Wang, Y. Li, Keyword elicitation for patent retrieval by using bibliographic information, in: *The 3rd International Conference on Data Mining and Intelligent Information Technology Applications, IEEE*, 2011, pp. 163–167.
- [95] A. Fujii, Enhancing patent retrieval by citation analysis, in: *Proceedings of the 30th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2007, pp. 793–794, <http://dx.doi.org/10.1145/1277741.1277912>.
- [96] P. Mahdabi, F. Crestani, Query-driven mining of citation networks for patent citation retrieval and recommendation, in: *Proceedings of the 23rd ACM International Conference on Conference on Information and Knowledge Management*, 2014, pp. 1659–1668, <http://dx.doi.org/10.1145/2661829.2661899>.
- [97] L. Siddharth, G. Li, J. Luo, Enhancing patent retrieval using text and knowledge graph embeddings: a technical note, *J. Eng. Des.* 33 (8–9) (2022) 670–683, <http://dx.doi.org/10.1080/09544828.2022.2144714>.
- [98] C.G. Harris, R. Arens, P. Srinivasan, Using classification code hierarchies for patent prior art searches, *Curr. Challenges Pat. Inf. Retr.* (2011) 287–304, http://dx.doi.org/10.1007/978-3-642-19231-9_14.
- [99] I.-S. Kang, S.-H. Na, J. Kim, J.-H. Lee, Cluster-based patent retrieval, *Inf. Process. Manage.* 43 (5) (2007) 1173–1182, <http://dx.doi.org/10.1016/j.ipm.2006.11.006>.
- [100] Y.-L. Chen, Y.-T. Chiu, An IPC-based vector space model for patent retrieval, *Inf. Process. Manage.* 47 (3) (2011) 309–322, <http://dx.doi.org/10.1016/j.ipm.2010.06.001>.
- [101] M. Verma, V. Varma, Exploring keyphrase extraction and IPC classification vectors for prior art search, in: *CLEF (Notebook Papers/Labs/Workshop)*, 2011.
- [102] D. Serafino, Survey of patent pools demonstrates variety of purposes and management structures, *Knowl. Ecol. Int.* (2007) <http://keionline.org/content/view/69/1>.
- [103] S. Robertson, S. Walker, S. Jones, M. Hancock-Beaulieu, M. Gatford, Okapi at TREC-3, 1994, p. 0.
- [104] L. Page, S. Brin, R. Motwani, T. Winograd, The PageRank Citation Ranking: Bringing Order to the Web, Technical Report 1999-66, Stanford InfoLab, 1999, URL <http://ilpubs.stanford.edu:8090/422/>.
- [105] B. Song, J. Luo, Mining patent precedents for data-driven design: the case of spherical rolling robots, *J. Mech. Des.* 139 (11) (2017) 111420, <http://dx.doi.org/10.1115/1.4037613>.
- [106] European Patent Office (EPO), Cooperative patent classification (CPC), 2025, <https://www.epo.org/en/searching-for-patents/helpful-resources/first-time-here/classification/cpc>. (Accessed 01 August 2025).

- [107] Cooperative patent classification (CPC) – official website, 2026, <https://www.cooperativepatentclassification.org/home>. (Accessed 1 February 2026).
- [108] M. Lupu, F. Piroi, X. Huang, J. Zhu, J. Tait, Overview of the TREC 2009 chemical IR track, in: TREC, 2009.
- [109] Y. Mejova, V.H. Thuc, S. Foster, C. Harris, B. Arens, P. Srinivasan, TREC blog and TREC chem: A view from the corn fields, 2009.
- [110] T. Strohman, D. Metzler, H. Turtle, W.B. Croft, Indri: A language model-based search engine for complex queries, in: Proceedings of the International Conference on Intelligent Analysis, vol. 2, Washington, DC., 2005, pp. 2–6.
- [111] F. Piroi, MAREC/IREC: the matrixware research collection / information retrieval facility research collection, 2021, <http://dx.doi.org/10.48436/2zx6e-5pr64>.
- [112] F. Kreuchau, V. Korzinov, A patent search strategy based on machine learning for the emerging field of service robotics, *Scientometrics* 111 (2) (2017) 743–772, <http://dx.doi.org/10.1007/s11192-017-2268-3>.
- [113] R. Setchi, I. Spasić, J. Morgan, C. Harrison, R. Corken, Artificial intelligence for patent prior art searching, *World Pat. Inf.* 64 (2021) 102021, <http://dx.doi.org/10.1016/j.wpi.2021.102021>.
- [114] A. Kravets, N. Lebedev, M. Legenchenko, Patents images retrieval and convolutional neural network training dataset quality improvement, in: IV International Research Conference “Information Technologies in Science, Management, Social Sphere and Medicine”, ITSMSSM 2017, Atlantis Press, 2017, pp. 287–293, <http://dx.doi.org/10.2991/itsmssm-17.2017.59>.
- [115] M. Kucer, D. Oyen, J. Castorena, J. Wu, Deepatent: Large scale patent drawing recognition and retrieval, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2022, pp. 2309–2318, <http://dx.doi.org/10.1109/WACV51458.2022.00063>.
- [116] K. Higuchi, Y. Honbu, K. Yanai, Patent image retrieval using cross-entropy-based metric learning, in: Proc. of the 29th International Workshop on Frontiers of Computer Vision, 2023, <http://dx.doi.org/10.1016/j.wpi.2023.102217>.
- [117] H. Wang, Y. Zhang, Learning efficient representations for image-based patent retrieval, in: Chinese Conference on Pattern Recognition and Computer Vision, PRCV, Springer, 2023, pp. 15–26, http://dx.doi.org/10.1007/978-981-99-8540-1_2.
- [118] H.-C. Lo, J.-M. Chu, J. Hsiang, C.-C. Cho, Large language model informed patent image retrieval, 2024, <http://dx.doi.org/10.48550/arXiv.2404.19360>, arXiv preprint [arXiv:2404.19360](https://arxiv.org/abs/2404.19360).
- [119] K. Kavimandan, A. Nalmpantis, E. Beauxis-Aussalet, R.-J. Sips, Hierarchical multi-positive contrastive learning for patent image retrieval, 2025, <http://dx.doi.org/10.48550/arXiv.2506.13496>, arXiv preprint [arXiv:2506.13496](https://arxiv.org/abs/2506.13496).
- [120] D.M. Kang, C.C. Lee, S. Lee, W. Lee, Patent prior art search using deep learning language model, in: Proceedings of the 24th Symposium on International Database Engineering & Applications, 2020, pp. 1–5, <http://dx.doi.org/10.1145/3410566.3410597>.
- [121] K. Pustularen, G. Bruns, R. Ewerth, A multimodal approach for semantic patent image retrieval, in: PatentSemTech 2021: Patent Text Mining and Semantic Technologies 2021: Proceedings of the 2nd Workshop on Patent Text Mining and Semantic Technologies (PatentSemTech) 2021, Co-Located with the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2021), vol. 2909, Aachen, Germany: RWTH Aachen, 2021, pp. 45–49, <http://dx.doi.org/10.15488/16876>.
- [122] K. Higuchi, K. Yanai, Patent image retrieval using transformer-based deep metric learning, *World Pat. Inf.* 74 (2023) 102217, <http://dx.doi.org/10.1016/j.wpi.2023.102217>.
- [123] R. Chikkamath, A. Sharma, C. Hewel, M. Endres, ChatGPT vs. Google gemini: Assessing AI frontiers for patent prior art search using European search reports, 2022.
- [124] K.-Y. Lee, J. Bai, PAI-NET: Retrieval-augmented generation patent network using prior art information, *Systems* 13 (4) (2025) 259, <http://dx.doi.org/10.3390/systems13040259>.
- [125] K. Daniell, I. Buzhinsky, S. Björkqvist, Efficient patent searching using graph transformers, 2025, <http://dx.doi.org/10.48550/arXiv.2508.10496>, arXiv preprint [arXiv:2508.10496](https://arxiv.org/abs/2508.10496).
- [126] S. Awale, E. Müller-Budack, R. Delaviz, R. Ewerth, Exploring patents visually: An interactive search system for multimodal patent image search and interpretation, 2025.
- [127] C. Gormley, Z. Tong, Elasticsearch: the Definitive Guide: a Distributed Real-Time Search and Analytics Engine, O'Reilly Media, Inc., 2015.
- [128] Freepatent, 2025, <https://www.freepatent.ru/>. (Accessed 07 July 2025).
- [129] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778, <http://dx.doi.org/10.48550/arXiv.1512.03385>.
- [130] F. Radenović, G. Tolias, O. Chum, Fine-tuning CNN image retrieval with no human annotation, *IEEE Trans. Pattern Anal. Mach. Intell.* 41 (7) (2018) 1655–1668, <http://dx.doi.org/10.1109/TPAMI.2018.2846566>.
- [131] Q. Yu, Y. Yang, F. Liu, Y.-Z. Song, T. Xiang, T.M. Hospedales, Sketch-a-net: A deep neural network that beats humans, *Int. J. Comput. Vis.* 122 (3) (2017) 411–425, <http://dx.doi.org/10.1007/s11263-016-0932-3>.
- [132] P. Sangkloy, N. Burnell, C. Ham, J. Hays, The sketchy database: learning to retrieve badly drawn bunnies, *ACM Trans. Graph.* 35 (4) (2016) 1–12, <http://dx.doi.org/10.1145/2897824.292595>.
- [133] S. Jiang, J. Luo, G. Ruiz-Pava, J. Hu, C.L. Magee, Deriving design feature vectors for patent images using convolutional neural networks, *J. Mech. Des.* 143 (6) (2021) 061405, <http://dx.doi.org/10.1115/1.4049214>.
- [134] A.v.d. Oord, Y. Li, O. Vinyals, Representation learning with contrastive predictive coding, 2018, <http://dx.doi.org/10.48550/arXiv.1807.03748>, arXiv preprint [arXiv:1807.03748](https://arxiv.org/abs/1807.03748).
- [135] J. Deng, J. Guo, N. Xue, S. Zafeiriou, Arcface: Additive angular margin loss for deep face recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 4690–4699, <http://dx.doi.org/10.48550/arXiv.1801.07698>.
- [136] H. Xuan, A. Stylianou, X. Liu, R. Pless, Hard negative examples are hard, but useful, in: European Conference on Computer Vision, Springer, 2020, pp. 126–142, http://dx.doi.org/10.1007/978-3-030-58568-6_8.
- [137] J. Johnson, M. Douze, H. Jégou, Billion-scale similarity search with GPUs, *IEEE Trans. Big Data* 7 (3) (2019) 535–547, <http://dx.doi.org/10.1109/TBDA.2019.2921572>.
- [138] M. Tan, Q. Le, Efficientnet: Rethinking model scaling for convolutional neural networks, in: International Conference on Machine Learning, PMLR, 2019, pp. 6105–6114, <http://dx.doi.org/10.48550/arXiv.1905.11946>.
- [139] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, 2020, arXiv preprint [arXiv:2010.11929](https://arxiv.org/abs/2010.11929).
- [140] Z. Zhong, L. Zheng, G. Kang, S. Li, Y. Yang, Random erasing data augmentation, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, 2020, pp. 13001–13008, <http://dx.doi.org/10.1609/aaai.v34i07.7000>.
- [141] P. Chen, S. Liu, H. Zhao, X. Wang, J. Jia, Gridmask data augmentation, 2020, <http://dx.doi.org/10.48550/arXiv.2001.04086>, arXiv preprint [arXiv:2001.04086](https://arxiv.org/abs/2001.04086).
- [142] K. Ajayi, J. Wei, M. Gryder, W. Shields, J. Wu, S.M. Jones, M. Kucer, D. Oyen, Deepatent2: A large-scale benchmarking corpus for technical drawing understanding, *Sci. Data* 10 (1) (2023) 772, <http://dx.doi.org/10.1038/s41597-023-02653-7>.
- [143] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), 2019, pp. 4171–4186.
- [144] M. Kang, S. Lee, W. Lee, Prior art search using multi-modal embedding of patent documents, in: 2020 IEEE International Conference on Big Data and Smart Computing, BigComp, IEEE, 2020, pp. 548–550, <http://dx.doi.org/10.1109/BigComp48618.2020.000-6>.
- [145] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: International Conference on Machine Learning, PmlR, 2021, pp. 8748–8763, <http://dx.doi.org/10.48550/arXiv.2103.00020>.
- [146] Y. Baek, B. Lee, D. Han, S. Yun, H. Lee, Character region awareness for text detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 9365–9374, <http://dx.doi.org/10.1109/CVPR.2019.00959>.
- [147] J. Baek, G. Kim, J. Lee, S. Park, D. Han, S. Yun, S.J. Oh, H. Lee, What is wrong with scene text recognition model comparisons? dataset and model analysis, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 4715–4723, <http://dx.doi.org/10.1109/ICCV.2019.00481>.
- [148] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, V. Stoyanov, Roberta: A robustly optimized bert pretraining approach, 2019, <http://dx.doi.org/10.48550/arXiv.1907.11692>, arXiv preprint [arXiv:1907.11692](https://arxiv.org/abs/1907.11692).
- [149] M. Freunek, A. Bodmer, BERT based patent novelty search by training claims to their own description, 2021, <http://dx.doi.org/10.48550/arXiv.2103.01126>, arXiv preprint [arXiv:2103.01126](https://arxiv.org/abs/2103.01126).
- [150] M. Freunek, A. Bodmer, BERT based freedom to operate patent analysis, 2021, <http://dx.doi.org/10.48550/arXiv.2105.00817>, arXiv preprint [arXiv:2105.00817](https://arxiv.org/abs/2105.00817).
- [151] K. Vowinkel, V.D. Hähnke, SEARCHFORMER: Semantic patent embeddings by siamese transformers for prior art search, *World Pat. Inf.* 73 (2023) 102192, <http://dx.doi.org/10.1016/j.wpi.2023.102192>.
- [152] R. Srebrovic, J. Yonamine, Leveraging the BERT algorithm for patents with TensorFlow and BigQuery, 2020, White paper.
- [153] A. Supraja, S. Archana, S. Suvetha, T. Geetha, Patent search and trend analysis, in: 2015 IEEE International Advance Computing Conference, IACC, IEEE, 2015, pp. 501–506, <http://dx.doi.org/10.1109/IADCC.2015.7154759>.
- [154] S. Sarica, B. Song, E. Low, J. Luo, Engineering knowledge graph for keyword discovery in patent search, in: Proceedings of the Design Society: International Conference on Engineering Design, vol. 1, Cambridge University Press, 2019, pp. 2249–2258, <http://dx.doi.org/10.1017/dsi.2019.231>.

- [155] S. Vrochidis, A. Moutzidou, I. Kompatsiaris, Concept-based patent image retrieval, *World Pat. Inf.* 34 (4) (2012) 292–303, <http://dx.doi.org/10.1016/j.wpi.2012.07.002>.
- [156] L. Andersson, M. Lupu, J. Palotti, A. Hanbury, A. Rauber, When is the time ripe for natural language processing for patent passage retrieval? in: *Proceedings of the 25th ACM International on Conference on Information and Knowledge Management*, 2016, pp. 1453–1462, <http://dx.doi.org/10.1145/2983323.2983858>.
- [157] R. Chikkamath, M. Endres, L. Bayyapu, C. Hewel, An empirical study on patent novelty detection: A novel approach using machine learning and natural language processing, in: *2020 Seventh International Conference on Social Networks Analysis, Management and Security, SNAMS, IEEE*, 2020, pp. 1–7, <http://dx.doi.org/10.1109/SNAMS52053.2020.9336557>.
- [158] S. Björkqvist, J. Kallio, Building a graph-based patent search engine, in: *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2023, pp. 3300–3304, <http://dx.doi.org/10.1145/3539618.3591842>.
- [159] V. Stamatis, M. Salampasis, K. Diamantaras, A novel re-ranking architecture for patent search, *World Pat. Inf.* 78 (2024) 102282, <http://dx.doi.org/10.1016/j.wpi.2024.102282>.
- [160] A. Ali, M.A. Humayun, L.C.D. Silva, P.E. Abas, Optimizing patent prior art search: An approach using patent abstract and key terms, *Information* 16 (2) (2025) 145, <http://dx.doi.org/10.3390/info16020145>.
- [161] M. Buckley, S. Althammer, A. Qoku, *PatentExplorer: Refining patent search with domain-specific topic models*, 2021.
- [162] D.M. Blei, A.Y. Ng, M.I. Jordan, Latent dirichlet allocation, *J. Mach. Learn. Res.* 3 (Jan) (2003) 993–1022.
- [163] R. Tandon, S. Sra, Sparse nonnegative matrix approximation: new formulations and algorithms, 2010.
- [164] P. Sidiropoulos, S. Vrochidis, I. Kompatsiaris, Content-based binary image retrieval using the adaptive hierarchical density histogram, *Pattern Recognit.* 44 (4) (2011) 739–750, <http://dx.doi.org/10.1016/j.patcog.2010.09.014>.
- [165] S. Vrochidis, A. Mountzidou, I. Kompatsiaris, PatMedia patent image search engine, 2025, <http://mklab-services.iti.gr/patmedia/index.php?login=true>. (Accessed 04 February 2025).
- [166] C. Li, W. Li, Y. Hong, H. Xiang, A patent retrieval method and system based on double classification, *Inform. Sci.* 672 (2024) 120659, <http://dx.doi.org/10.1016/j.ins.2024.120659>.
- [167] T. Mikolov, I. Sutskever, K. Chen, G.S. Corrado, J. Dean, Distributed representations of words and phrases and their compositionality, *Adv. Neural Inf. Process. Syst.* 26 (2013).
- [168] J. Pennington, R. Socher, C.D. Manning, Glove: Global vectors for word representation, in: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing, EMNLP*, 2014, pp. 1532–1543.
- [169] P. Bojanowski, E. Grave, A. Joulin, T. Mikolov, Enriching word vectors with subword information, *Trans. Assoc. Comput. Linguist.* 5 (2017) 135–146, http://dx.doi.org/10.1162/tac1a_00051.
- [170] J. Risch, R. Krestel, Domain-specific word embeddings for patent classification, *Data Technol. Appl.* 53 (1) (2019) 108–122, <http://dx.doi.org/10.1108/DTA-01-2019-0002>.
- [171] J. Risch, N. Alder, C. Hewel, R. Krestel, Patentmatch: a dataset for matching patent claims & prior art, 2020, <http://dx.doi.org/10.48550/arXiv.2012.13919>, arXiv preprint [arXiv:2012.13919](https://arxiv.org/abs/2012.13919).
- [172] G. Kumaravel, S. Sankaranarayanan, PQPS: Prior-art query-based patent summarizer using RBM and bi-LSTM, *Mob. Inf. Syst.* 2021 (1) (2021) 2497770, <http://dx.doi.org/10.1155/2021/2497770>.
- [173] M. Honnibal, I. Montani, S.V. Landeghem, A. Boyd, spaCy: Industrial-strength natural language processing in python, 2020, Zenodo. <https://spacy.io>.
- [174] K.S. Tai, R. Socher, C.D. Manning, Improved semantic representations from tree-structured long short-term memory networks, 2015, <http://dx.doi.org/10.48550/arXiv.1503.00075>, arXiv preprint [arXiv:1503.00075](https://arxiv.org/abs/1503.00075).
- [175] H. Gurulingappa, B. Müller, R. Klinger, H.-T. Mevissen, M. Hofmann-Apitius, C.M. Friedrich, J. Fluck, Prior art search in chemistry patents based on semantic concepts and co-citation analysis, in: *TREC*, 2010.
- [176] E. Pasche, J. Gobeill, D. Teodoro, A. Gaudinat, D. Vishnykova, C. Lovis, P. Ruch, An advanced search engine for patent analytics in medicinal chemistry, in: *Quality of Life Through Quality of Information*, IOS Press, 2012, pp. 204–209, <http://dx.doi.org/10.3233/978-1-61499-101-4-204>.
- [177] E. Pasche, J. Gobeill, O. Kreim, F. Oezdemir-Zaeach, T. Vachon, C. Lovis, P. Ruch, Development and tuning of an original search engine for patent libraries in medicinal chemistry, *BMC Bioinformatics* 15 (Suppl. 1) (2014) S15, <http://dx.doi.org/10.1186/1471-2105-15-S1-S15>.
- [178] L. Morin, V. Weber, G.I. Meijer, F. Yu, P.W. Staar, PatCID: an open-access dataset of chemical structures in patent documents, *Nat. Commun.* 15 (1) (2024) 6532, <http://dx.doi.org/10.1038/s41467-024-50779-y>.
- [179] L. Tu, S. Lv, H. Shi, Research on the application of natural language processing technology in medicine patent retrieval, in: *2024 5th International Conference on Information Science, Parallel and Distributed Systems, ISPPDS, IEEE*, 2024, pp. 176–181, <http://dx.doi.org/10.1109/ISPPDS62779.2024.10667520>.
- [180] National Institute of Standards and Technology (NIST), TREC chemical IR track (TREC-chem), 2009–2011, Last updated: August 29, URL <https://trec.nist.gov/data/chem-ir.html>. (Accessed 28 July 2025).
- [181] Information Retrieval Facility, PatOlympics – IRF symposium 2011, 2011, <https://www.ir-facility.org/events/irf-symposium/irf-symposium-2011/patolympics>. (Accessed 28 July 2025).
- [182] M. Hofmann-Apitius, J. Fluck, L. Furlong, O. Fornes, C. Kolářik, S. Hanser, M. Boeker, S. Schulz, F. Sanz, R. Klinger, et al., Knowledge environments representing molecular entities for the virtual physiological human, *Philos. Trans. R. Soc. A: Math. Phys. Eng. Sci.* 366 (1878) (2008) 3091–3110, <http://dx.doi.org/10.1098/rsta.2008.0099>.
- [183] P.D. Duek, A. Gleizes, C. Zwahlen, A. Mottaz, A. Bairoch, L. Lane, CALOHA: A new human anatomical ontology as a support for complex queries and tissue expression display in neXtProt, *Bio-Ontologies* 2011 (2011).
- [184] I. Ounis, G. Amati, V. Plachouras, B. He, C. Macdonald, D. Johnson, Terrier information retrieval platform, in: *Advances in Information Retrieval*, in: *Lecture Notes in Computer Science*, vol. 3408, Springer, 2005, pp. 517–519, http://dx.doi.org/10.1007/978-3-540-31865-1_37.
- [185] G. Amati, C.J. Van Rijsbergen, Probabilistic models of information retrieval based on measuring the divergence from randomness, *ACM Trans. Inf. Syst. (TOIS)* 20 (4) (2002) 357–389, <http://dx.doi.org/10.1145/582415.582416>.
- [186] A. Lourenço, R. Carreira, S. Carneiro, P. Maia, D. Glez-Pena, F. Fdez-Riverola, E.C. Ferreira, I. Rocha, M. Rocha, @ note: a workbook for biomedical text mining, *J. Biomed. Inform.* 42 (4) (2009) 710–720, <http://dx.doi.org/10.1016/j.jbi.2009.04.002>.
- [187] S. Taduri, K.H. Law, J.P. Kesan, R.D. Sriram, Utilization of bio-ontologies for enhancing patent information retrieval, *COMPSAC*, in: *2019 IEEE 43rd Annual Computer Software and Applications Conference*, vol. 2, IEEE, 2019, pp. 91–96, <http://dx.doi.org/10.1109/COMPSAC.2019.10189>.
- [188] N.F. Noy, N.H. Shah, P.L. Whetzel, B. Dai, M. Dorf, N. Griffith, C. Jonquet, D.L. Rubin, M.-A. Storey, C.G. Chute, et al., BioPortal: ontologies and integrated data resources at the click of a mouse, *Nucleic Acids Res.* 37 (Suppl. 2) (2009) W170–W173, <http://dx.doi.org/10.1093/nar/gkp440>.
- [189] J. Golbeck, G. Fragoso, F. Hartel, J. Hendler, J. Oberthaler, B. Parsia, The national cancer institute's thesaurus and ontology, *J. Web Semant.* 1 (1) (2003) 75–80, <http://dx.doi.org/10.1016/j.websem.2003.07.007>.
- [190] C.E. Lipscomb, Medical subject headings (MeSH), *Bull. Med. Libr. Assoc.* 88 (3) (2000) 265.
- [191] M. Ashburner, C.A. Ball, J.A. Blake, D. Botstein, H. Butler, J.M. Cherry, A.P. Davis, K. Dolinski, S.S. Dwight, J.T. Eppig, et al., Gene ontology: tool for the unification of biology, *Nature Genet.* 25 (1) (2000) 25–29, <http://dx.doi.org/10.1038/75556>.
- [192] S.H. Brown, P.L. Elkin, C. Husser, B.A. Bauer, M.J. Lincoln, J. Carter, M. Erlbaum, M.S. Tuttle, et al., VA national drug file reference terminology: a cross-institutional content coverage study, in: *MEDINFO 2004*, IOS Press, 2004, pp. 477–481, <http://dx.doi.org/10.3233/978-1-60750-949-3-477>.
- [193] Neo4j graph database, 2025, <https://neo4j.com>. (Accessed 08 November 2025).
- [194] Y. Gadiya, A. Zaliani, P. Gribbon, M. Hofmann-Apitius, PEMT: a patent enrichment tool for drug discovery, *Bioinformatics* 39 (1) (2023) btac716, <http://dx.doi.org/10.1093/bioinformatics/btac716>.
- [195] A. Gaulton, L.J. Bellis, A.P. Bento, J. Chambers, M. Davies, A. Hersey, Y. Light, S. McGlinchey, D. Michalovich, B. Al-Lazikani, et al., ChEMBL: a large-scale bioactivity database for drug discovery, *Nucleic Acids Res.* 40 (D1) (2012) D1100–D1107, <http://dx.doi.org/10.1093/nar/gkr777>.
- [196] G. Papadatos, M. Davies, N. Dedman, J. Chambers, A. Gaulton, J. Siddle, R. Koks, S.A. Irvine, J. Pettersson, N. Goncharoff, et al., SureChEMBL: a large-scale, chemically annotated patent document database, *Nucleic Acids Res.* 44 (D1) (2016) D1220–D1228, <http://dx.doi.org/10.1093/nar/gkv1253>.
- [197] Orphanet, 2025, <https://www.orpha.net>. (Accessed 07 November 2025).
- [198] Y. Shimizu, M. Ohta, S. Ishida, K. Terayama, M. Osawa, T. Honma, K. Ikeda, AI-driven molecular generation of not-patented pharmaceutical compounds using world open patent data, *J. Cheminformatics* 15 (1) (2023) 120, <http://dx.doi.org/10.1186/s13321-023-00791-z>.
- [199] S.R. Heller, A. McNaught, I. Pletnev, S. Stein, D. Tchekhovskoi, InChI, the IUPAC international chemical identifier, *J. Cheminformatics* 7 (1) (2015) 23, <http://dx.doi.org/10.1186/s13321-015-0068-4>.
- [200] K. Rajan, H.O. Brinkhaus, M. Sorokina, A. Zieslesny, C. Steinbeck, DECIMER-segmentation: Automated extraction of chemical structure depictions from scientific literature, *J. Cheminformatics* 13 (1) (2021) 20, <http://dx.doi.org/10.1186/s13321-021-00496-1>.
- [201] L. Morin, M. Danelljan, M.I. Agea, A. Nassar, V. Weber, I. Meijer, P. Staar, F. Yu, MolGrapher: graph-based visual recognition of chemical structures, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 19552–19561, <http://dx.doi.org/10.1109/iccvc51070.2023.01791>.
- [202] C.W. Kosonocky, C.O. Wilke, E.M. Marcotte, A.D. Ellington, Mining patents with large language models elucidates the chemical function landscape, *Digit. Discov.* 3 (6) (2024) 1150–1159, <http://dx.doi.org/10.1039/D4DD00011K>.
- [203] OpenAI, Text-embedding-ada-002, 2025, <https://platform.openai.com/docs/models/text-embedding-ada-002>. (Accessed 10 November 2025).

- [204] A.S. Geissler, J. Gorodkin, S.E. Seemann, Patent data-driven analysis of literature associations with changing innovation trends, *Front. Res. Metrics Anal.* 9 (2024) 1432673, <http://dx.doi.org/10.3389/frma.2024.1432673>.
- [205] S. Frihat, N. Fuhr, Integration of biomedical concepts for enhanced medical literature retrieval, *Int. J. Data Sci. Anal.* (2025) 1–14, <http://dx.doi.org/10.1007/s41060-025-00724-z>.
- [206] National Library of Medicine, Medical subject headings 2025, 2025, URL <https://meshb.nlm.nih.gov/>.
- [207] R. Rodriguez-Esteban, M. Bundschuh, Text mining patents for biomedical knowledge, *Drug Discov. Today* 21 (6) (2016) 997–1002, <http://dx.doi.org/10.1016/j.drudis.2016.05.002>.
- [208] K. Degtyarenko, P. De Matos, M. Ennis, J. Hastings, M. Zbinden, A. McNaught, R. Alcántara, M. Darsow, M. Guedj, M. Ashburner, ChEBI: a database and ontology for chemical entities of biological interest, *Nucleic Acids Res.* 36 (Suppl. 1) (2007) D344–D350, <http://dx.doi.org/10.1093/nar/gkm791>.
- [209] T. Grego, P. Pék, F.M. Couto, D. Rebholz-Schuhmann, Identification of chemical entities in patent documents, in: *International Work-Conference on Artificial Neural Networks*, Springer, 2009, pp. 942–949, http://dx.doi.org/10.1007/978-3-642-02481-8_144.
- [210] M. Kiss, Á. Nagy, V. Vincze, A. Almási, Z. Alexin, J. Csirik, A manually annotated corpus of pharmaceutical patents, in: *International Conference on Text, Speech and Dialogue*, Springer, 2012, pp. 135–142, http://dx.doi.org/10.1007/978-3-642-32790-2_16.
- [211] M. Krallinger, O. Rabal, A. Lourenço, M.P. Perez, G.P. Rodriguez, M. Vazquez, F. Leitner, J. Oyarzabal, A. Valencia, Overview of the CHEMDNER patents task, in: *Proceedings of the Fifth BioCreative Challenge Evaluation Workshop*, 2015, pp. 63–75.
- [212] K.M. Hettne, R.H. Stierum, M.J. Schuemie, P.J. Hendriksen, B.J. Schijvennaars, E.M.v. Mulligen, J. Kleinjans, J.A. Kors, A dictionary to identify small molecules and drugs in free text, *Bioinformatics* 25 (22) (2009) 2983–2991, <http://dx.doi.org/10.1093/bioinformatics/btp535>.
- [213] D.S. Wishart, C. Knox, A.C. Guo, S. Shrivastava, M. Hassanali, P. Stothard, Z. Chang, J. Woolsey, DrugBank: a comprehensive resource for in silico drug discovery and exploration, *Nucleic Acids Res.* 34 (Suppl. 1) (2006) D668–D672, <http://dx.doi.org/10.1093/nar/gkj067>.
- [214] S. Eltyeb, N. Salim, Chemical named entities recognition: a review on approaches and applications, *J. Cheminformatics* 6 (1) (2014) 17, <http://dx.doi.org/10.1186/1758-2946-6-17>.
- [215] M. Narayanaswamy, K. Ravikumar, K. Vijay-Shanker, A biological named entity recognizer, in: *Biocomputing 2003*, World Scientific, 2002, pp. 427–438, http://dx.doi.org/10.1142/9789812776303_0040.
- [216] A. McCallum, D. Freitag, F.C. Pereira, et al., Maximum entropy Markov models for information extraction and segmentation, in: *Icml*, vol. 17, 2000, pp. 591–598.
- [217] T. Grego, C. Pesquita, H.P. Bastos, F.M. Couto, Chemical entity recognition and resolution to ChEBI, *Int. Sch. Res. Not.* 2012 (1) (2012) 619427, <http://dx.doi.org/10.5402/2012/619427>.
- [218] B. Settles, ABNER: an open source tool for automatically tagging genes, proteins and other entity names in text, *Bioinformatics* 21 (14) (2005) 3191–3192, <http://dx.doi.org/10.1093/bioinformatics/bti475>.
- [219] R. Leaman, G. Gonzalez, BANNER: an executable survey of advances in biomedical named entity recognition, in: *Biocomputing 2008*, World Scientific, 2008, pp. 652–663, http://dx.doi.org/10.1142/9789812776136_0062.
- [220] T. Rocktäschel, M. Weidlich, U. Leser, ChemSpot: a hybrid system for chemical named entity recognition, *Bioinformatics* 28 (12) (2012) 1633–1640, <http://dx.doi.org/10.1093/bioinformatics/bts183>.
- [221] S.A. Akhondi, E. Pons, Z. Afzal, H. van Haagen, B.F. Becker, K.M. Hettne, E.M. van Mulligen, J.A. Kors, Chemical entity recognition in patents by combining dictionary-based and statistical approaches, *Database* 2016 (2016) baw061, <http://dx.doi.org/10.1093/database/baw061>.
- [222] T.H. Dang, H.-Q. Le, T.M. Nguyen, S.T. Vu, D3NER: biomedical named entity recognition using CRF-biLSTM improved with fine-tuned embeddings of various linguistic information, *Bioinformatics* 34 (20) (2018) 3539–3546, <http://dx.doi.org/10.1093/bioinformatics/bty356>.
- [223] F. Saad, Named entity recognition for biomedical patent text using bi-lstm variants, in: *Proceedings of the 21st International Conference on Information Integration and Web-Based Applications & Services*, 2019, pp. 617–621, <http://dx.doi.org/10.1145/3366030.3366104>.
- [224] Z. Zhai, D.Q. Nguyen, S.A. Akhondi, C. Thorne, C. Druckenbrodt, T. Cohn, M. Gregory, K. Verspoor, Improving chemical named entity recognition in patents with contextualized word embeddings, 2019, <http://dx.doi.org/10.18653/v1/W19-5035>, arXiv preprint arXiv:1907.02679.
- [225] Y. Hu, S. Verberne, Named entity recognition for Chinese biomedical patents, in: *Proceedings of the 28th International Conference on Computational Linguistics*, 2020, pp. 627–637, <http://dx.doi.org/10.18653/v1/2020.coling-main.54>.
- [226] F. Saad, H. Aras, R. Hackl-Sommer, Improving named entity recognition for biomedical and patent data using bi-LSTM deep neural network models, in: *International Conference on Applications of Natural Language To Information Systems*, Springer, 2020, pp. 25–36, http://dx.doi.org/10.1007/978-3-030-51310-8_3.
- [227] D. Mahendran, G. Gurdin, N. Lewinski, C. Tang, B.T. McInnes, Identifying chemical reactions and their associated attributes in patents, *Front. Res. Metrics Anal.* 6 (2021) 688353, <http://dx.doi.org/10.3389/frma.2021.688353>.
- [228] J. Copara, N. Naderi, J. Knafo, P. Ruch, D. Teodoro, Named entity recognition in chemical patents using ensemble of contextual language models, in: *Proceedings of the CLEF 2020 Conference*, 22–25 September 2020, 2020, <http://dx.doi.org/10.48550/arXiv.2007.12569>.
- [229] C. Malarkodi, P.R. Rao, S.L. Devi, CLRG ChemNER: A chemical named entity recognizer@ ChEMU CLEF 2020., in: *CLEF (Working Notes)*, 2020.
- [230] I.V. Filippov, M.C. Nicklaus, Optical structure recognition software to recover chemical information: OSRA, an open source solution, 2009, <http://dx.doi.org/10.1021/ci800067r>.
- [231] M. Zimmermann, Chemical structure reconstruction with chemoCR, in: *TREC*, 2011.
- [232] M. Oldenhof, A. Arany, Y. Moreau, J. Simm, ChemGrapher: optical graph recognition of chemical compounds by deep learning, *J. Chem. Inf. Model.* 60 (10) (2020) 4506–4517, <http://dx.doi.org/10.1021/acs.jcim.0c00459>.
- [233] J. Park, G.R. Rosania, K.A. Shedden, M. Nguyen, N. Lyu, K. Saitou, Automated extraction of chemical structure information from digital raster images, *Chem. Central J.* 3 (1) (2009) 4, <http://dx.doi.org/10.1186/1752-153X-3-4>.
- [234] A.T. Valko, A.P. Johnson, CLiDE pro: the latest generation of CLiDE, a tool for optical chemical structure recognition, *J. Chem. Inf. Model.* 49 (4) (2009) 780–787, <http://dx.doi.org/10.1021/ci800449t>.
- [235] J. Staker, K. Marshall, R. Abel, C.M. McQuaw, Molecular structure extraction from documents using deep learning, *J. Chem. Inf. Model.* 59 (3) (2019) 1017–1029, <http://dx.doi.org/10.1021/acs.jcim.8b00669>.
- [236] E.J. Beard, J.M. Cole, ChemSchematicResolver: a toolkit to decode 2D chemical diagrams with labels and R-groups into annotated chemical named entities, *J. Chem. Inf. Model.* 60 (4) (2020) 2059–2072, <http://dx.doi.org/10.1021/acs.jcim.0c00042>.
- [237] M.C. Swain, J.M. Cole, ChemDataExtractor: a toolkit for automated extraction of chemical information from the scientific literature, *J. Chem. Inf. Model.* 56 (10) (2016) 1894–1904, <http://dx.doi.org/10.1021/acs.jcim.6b00207>.
- [238] J. Wang, Z. Shen, Y. Liao, Z. Yuan, S. Li, G. He, M. Lan, X. Qian, K. Zhang, H. Li, Multi-modal chemical information reconstruction from images and texts for exploring the near-drug space, *Brief. Bioinform.* 23 (6) (2022) bbac461, <http://dx.doi.org/10.1093/bib/bbac461>.
- [239] Our World in Data, Number of entries in biological sequence databases, 2025, <https://ourworldindata.org/grapher/number-of-entries-in-biological-sequence-databases>. (Accessed 10 November 2025).
- [240] S.F. Altschul, W. Gish, W. Miller, E.W. Myers, D.J. Lipman, Basic local alignment search tool, *J. Mol. Biol.* 215 (3) (1990) 403–410, [http://dx.doi.org/10.1016/S0022-2836\(05\)80360-2](http://dx.doi.org/10.1016/S0022-2836(05)80360-2).
- [241] R. Casey, BLAST sequences aid in genomics and proteomics, *Bus. Intell. Netw.* (2005).
- [242] University of California, Berkeley Library, BLAST: Basic local alignment search tool, 2025, <https://guides.lib.berkeley.edu/ncbi/blast>. (Accessed 11 November 2025).
- [243] S.F. Altschul, T.L. Madden, A.A. Schäffer, J. Zhang, Z. Zhang, W. Miller, D.J. Lipman, Gapped BLAST and PSI-BLAST: a new generation of protein database search programs, *Nucleic Acids Res.* 25 (17) (1997) 3389–3402, <http://dx.doi.org/10.1093/nar/25.17.3389>.
- [244] D.J. Lipman, W.R. Pearson, Rapid and sensitive protein similarity searches, *Science* 227 (4693) (1985) 1435–1441, <http://dx.doi.org/10.1126/science.2983426>.
- [245] European Bioinformatics Institute (EMBL-EBI), FASTA – EMBL-EBI job dispatcher service, 2025, https://www.ebi.ac.uk/jdispatcher/sss/fasta?database=uniprotkb_swissprot. (Accessed 11 November 2025).
- [246] W.J. Kent, BLAT—the BLAST-like alignment tool, *Genome Res.* 12 (4) (2002) 656–664, <http://dx.doi.org/10.1101/gr.229202>.
- [247] Science Foundation Ireland, Clustal: Multiple sequence alignment, 2025, <http://www.clustal.org/>. (Accessed 11 November 2025).
- [248] P.J. Andree, M.F. Harper, S. Nauche, R.A. Poolman, J. Shaw, J.C. Swinkels, S. Wycherley, A comparative study of patent sequence databases, *World Pat. Inf.* 30 (4) (2008) 300–308, <http://dx.doi.org/10.1016/j.wpi.2008.04.005>.
- [249] G.G. Xu, A. Webster, E. Doran, Patent sequence databases, *World Pat. Inf.* 24 (2) (2002) 95–101, [http://dx.doi.org/10.1016/S0172-2190\(02\)00004-2](http://dx.doi.org/10.1016/S0172-2190(02)00004-2).
- [250] W. Li, H. McWilliam, A.R. de la Torre, A. Grodowski, I. Benediktovich, M. Goujon, S. Nauche, R. Lopez, Non-redundant patent sequence databases with value-added annotations at two levels, *Nucleic Acids Res.* 38 (Suppl. 1) (2010) D52–D56, <http://dx.doi.org/10.1093/nar/gkp960>.
- [251] W. Li, B. Kondratowicz, H. McWilliam, S. Nauche, R. Lopez, The annotation-enriched non-redundant patent sequence databases, *Database* 2013 (2013) bat005, <http://dx.doi.org/10.1093/database/bat005>.
- [252] R.J. Rouse, J. Castagnetto, R.H. Niedner, PatGen—a consolidated resource for searching genetic patent sequences, *Bioinformatics* 21 (8) (2005) 1707–1708, <http://dx.doi.org/10.1093/bioinformatics/bti202>.
- [253] B. Lee, T. Kim, S.-K. Kim, K.H. Lee, D. Lee, Patome: a database server for biological sequence annotation and analysis in issued patents and published patent applications, *Nucleic Acids Res.* 35 (Suppl. 1) (2007) D47–D50, <http://dx.doi.org/10.1093/nar/gkl807>.

- [254] O.A. Jefferson, D. Köllhofer, P. Ajjikuttira, R.A. Jefferson, Public disclosure of biological sequences in global patent practice, *World Pat. Inf.* 43 (2015) 12–24, <http://dx.doi.org/10.1016/j.wpi.2015.08.005>.
- [255] S. Senger, L. Bartek, G. Papadatos, A. Gaulton, Managing expectations: assessment of chemistry databases generated by automated extraction of chemical structures from patents, *J. Cheminformatics* 7 (1) (2015) 49, <http://dx.doi.org/10.1186/s13321-015-0097-z>.
- [256] S. Chithrananda, G. Grand, B. Ramsundar, Chemberta: large-scale self-supervised pretraining for molecular property prediction, 2020, <http://dx.doi.org/10.48550/arXiv.2010.09885>, arXiv preprint arXiv:2010.09885.
- [257] G. Kallergis, E. Asgari, M. Empting, A.K. Hirsch, F. Klawonn, A.C. McHardy, Domain adaptable language modeling of chemical compounds identifies potent pathoblockers for *Pseudomonas aeruginosa*, *Commun. Chem.* 8 (1) (2025) 114, <http://dx.doi.org/10.1038/s42004-025-01484-4>.
- [258] Y. Ji, Z. Zhou, H. Liu, R.V. Davuluri, DNABERT: pre-trained bidirectional encoder representations from transformers model for DNA-language in genome, *Bioinformatics* 37 (15) (2021) 2112–2120, <http://dx.doi.org/10.1093/bioinformatics/btab083>.
- [259] A. Elnaggar, M. Heinzinger, C. Dallago, G. Rehawi, Y. Wang, L. Jones, T. Gibbs, T. Feher, C. Angerer, M. Steinegger, et al., Prottrans: Toward understanding the language of life through self-supervised learning, *IEEE Trans. Pattern Anal. Mach. Intell.* 44 (10) (2021) 7112–7127, <http://dx.doi.org/10.1109/TPAMI.2021.3095381>.
- [260] C. Edwards, C. Zhai, H. Ji, Text2mol: Cross-modal molecule retrieval with natural language queries, in: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021, pp. 595–607, <http://dx.doi.org/10.18653/v1/2021.emnlp-main.47>.
- [261] X. Tang, A. Tran, J. Tan, M.B. Gerstein, Mollm: a unified language model for integrating biomedical text with 2d and 3D molecular representations, *Bioinformatics* 40 (Suppl. 1) (2024) i357–i368, <http://dx.doi.org/10.1093/bioinformatics/btae260>.
- [262] E. Wang, S. Schmidgall, P.F. Jaeger, F. Zhang, R. Pilgrim, Y. Matias, J. Barral, D. Fleet, S. Azizi, Txxgemma: Efficient and agentic llms for therapeutics, 2025, <http://dx.doi.org/10.48550/arXiv.2504.06196>, arXiv preprint arXiv:2504.06196.
- [263] Google DeepMind, Txxgemma, 2025, <https://deepmind.google/models/gemma/txxgemma/>. (Accessed 27 January 2025).
- [264] B.H. Hall, Patents and patent policy, *Oxf. Rev. Econ. Policy* 23 (4) (2007) 568–587, <http://dx.doi.org/10.1093/oxrep/grm037>.
- [265] S. Karapapa, L. McDonagh, *Intellectual Property Law*, Oxford University Press, Oxford, 2019, <http://dx.doi.org/10.1093/he/9780198747697.001.0001>.
- [266] World Intellectual Property Organization (WIPO), International patent classification (IPC), 2025, <https://www.wipo.int/en/web/classification-ipc>. (Accessed 01 August 2025).
- [267] S. Li, J. Hu, Y. Cui, J. Hu, DeepPatent: patent classification with convolutional neural networks and word embedding, *Scientometrics* 117 (2) (2018) 721–744, <http://dx.doi.org/10.1007/s11192-018-2905-5>.
- [268] European Patent Office (EPO), Classification search - espacenet, patent search, 2025, <https://worldwide.espacenet.com/patent/cpc-browser>. (Accessed 27 October 2025).
- [269] J. Yun, Y. Geum, Automated classification of patents: A topic modeling approach, *Comput. Ind. Eng.* 147 (2020) 106636, <http://dx.doi.org/10.1016/j.cie.2020.106636>.
- [270] H. Bekamiri, D.S. Hain, R. Jurowetzki, Patentsberta: A deep nlp based hybrid model for patent distance and classification using augmented sbert, *Technol. Forecast. Soc. Change* 206 (2024) 123536, <http://dx.doi.org/10.1016/j.techfore.2024.123536>.
- [271] J. Wang, S.K.R. Mudhiganti, M. Sharma, Patentformer: A novel method to automate the generation of patent applications, in: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 2024, pp. 1361–1380, <http://dx.doi.org/10.18653/v1/2024.emnlp-industry.101>.
- [272] D. Christofidellis, A.B. Torres, A. Dave, M. Roveri, K. Schmidt, S. Swaminathan, H. Vandierendonck, D. Zubarev, M. Manica, Pgt: a prompt based generative transformer for the patent domain, in: *ICML 2022 Workshop on Knowledge Retrieval and Language Models*, 2022.
- [273] F.-C. Chen, C.-L. Pan, A.D. Team, Evaluating application of large language models to biomedical patent claim generation, *World Pat. Inf.* 80 (2025) 102339, <http://dx.doi.org/10.1016/j.wpi.2025.102339>.

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